Let the Machine Reveal Itself

Machine formation and equilibrium

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Compression sample v4 — October 2026

We begin with a physical event: two light nuclei fuse and release energy. Each step between that event and a working magnetic confinement system introduces a new demand. Following those demands will give us the tokamak's components, the quantities that describe them, and the equations that tie them together.

1. The first requirement: make nuclei approach

The fuel considered here is an equal mixture of deuterium and tritium, the two heavy isotopes of hydrogen. Their reaction is

$$^2\mathrm{H} + {}^3\mathrm{H} \longrightarrow {}^4\mathrm{He}\ (3.5\,\mathrm{MeV}) + n\ (14.1\,\mathrm{MeV}), \qquad Q = 17.6\,\mathrm{MeV}. \tag{1}$$

The energy comes from the decrease in rest mass. In the centre-of-mass frame, the two products carry equal and opposite momenta, so their kinetic energies are inversely proportional to their masses. The neutron takes about four fifths of the released energy; the helium nucleus, or alpha particle, takes the rest. That division will later determine how much energy heats the fuel and how much must be collected outside it.

Both reacting nuclei carry charge $+e$. Their electrostatic repulsion is

$$U(r) = \frac{e^2}{4\pi\epsilon_0 r}. \tag{2}$$

At a separation of 2 fm this is about 0.72 MeV. Quantum tunnelling allows fusion at much lower collision energies. At temperatures of order 10–20 keV, collisions in the energetic part of the ion distribution have a useful probability of penetrating the barrier. We use temperature in energy units throughout: T denotes $k_B T_K$, so 10 keV corresponds to 1.16×10^8 K.

Fuel at this temperature must be kept away from material surfaces. Contact would remove its energy, erode the surface and introduce impurities into the fuel. A *vacuum vessel* provides an evacuated chamber that limits contact with cold gas; the hot fuel inside it must be confined by electromagnetic fields. Reaction energies and the reactor-design framework used here follow Freidberg, Mangiarotti and Minervini.[1]

> **Design consequence.** An evacuated vessel limits cold-gas contact with the hot fuel. Keeping the plasma clear of its wall requires magnetic confinement.

[1] J. P. Freidberg, F. J. Mangiarotti and J. Minervini, "Designing a Tokamak Fusion Reactor—How Does Plasma Physics Fit In?" (2015), open manuscript. This is the reference for the reduced reactor model developed later.

2. Charged particles give us a handle

At these temperatures the fuel is ionized: electrons and nuclei form a plasma. On lengths large compared with its electrostatic screening length, the positive and negative charge densities nearly cancel. This property is called *quasineutrality*. Each constituent still responds to the Lorentz force,

$$m\frac{d\mathbf{v}}{dt} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \tag{3}$$

Set $\mathbf{E} = 0$ and $\mathbf{B} = B\hat{\mathbf{z}}$. Since $\mathbf{v} \cdot (\mathbf{v} \times \mathbf{B}) = 0$, the magnetic force does no work: it preserves the particle's kinetic energy while turning its velocity. With $w = v_x + iv_y$, the perpendicular motion satisfies $\dot{w} = -i(qB/m)w$. Thus $w = w(0)e^{-iqBt/m}$, and integration gives

$$\omega_c = \frac{|q|B}{m}, \qquad r_L = \frac{v_\perp}{\omega_c} = \frac{mv_\perp}{|q|B}. \tag{4}$$

Here ω_c is the gyrofrequency and r_L the gyroradius. Parallel motion remains free, so the orbit is a helix around a magnetic field line. A 10 keV deuteron with $v_\perp = \sqrt{2T/m_D} = 9.8 \times 10^5\,\mathrm{m\,s^{-1}}$ has $r_L = 6.8\,\mathrm{mm}$ in a 3 T field. A metre-scale chamber can therefore contain small transverse orbits. Its next problem is escape along the field.

> **Design consequence.** A magnetic field bounds transverse motion while leaving particles free to stream along it. Confinement must therefore address parallel escape.

3. A straight magnetic corridor

A coil is a conductor wound into turns so that their magnetic fields add. For a long solenoid, the turns surround a cylindrical volume. Ampère's law gives

$$\oint_C \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 I_{\mathrm{linked}}. \tag{5}$$

Choose a rectangular path with length L inside the solenoid and its return far outside. *Model used here.* For a solenoid long compared with its radius, the exterior field is small. With n_{turn} turns per unit length, each carrying current I_c,

$$BL = \mu_0(n_{\mathrm{turn}}L)I_c, \qquad B \simeq \mu_0 n_{\mathrm{turn}} I_c. \tag{6}$$

This field guides particles down the tube. At $v_\parallel \sim 10^6\,\mathrm{m\,s^{-1}}$, a particle traverses a 10 m device in about $10\,\mu$s. Increasing B reduces the gyroradius but leaves this parallel transit time unchanged. Magnetic mirrors can reflect a range of particle pitch angles; here we pursue another way to remove direct end loss: join the ends of the corridor.

> **Design consequence.** The straight solenoid leaves a direct escape path at each end. Closing that path requires a different geometry.

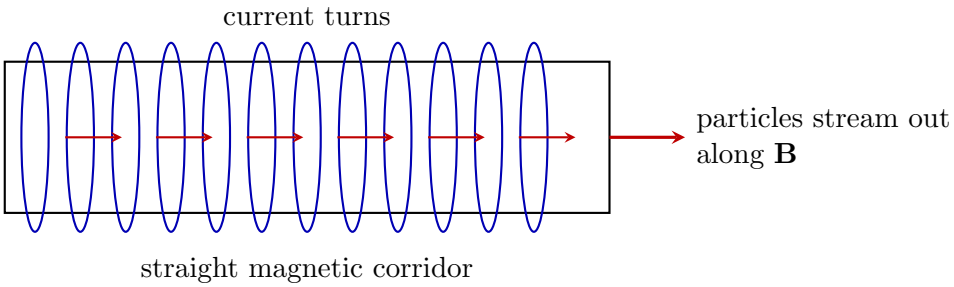


Figure 1. The axial field guides particles through the open ends.

4. Closing the corridor creates the torus

The resulting tube is a torus. Its long-way direction is *toroidal*; the short way around a tube cross-section is *poloidal*. The major radius R_0 measures from the symmetry axis to the cross-section's centre. The minor radius a measures the horizontal extent of the plasma from that centre. Their ratio is

$$A \equiv \frac{R_0}{a} \qquad \text{(aspect ratio)}. \tag{7}$$

Toroidal-field (TF) coils wrap around the tube at successive positions around the ring. Approximating them by a continuous, axisymmetric winding, a circular Ampère path of radius R links N turns carrying I_c:

$$2\pi R B_\varphi(R) = \mu_0 N I_c, \qquad B_\varphi(R) = \frac{\mu_0 N I_c}{2\pi R}. \tag{8}$$

Model used here. The continuous-winding approximation is axisymmetric. Discrete TF coils introduce a periodic field variation called *ripple*. The field is stronger on the *inboard* side, nearest the symmetry axis, than on the *outboard* side:

$$\frac{B_{\text{in}}}{B_{\text{out}}} = \frac{R_0 + a}{R_0 - a} = \frac{A+1}{A-1}. \tag{9}$$

The closed path removes the ends, but its curvature and field gradient change the particle orbits. We must calculate that change before accepting the magnetic bottle.

> **Design consequence.** Closing the corridor creates field curvature and a gradient, $B_\varphi \propto 1/R$. Their effect on particle orbits must now be checked.

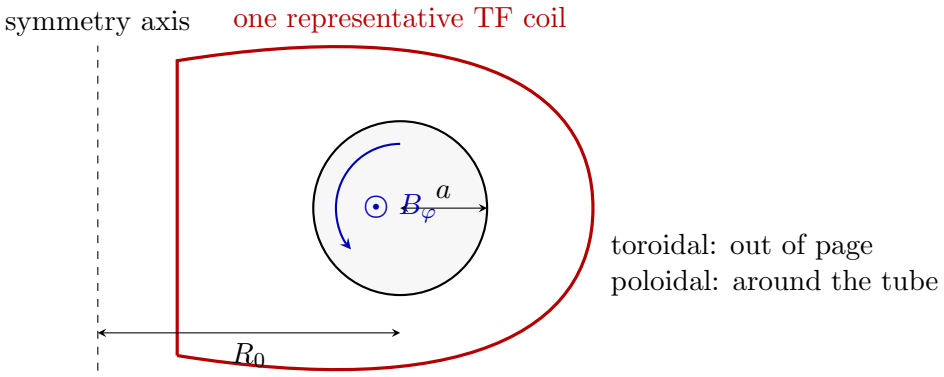


Figure 2. Torus geometry in a poloidal section: the TF winding surrounds the plasma tube; R_0 locates its centre and a measures its minor radius. The blue arrow marks the poloidal direction.

5. Why the toroidal bottle still leaks

When $r_L/R_0 \ll 1$, we can average over gyromotion and follow the orbit's centre, called the *guiding centre*. For a slowly varying perpendicular force $\mathbf{F}$, seek a constant drift velocity satisfying $q\mathbf{v}_F \times \mathbf{B} + \mathbf{F} = 0$. Crossing with $\mathbf{B}$ yields

$$\mathbf{v}_F = \frac{\mathbf{F} \times \mathbf{B}}{qB^2}. \tag{10}$$

Two effective forces arise in a toroidal field. The magnetic-moment magnitude of a gyro-orbit follows from its circulating charge:

$$\mu = \frac{|q|\omega_c}{2\pi}\,\pi r_L^2 = \frac{mv_\perp^2}{2B}.$$

For fields varying slowly over a gyroperiod and gyroradius, μ is an adiabatic invariant. The perpendicular energy μB then gives a gradient force $-\mu\nabla B$. Following a curved field line also gives the effective outward curvature force $(mv_\parallel^2/R)\hat{\mathbf{R}}$. The guiding-centre ordering underlying these expressions is developed systematically by Brizard and Hahm.[2]

Because $B \propto 1/R$, we have $\nabla B = -(B/R)\hat{\mathbf{R}}$. Both effective forces point outward. With $\mathbf{B} = B\hat{\boldsymbol{\varphi}}$ and $\hat{\mathbf{R}} \times \hat{\boldsymbol{\varphi}} = \hat{\mathbf{Z}}$,

$$\mathbf{v}_{\nabla B} = \frac{mv_\perp^2}{2qBR}\hat{\mathbf{Z}}, \qquad \mathbf{v}_c = \frac{mv_\parallel^2}{qBR}\hat{\mathbf{Z}}, \qquad \mathbf{v}_d = \frac{m}{qBR}\left(v_\parallel^2 + \frac{v_\perp^2}{2}\right)\hat{\mathbf{Z}}. \tag{11}$$

Ions and electrons drift vertically in opposite directions. The resulting charge separation creates a vertical electric field. Setting $\mathbf{F} = q\mathbf{E}$ in Eq. (10) gives

$$\mathbf{v}_E = \frac{\mathbf{E} \times \mathbf{B}}{B^2}. \tag{12}$$

This drift is common to both species and carries the plasma outward toward the wall. A purely toroidal field therefore leaves a systematic escape mechanism.

The remedy is to make field lines visit the top and bottom of the tube. Then a vertical drift alternates between outward and inward motion relative to the local magnetic surface. Adding a poloidal field provides this rotation and allows bounded orbit excursions in the axisymmetric confinement picture.

> **Design consequence.** Toroidal drifts separate charge and drive outward motion. A poloidal field is needed to carry particles around the cross-section.

6. Plasma current supplies the second field

External helical windings can produce the required twist; that is the stellarator route. The tokamak uses a toroidal current in the plasma to generate the principal poloidal field. For a circular cross-section at large aspect ratio, Ampère's law gives

$$B_\theta(r) \simeq \frac{\mu_0 I_{\mathrm{enc}}(r)}{2\pi r}, \tag{13}$$

where $I_{\mathrm{enc}}(r)$ is the plasma current inside radius r. The combination $\mathbf{B} = B_\varphi\hat{\boldsymbol{\varphi}} + B_\theta\hat{\boldsymbol{\theta}}$ produces helical field lines.

[2] A. J. Brizard and T. S. Hahm, "Foundations of nonlinear gyrokinetic theory," *Reviews of Modern Physics* **79**, 421–468 (2007), doi:10.1103/RevModPhys.79.421.

To initiate that current, place a solenoid through the central hole and change its magnetic flux. Faraday's law,

$$\oint \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d\Phi}{dt}, \tag{14}$$

creates a toroidal electric field. The central solenoid is the primary winding of a transformer; the plasma ring acts as its one-turn secondary. This gives the *plasma current* I_p a physical source. The TF coils provide the long-way field, and the induced plasma current provides the short-way field.

Design consequence. Changing central-solenoid flux drives plasma current and its poloidal field. The resulting helical field needs a measure of its pitch.

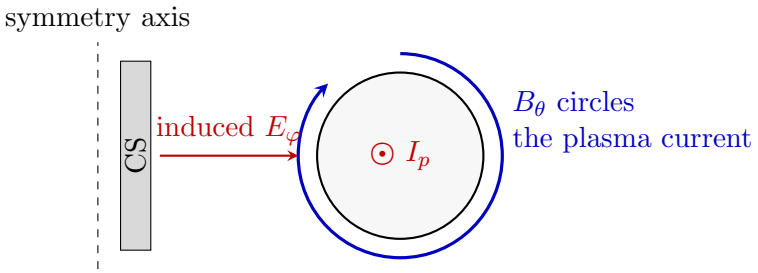


Figure 3. Induced toroidal electric field drives current; its poloidal field circles the plasma.

7. The helix has a pitch: the safety factor

A field-line displacement is parallel to $\mathbf{B}$. On a circular surface its toroidal and poloidal lengths therefore satisfy

$$\frac{R\,d\varphi}{r\,d\theta} = \frac{B_\varphi}{B_\theta}. \tag{15}$$

At large aspect ratio take $R \simeq R_0$ and $B_\varphi \simeq B_0$. During one poloidal circuit the field line advances toroidally through $2\pi q$, where

$$q(r) \equiv \frac{\Delta\varphi}{2\pi} \simeq \frac{rB_0}{R_0 B_\theta(r)}. \tag{16}$$

The *safety factor* q counts toroidal turns per poloidal turn. Its historical name reflects its use in stability analysis. Substituting Eq. (13) gives

$$q(r) \simeq \frac{2\pi r^2 B_0}{\mu_0 R_0 I_{\mathrm{enc}}(r)}, \qquad q_a \simeq \frac{2\pi a^2 B_0}{\mu_0 R_0 I_p}. \tag{17}$$

Thus an imposed lower bound on the edge safety factor becomes a ceiling on current:

$$I_p \simeq \frac{2\pi a^2 B_0}{\mu_0 R_0 q_a}. \tag{18}$$

Increasing plasma current strengthens the poloidal field and reduces the number of toroidal turns per poloidal circuit. Size, current, field and pitch are now coupled. For $q = m/n$ in lowest terms, the ideal field line closes after n poloidal turns and m toroidal turns. An irrational q produces a line that densely samples its surface.

Design consequence. An adopted minimum safety factor bounds plasma current at fixed size and toroidal field. Force balance must also be satisfied.

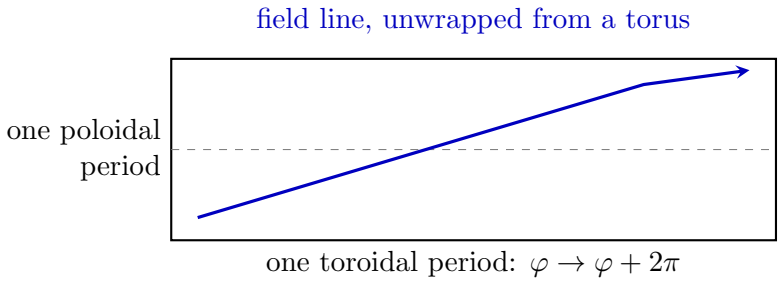


Figure 4. Unwrapping the magnetic surface shows the field-line pitch.

8. The plasma pushes back: pressure and beta

Electrons and ions both contribute to pressure. With singly charged fuel, $n_e \simeq n_i \equiv n$, and equal temperatures,

$$p = n_e T_e + n_i T_i \simeq 2nT. \tag{19}$$

At $n = 10^{20}\,\mathrm{m}^{-3}$ and $T = 10\,\mathrm{keV}$ this gives $p \simeq 3.2 \times 10^5\,\mathrm{Pa}$. The pressure gradient drives expansion. On scales large compared with particle orbits, a single-fluid model with scalar pressure gives

$$\rho_m \frac{d\mathbf{u}}{dt} = -\nabla p + \mathbf{j} \times \mathbf{B}. \tag{20}$$

Here ρ_m is mass density, $\mathbf{u}$ bulk velocity and $\mathbf{j}$ current per unit area. We use static magnetohydrodynamic (MHD) equilibrium: negligible bulk acceleration, isotropic pressure and negligible viscous force. Then

$$\mathbf{j} \times \mathbf{B} = \nabla p. \tag{21}$$

Dotting with $\mathbf{B}$ immediately gives

$$\mathbf{B} \cdot \nabla p = 0. \tag{22}$$

Pressure is constant along each field line, and on a nested surface densely sampled by those lines. Magnetic geometry must support this pressure distribution.

A natural pressure scale follows from the work needed to establish a field. For a long solenoid of N turns, length L and area S, its linked flux is NBS and $B = \mu_0 N I_c / L$. Integrating $dW_B = I_c\, d(NBS)$ gives

$$W_B = \frac{B^2}{2\mu_0} SL. \tag{23}$$

The energy density $B^2/(2\mu_0)$ has pressure units. Its ratio to plasma pressure defines the dimensionless parameter

$$\beta \equiv \frac{p}{B^2/(2\mu_0)}. \tag{24}$$

Beta measures how much plasma pressure is supported relative to the magnetic energy density. Later global estimates use representative or volume-averaged pressure and B_0, with that convention stated in the calculation. The allowable beta requires a stability model; the definition itself sets no universal limit.

Design consequence. Plasma pressure and current produce forces that require external equilibrium fields. The next task is to identify the field that balances radial expansion.

9. External coils hold the plasma in position

The plasma ring's pressure and self-generated magnetic field create an outward expansion tendency. A vertical field supplies an adjustable radial force on its toroidal current. For a ring element,

$$d\mathbf{F} = I_p R_0 \, d\varphi \, \hat{\boldsymbol{\varphi}} \times B_v \hat{\mathbf{Z}} = I_p R_0 B_v \, d\varphi \, \hat{\mathbf{R}}. \tag{25}$$

Changing the sign of B_v reverses the local force. The circumferentially integrated *radial load* is $2\pi R_0 I_p B_v$; vector forces around a complete symmetric ring cancel in translation while still expanding or compressing the ring.

External *poloidal-field coils*, or PF coils, are circular windings around the symmetry axis, placed at selected radii and heights. Their fields lie in the R–Z plane. By changing their currents we can supply the vertical field that balances radial expansion, and later adjust plasma height and shape. A power supply is the circuit that controls each coil's current; the coil is the magnetic actuator. The necessary field magnitude depends on the plasma pressure and current profiles, which enter the equilibrium equation derived below.

The coil roles are now distinct. TF coils generate B_φ. The central solenoid induces I_p, whose field supplies the main poloidal twist. The external PF system positions and shapes the resulting plasma.

Design consequence. A controlled vertical field supplies radial force on the plasma ring. PF coils must control its cross-sectional boundary as well as its position.

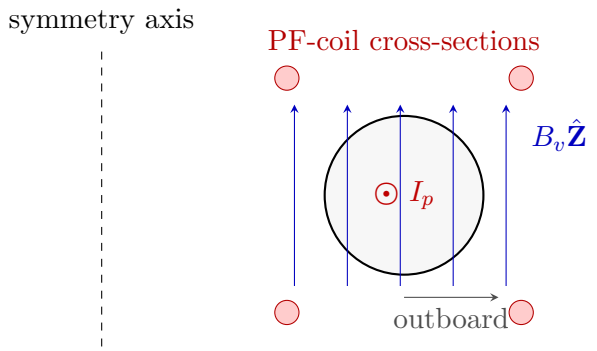


Figure 5. A vertical field acts radially on the toroidal plasma current.

10. Flux surfaces define the magnetic boundary

In cylindrical coordinates (R, φ, Z), axisymmetry means $\partial/\partial\varphi = 0$. The divergence-free condition becomes

$$\partial_R(RB_R) + \partial_Z(RB_Z) = 0. \tag{26}$$

Introduce a flux function $\psi(R, Z)$ through

$$B_R = -\frac{1}{R}\partial_Z\psi, \qquad B_Z = \frac{1}{R}\partial_R\psi, \qquad \mathbf{B}_p = \frac{\nabla\psi \times \hat{\boldsymbol{\varphi}}}{R}. \tag{27}$$

The mixed derivatives cancel in Eq. (26). Also,

$$\mathbf{B} \cdot \nabla\psi = 0, \tag{28}$$

so field lines lie on surfaces of constant ψ. With this convention, 2π times a difference in ψ is a poloidal magnetic-flux difference. Rotating a closed contour of ψ about the symmetry axis constructs a magnetic surface.

The outermost closed surface defines the confined plasma boundary. The vacuum vessel surrounds this magnetic configuration. PF-coil currents and plasma current together set its flux contours; controlling these currents lets us change the boundary's position and shape. The point inside the nested contours about which they contract is the *magnetic axis*.

This flux description is the starting point for the Grad–Shafranov equilibrium calculation in §§21–24. An analytic treatment is given by Cerfon and Freidberg.[3]

> **Design consequence.** The magnetic boundary is a contour of ψ, controlled by PF currents. The vessel must leave room around that confined plasma.

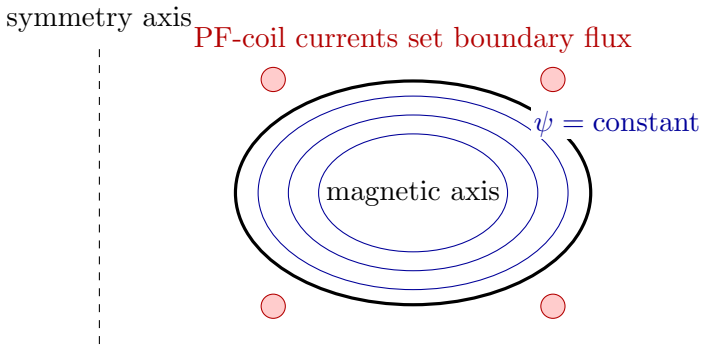


Figure 6. PF currents set the flux contours that form the plasma boundary.

11. The transformer has a finite flux budget

Faraday's law around the plasma ring defines the loop voltage:

$$V_{\text{loop}} \equiv \oint \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d\Phi_{\text{link}}}{dt}. \tag{29}$$

Over a discharge,

$$\int V_{\text{loop}}\, dt = -\Delta\Phi_{\text{link}}. \tag{30}$$

One volt-second equals one weber of linked flux. The central solenoid can swing only through a finite flux range before reaching its current, field or mechanical limits. The flux budget pays for initiating and ramping the plasma current and for maintaining it against resistance. Sustained transformer-driven operation therefore has a finite pulse length. Longer operation needs another source of current or a new pulse after resetting the solenoid.

Electrical resistance also provides *ohmic heating*. In a resistive fluid model,

$$P_{\text{ohm}} = \int \eta j^2\, dV \simeq I_p^2 R_p, \tag{31}$$

where η is resistivity and R_p an effective plasma resistance. During a current ramp, part of the electrical input changes magnetic energy. For a simple circuit with fixed inductance L_p, $V_{\text{loop}} = L_p \dot{I}_p + R_p I_p$; only when the inductive term is small does $I_p V_{\text{loop}} \simeq P_{\text{ohm}}$.

[3] A. J. Cerfon and J. P. Freidberg, "One size fits all" analytic solutions to the Grad–Shafranov equation, *Physics of Plasmas* **17**, 032502 (2010), doi:10.1063/1.3328818.

In a fully ionized collisional plasma, classical Spitzer resistivity scales as

$$\eta \propto T_e^{-3/2} \tag{32}$$

for fixed composition and approximately fixed Coulomb logarithm.[4] At comparable current density, heating falls as the electrons get hotter. Resistance is useful during startup, but reaching fusion temperature calls for auxiliary heating.

Design consequence. The transformer has a finite volt-second budget, while ohmic heating weakens as temperature rises. Additional heating must sustain the hot plasma.

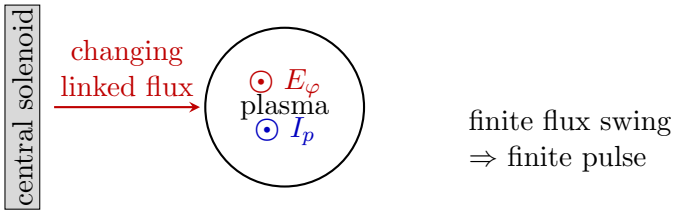


Figure 7. The transformer couples changing linked flux to the plasma ring.

12. Heating must balance energy loss

A Maxwellian species has thermal energy density $3n_sT_s/2$. For approximately uniform fuel at equal ion and electron temperatures,

$$W \simeq 3nTV, \qquad P_{\text{loss}} \equiv W/\tau_E. \tag{33}$$

Here τ_E is the energy-confinement time. In this section P_{loss} includes transport and radiation losses together; each loss channel is counted once. The thermal-energy balance is

$$\frac{dW}{dt} = P_{\text{ohm}} + P_{\text{aux}} + P_\alpha - P_{\text{loss}}. \tag{34}$$

The powers on the right are deposited in the plasma. Auxiliary heating must raise W against losses while fusion power is small. Later, confined alpha particles from Eq. (1) transfer their energy to the fuel, providing P_α. Neutrons carry their energy out of the magnetically confined plasma.

A *neutral-beam injector* accelerates ions and neutralizes them before injection. The fast neutral atoms cross the confining magnetic field, become ionized inside the plasma, and transfer energy and momentum through collisions. A *radio-frequency* (RF) system instead sends electromagnetic waves through an antenna or launcher; suitable frequencies and wave patterns transfer energy to electrons or ions. These systems require access openings, or ports, through the vessel. With suitable injection or wave conditions, they can also drive plasma current without transformer induction.

We now have the equipment that produces a hot current-carrying plasma. Its useful volume and stable pressure depend on the shape that the PF system can create and hold.

Design consequence. Beam injectors or RF launchers supply auxiliary heat through vessel ports. Their deposited power must cover losses and any increase in stored energy.

[4] Naval Research Laboratory, *NRL Plasma Formulary* (2019), collisional transport and resistivity relations, open formulary.

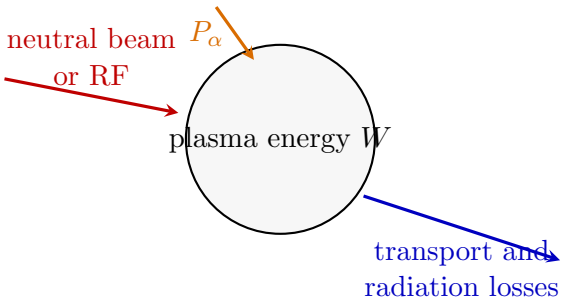


Figure 8. Deposited heating competes with energy carried out of the plasma.

13. Elongation buys volume

An elliptical cross-section has horizontal semi-axis a and vertical semi-axis κa, where

$$\kappa \equiv \frac{\text{vertical semi-axis}}{\text{horizontal semi-axis}} \tag{35}$$

is the *elongation*. A convenient boundary parameterization is $R = R_0 + a\cos\theta$, $Z = \kappa a \sin\theta$. Its area is $\pi\kappa a^2$. Rotating the ellipse about the symmetry axis gives

$$V = 2\pi R_0 \, \pi\kappa a^2 = 2\pi^2 R_0 a^2 \kappa. \tag{36}$$

Scope. The volume formula is exact for the specified ellipse with $R_0 > a$. At fixed R_0 and a, increasing κ increases volume in direct proportion. At a fixed plasma state, that volume can contain more fuel and produce more fusion power.

Creating the ellipse requires a spatially varying PF field. Coil locations and currents are selected so that the combined plasma and external fields support a tall boundary. Shape is therefore an additional task for the same PF system that holds the plasma's radial position. Elongation also changes its response to displacement.

> **Design consequence.** Elongation increases volume at fixed R_0 and a. Its benefit must be assessed together with the response to vertical displacement.

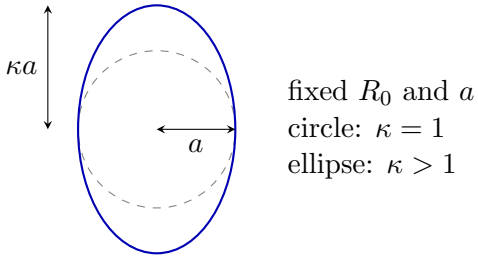


Figure 9. Elongation increases cross-sectional area from πa^2 to $\pi\kappa a^2$. The added height also changes the vertical-stability problem.

14. A taller plasma needs position feedback

To expose the stability question, represent a small vertical displacement by z. Expand its force about equilibrium:

$$F_Z(z) = F_Z(0) + K_z z + \cdots, \qquad K_z = \left.\frac{\partial F_Z}{\partial z}\right|_0. \tag{37}$$

Since $F_Z(0) = 0$, a one-coordinate inertial model gives

$$M\ddot{z} = K_z z. \tag{38}$$

For $K_z < 0$ a displacement produces a restoring force. For $K_z > 0$ the growing solution is $z \propto e^{\sqrt{K_z/M}t}$. This model explains the sign of the feedback needed. *Scope.* A real vertical mode also couples to induced currents in nearby conducting structures and to the coil circuits; its growth rate requires those dynamics.

Elongation generally increases the difficulty of vertical stabilization.[5] Magnetic sensors measure signals from which plasma position is estimated. A controller compares that position with the requested one and commands power supplies to change control-coil currents. The resulting field must oppose the displacement quickly enough to keep the plasma clear of the vessel. Conducting structures can slow the motion and give the active system time to respond.

Design consequence. An unstable vertical displacement requires active correction. Sensors, a controller and powered coils must close the restoring feedback loop.

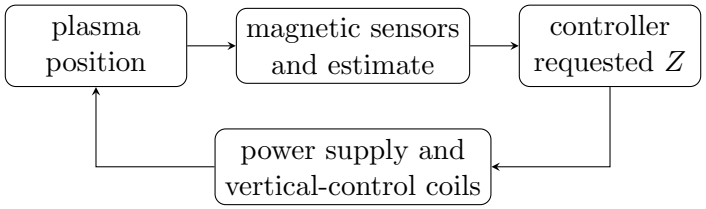


Figure 10. The feedback loop turns a measured displacement into a corrective magnetic force. Sensor, circuit and controller response times enter the achievable elongation.

15. Shape also changes the current allowance

The safety-factor relation from §7 can be generalized to the elongated geometry. The reduced reactor model of Freidberg, Mangiarotti and Minervini adopts

$$q_* = \frac{2\pi a^2 B_0}{\mu_0 R_0 I_p} \frac{1+\kappa^2}{2}. \tag{39}$$

The first factor is the circular edge estimate. The second accounts approximately for elongation in this large-aspect-ratio description. We take this shaped relation from the model, which uses it as a global measure of field-line pitch.

A *kink* is a collective displacement that bends the current-carrying plasma. The model represents its global current limit by requiring $q_* > q_K$, with an adopted threshold $q_K \simeq 2$. Substitution gives

$$I_p < \frac{2\pi a^2 B_0}{\mu_0 R_0 q_K} \frac{1+\kappa^2}{2}. \tag{40}$$

The shaped estimate and threshold are the closures used in Eqs. (30) and (39) of the 2015 reference. *Model used here.* Profiles, conducting-wall response and operating regime affect the physical stability boundary. The current ceiling here uses the stated large-aspect-ratio closure and $q_K \simeq 2$.

[5] J. P. Freidberg, A. Cerfon and J. P. Lee, "Tokamak elongation—how much is too much? Part 1. Theory," *Journal of Plasma Physics* **81**, 515810607 (2015), open manuscript.

Within that model, stronger toroidal field, greater minor radius and greater elongation permit more current. Increasing R_0 while holding the other quantities fixed reduces the allowance. This matters because the current that creates the poloidal confinement field also enters stability limits. Elongation offers both more volume and a larger current allowance, while demanding stronger position control. These consequences must be considered together.

> **Design consequence.** Elongation raises the current allowance in the adopted model. The chosen shape must also remain within the vertical-control capability.

16. The current profile determines the interior pitch

Total current is an area integral. In the circular approximation,

$$I_{\mathrm{enc}}(r) = 2\pi \int_0^r j_\varphi(r')r'\,dr', \qquad q(r) \simeq \frac{2\pi r^2 B_0}{\mu_0 R_0 I_{\mathrm{enc}}(r)}. \tag{41}$$

Two plasmas with the same $I_p = I_{\mathrm{enc}}(a)$ can therefore have different interior $q(r)$ if their current density is distributed differently. A centrally concentrated current places more enclosed current inside a small radius and lowers q there relative to a broader distribution with the same total current.

For example, a uniform $j_\varphi = j_0$ gives $I_{\mathrm{enc}} = \pi j_0 r^2$ and a constant $q \simeq 2B_0/(\mu_0 R_0 j_0)$ in this approximation. A varying profile changes that pitch from surface to surface. The radial variation of pitch is called *magnetic shear*. Both pitch and shear influence how disturbances couple across magnetic surfaces. Rational surfaces, where $q = m/n$, allow perturbations with matching periodicity to remain aligned with the field.

The transformer and auxiliary current-drive systems influence where current flows, as well as how much flows. Current control must account for $j_\varphi(r)$ and $q(r)$ alongside I_p.

> **Design consequence.** Total current alone does not fix $q(r)$. Current control must account for the distribution $j_\varphi(r)$ as well as I_p.

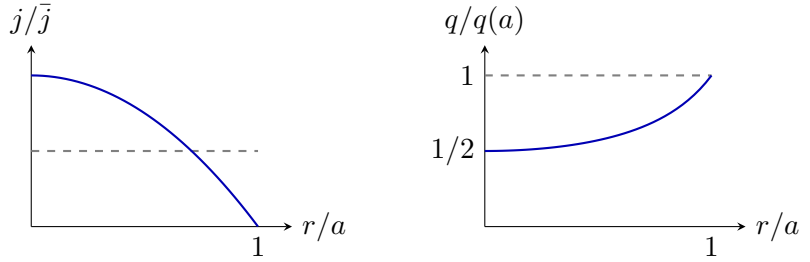


Figure 11. Illustrative circular, large-aspect-ratio profiles with the same total current: uniform $j = \bar{j}$ (dashed) and peaked $j = 2\bar{j}[1 - (r/a)^2]$ (blue), where $\bar{j} = I_p/(\pi a^2)$. The corresponding normalized pitches are 1 and $[2 - (r/a)^2]^{-1}$; the peaked profile lowers the interior pitch while preserving $q(a)$.

17. Why field and volume dominate a first power estimate

For an equal D–T mixture with total fuel-ion density n, $n_D = n_T = n/2$. In a uniform plasma the reaction rate per unit volume is $n_D n_T \langle\sigma v\rangle = n^2\langle\sigma v\rangle/4$. The *reactivity* $\langle\sigma v\rangle$ averages the cross-section

times relative speed over the ion distributions. At fixed temperature and fuel mixture,

$$P_{\text{fus}} = \frac{n^2}{4}\langle\sigma v\rangle Q V \propto n^2 V. \tag{42}$$

Using $p = 2nT$ and $p = \beta B_0^2/(2\mu_0)$ gives $n = \beta B_0^2/(4\mu_0 T)$, and hence

$$P_{\text{fus}} \propto \beta^2 B_0^4 V \propto \beta^2 B_0^4 R_0 a^2 \kappa. \tag{43}$$

Scope. This scaling holds temperature, composition and profile factors fixed, and uses B_0 as the reference magnetic field. For example, a 10% field increase gives a factor $1.1^4 \simeq 1.46$ in fusion power if beta and volume can be maintained. The fourth power explains the attraction of high field. Maintaining those other quantities requires compatible confinement, current and stability; producing the field also imposes loads on the magnet. The scaling identifies the incentive that later engineering calculations must price.

> **Design consequence.** The conditional $\beta^2 B_0^4 V$ scaling rewards field, pressure and volume. These quantities must be achievable together in one operating state.

18. The machine and its coupled variables

A component enters the ledger only after a section has forced it. We can now identify each component by the physical task that introduced it:

Component	Purpose
Vacuum vessel	Maintains the evacuated environment around the magnetically confined fuel.
TF coils	Produce the toroidal field that guides particle motion around the ring.
Central solenoid	Changes linked flux to initiate and ramp plasma current.
Plasma current	Generates the principal poloidal field and sets field-line pitch.
PF and control coils	Position and shape the plasma; apply corrective forces.
Sensors, controller, power supplies	Measure the state and command the coil response.
Beam or RF systems	Deposit auxiliary heat and, under suitable conditions, drive current.

The associated variables form a coupled set. Geometry (R_0, a, κ) determines volume; B_0 and I_p determine pitch through Eq. (39); the current profile sets interior $q(r)$. Density and temperature set pressure and reaction rate, while beta compares pressure with magnetic energy density. Finally, τ_E connects stored energy to the heating requirement.

The remaining task in this block is to put shape and force balance into one equation. We first add the D-shaped boundary parameter, then derive the equation that determines which boundary, pressure profile and current distribution can coexist.

> **Design consequence.** The components now have defined physical tasks. Their settings must support a common plasma state, requiring a shared description of shape and force balance.

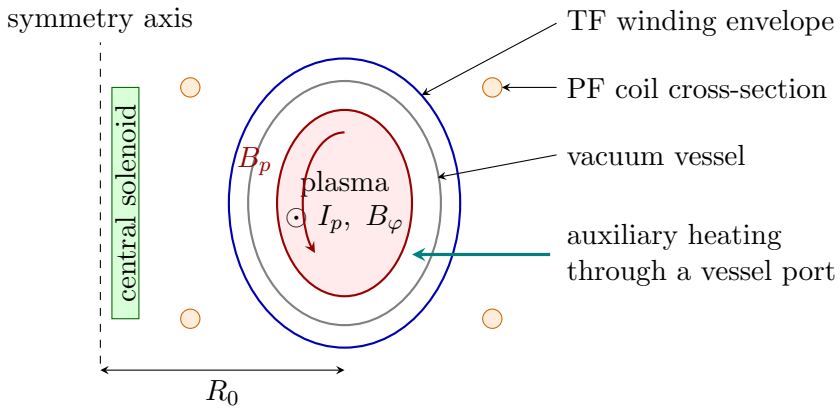


Figure 12. Component map in a poloidal cross-section. The TF winding wraps around the vessel; separate PF rings appear as small coil cross-sections. The central solenoid changes flux through the plasma ring. The winding and vessel outlines are schematic; a heating port is shown only by its access direction.

19. Triangularity adds a D-shaped boundary

Elongation specifies height relative to width. A second parameter measures the inward shift of the upper and lower tips of the cross-section. A useful boundary family is

$$\begin{aligned} R(\theta) &= R_0 + a\cos(\theta + x\sin\theta), \qquad x = \arcsin\delta, \\ Z(\theta) &= \kappa a \sin\theta. \end{aligned} \tag{44}$$

At the top, $\theta = \pi/2$, so $R_{\text{top}} = R_0 - a\sin x = R_0 - \delta a$. Thus

$$\delta = \frac{R_0 - R_{\text{top}}}{a} \tag{45}$$

is the *triangularity*. Positive δ pulls the top and bottom inward, producing a D-like cross-section. At $\delta = 0$ the boundary is the ellipse of §13. *Model used here.* The Miller parameterization specifies a boundary family. The small-δ harmonic expansion below retains terms through first order.

For small δ, expand $\cos(\theta + x\sin\theta)$ and use $\sin^2\theta = (1 - \cos 2\theta)/2$:

$$R(\theta) = R_0 - \frac{a\delta}{2} + a\cos\theta + \frac{a\delta}{2}\cos 2\theta + O(a\delta^2). \tag{46}$$

The added second angular harmonic and centring shift describe a shape change independent of elongation. The PF system must supply another adjustable spatial pattern to control it.

> **Design consequence.** Triangularity adds an independent boundary parameter. The PF system needs enough adjustable field patterns to control it alongside position and elongation.

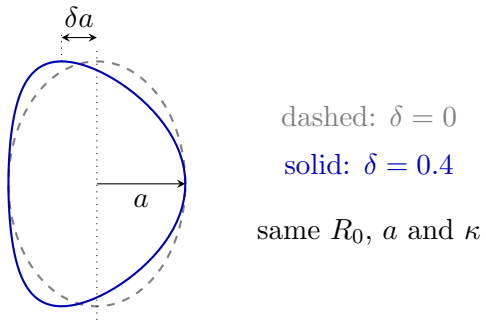


Figure 13. Triangularity shifts the upper and lower tips toward the central hole. Elongation and horizontal half-width are held fixed.

20. Coil placement controls the available shape patterns

In a current-free vacuum region away from the coils, $\nabla \times \mathbf{B} = 0$ and $\nabla \cdot \mathbf{B} = 0$. Locally we can write $\mathbf{B} = -\nabla \Phi_m$, so the magnetic scalar potential satisfies $\nabla^2 \Phi_m = 0$. A two-dimensional local example organizes the first patterns as

$$\Phi_1 = C_1 r \sin\theta, \qquad \Phi_2 = C_2 r^2 \sin 2\theta, \qquad \Phi_3 = C_3 r^3 \sin 3\theta.$$

Each is harmonic. With local Cartesian coordinates $x = r\cos\theta$, $z = r\sin\theta$, the first two become $C_1 z$ and $2C_2 xz$. Their fields are, respectively, uniform and linear in position. The third potential, $C_3(3x^2 z - z^3)$, gives a quadratic field.

A uniform vertical field changes the radial force on the plasma. The linear, *quadrupole* pattern contributes to elliptical shaping. The next, *hexapole* pattern supplies a further shaping degree of freedom useful for triangularity. These names describe successive spatial patterns of the field. The actual response of the boundary follows from force balance in toroidal geometry.

Each coil influences several patterns. Its position fixes that influence, while its power supply sets the current. Independent control of position and shape therefore requires enough independently adjustable coil circuits, with suitable placement and current range. More detailed shaping makes this an increasingly coupled control problem.

Design consequence. Each coil contributes to several field patterns. Coil placement and independently powered circuits must provide the required shaping freedom.

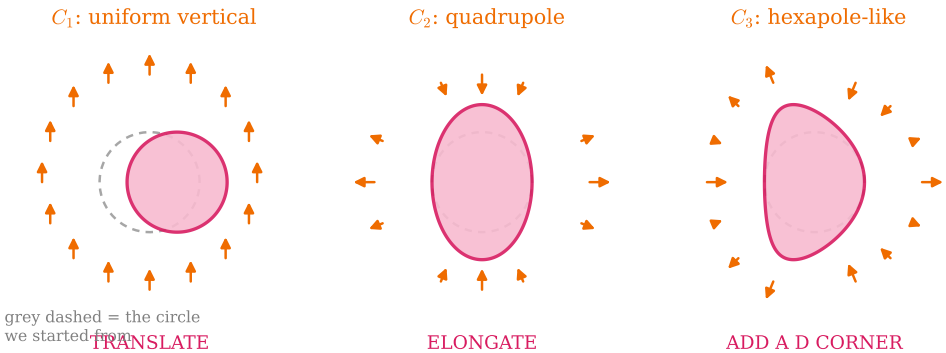


Figure 14. Successive field patterns supply additional shaping freedom.

21. Equilibrium joins pressure to magnetic surfaces

The static fluid balance derived in §8 is $\nabla p = \mathbf{j} \times \mathbf{B}$. Its projection along the field gave $\mathbf{B} \cdot \nabla p = 0$. In the nested-surface equilibrium considered here, we therefore write $p = p(\psi)$: one pressure value for each flux surface. The pressure profile specifies how that value changes from the hot interior to the edge.

This is the point where particle confinement becomes a collective design problem. Pressure, plasma current and the field from the coils must satisfy force balance at every point. *Model used here.* We retain the static, axisymmetric, isotropic-pressure model introduced in §8. Substantial flow or pressure anisotropy requires a generalized equilibrium. Each surface is labelled by its poloidal flux ψ, so $p = p(\psi)$ gives both a pressure value and, through $dp/d\psi$, the pressure change from one surface to the next. The spatial gradient is $\nabla p = (dp/d\psi)\nabla\psi$: its direction is normal to the surface and its magnitude depends on the spacing of the contours. This links a profile specified as a function of flux to the physical force density that the magnetic field must balance. Within the stated model, the geometry can therefore be reduced to a scalar differential equation.

Design consequence. Force balance makes pressure constant on connected magnetic surfaces. Describing those surfaces requires a common flux coordinate.

22. Two scalar functions describe the field

Combine the poloidal flux representation of §10 with a toroidal component written as $B_\varphi = F(R, Z)/R$:

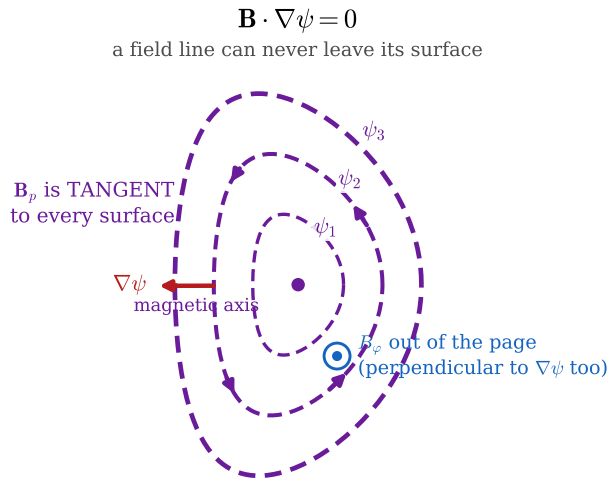
$$\mathbf{B} = \frac{1}{R}\nabla\psi \times \hat{\boldsymbol{\varphi}} + \frac{F}{R}\hat{\boldsymbol{\varphi}}. \tag{47}$$

Axisymmetry and $\nabla \cdot \mathbf{B} = 0$ permit this representation away from the coordinate axis. The contours of ψ locate the magnetic surfaces, and the function F sets the toroidal field strength. The factor $1/R$ retains the toroidal geometry.

We have already established $p = p(\psi)$. The toroidal component of force balance will now establish $F = F(\psi)$ as well. The two functions carry different information. Derivatives of $\psi(R, Z)$ give B_R and B_Z, so closely spaced contours mean a stronger poloidal field at the same R. Its contour shapes locate the magnetic surfaces and boundary. Meanwhile $F = RB_\varphi$ fixes the toroidal field; it has units of tesla-metres, whereas ψ has flux units. In a current-free toroidal-field region F is constant, recovering $B_\varphi \propto 1/R$. In the plasma, $F(\psi)$ can vary across surfaces and contributes to the current and force balance.

Specifying $p(\psi)$ and $F(\psi)$ thus supplies two surface profiles. It still leaves $\psi(R, Z)$ to be found: the map from those surface labels to real space. The Grad–Shafranov equation determines that map with the imposed boundary or external-coil conditions. Its solution simultaneously supplies the poloidal field, flux-surface geometry and toroidal current density.

Design consequence. The flux function specifies the surfaces and poloidal field. The next step is to solve for it using pressure and toroidal-field profiles.



Explanatory sketch. No hardware is drawn: this is the mathematical map that makes flux surfaces intelligible, not a picture of a machine.

Figure 15. Flux contours and their normal identify the poloidal field geometry.

23. Deriving the Grad–Shafranov equation

Insert Eq. (47) into $\mu_0\mathbf{j} = \nabla\times\mathbf{B}$. The toroidal current density comes from the curl of the poloidal field:

$$j_\varphi = -\frac{1}{\mu_0 R}\Delta^*\psi, \qquad \Delta^*\psi \equiv R\frac{\partial}{\partial R}\left(\frac{1}{R}\frac{\partial\psi}{\partial R}\right) + \frac{\partial^2\psi}{\partial Z^2}. \tag{48}$$

To check the sign, use $(\nabla\times\mathbf{B})_\varphi = \partial_Z B_R - \partial_R B_Z$ and the components in Eq. (27). The remaining components give

$$\mathbf{j}_p = \frac{1}{\mu_0 R}\nabla F\times\hat{\boldsymbol{\varphi}}.$$

Since pressure is axisymmetric, the toroidal component of $\mathbf{j}\times\mathbf{B}$ must vanish. Its value is proportional to $\mathbf{B}_p\cdot\nabla F$, so F is constant along the poloidal field. On the nested surfaces we can write $F = F(\psi)$ and hence $\mathbf{j}_p = F'(\psi)\mathbf{B}_p/\mu_0$.

Only two cross products remain: $j_\varphi\hat{\boldsymbol{\varphi}}\times\mathbf{B}_p$ and $\mathbf{j}_p\times B_\varphi\hat{\boldsymbol{\varphi}}$. Using $\hat{\boldsymbol{\varphi}}\times\mathbf{B}_p = \nabla\psi/R$, they give

$$\mathbf{j}\times\mathbf{B} = -\frac{\Delta^*\psi + FF'}{\mu_0 R^2}\nabla\psi, \qquad \nabla p = p'\nabla\psi.$$

Equating coefficients normal to a regular flux surface yields

$$\boxed{\Delta^*\psi = -\mu_0 R^2\frac{dp}{d\psi} - F\frac{dF}{d\psi}.} \tag{49}$$

This is the *Grad–Shafranov equation*. The pressure gradient and toroidal-field profile supply its source terms; the solution supplies the flux surfaces.

Scope. At a smooth magnetic axis or null, the equation is interpreted through the regular solution. The intermediate cancellation of $\nabla\psi$ applies where that gradient is nonzero.

The derivation reveals why geometry alone is insufficient to specify a tokamak. Changing pressure or current changes the differential equation that its magnetic surfaces must satisfy. Coil settings then have to support the resulting equilibrium.

24. From an equilibrium equation to an operating machine

There are two useful ways to pose the calculation. In a *fixed-boundary* calculation we prescribe the plasma boundary and the profiles $p(\psi)$ and $F(\psi)$, then solve for the interior flux surfaces and current. In a *free-boundary* calculation we include the external-coil fields and surrounding vacuum; the plasma boundary is part of the solution. A desired shape can thus be tested first, then translated into requirements on coil locations and currents. The analytic equilibria cited in §10 illustrate how profiles and boundary geometry enter this problem.

The magnetic axis generally shifts outward relative to the geometric centre of the boundary. This *Shafranov shift* reflects the pressure and current contributions to toroidal force balance. Its size depends on the profiles and geometry. When those profiles change during heating or a current ramp, the equilibrium changes, and the PF currents needed to maintain the requested boundary change with it.

A D-shaped plasma and a D-shaped TF coil have separate design purposes. Triangularity specifies a plasma boundary chosen through equilibrium and stability considerations. A D-like TF-coil shape can instead be chosen to carry magnetic loads mainly in tension and reduce conductor bending. The engineering design must accommodate both the required magnetic surfaces and the forces on the conductors that produce their field.

We have reached a working description of machine formation. Hot charged particles require a vacuum environment and magnetic guidance; closing the field introduces curvature drifts; plasma current supplies the poloidal twist; pressure demands external equilibrium fields; transformer limits demand auxiliary heating and current drive; useful shaping demands feedback. The Grad–Shafranov equation joins these choices into a calculable equilibrium. The next block asks what fuel state and confinement this machine must sustain to produce useful fusion power.

Design consequence. Pressure and current evolution change the equilibrium. Each operating state needs compatible profiles, a magnetic boundary and PF-coil settings.

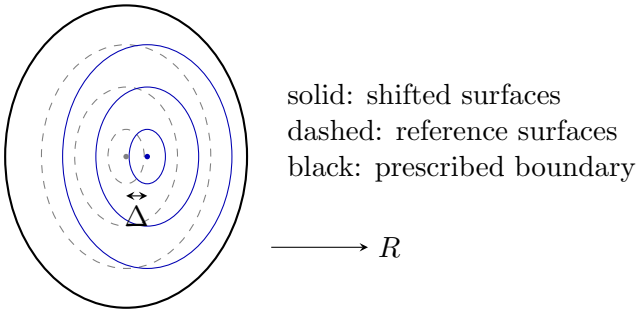


Figure 16. Schematic outward displacement of the magnetic axis relative to a fixed boundary. The interior contours illustrate the shift only; an equilibrium calculation determines their deformation and spacing for the specified profiles.

Let the Machine Reveal Itself

Fusion power, operating limits and radial build

Jony (TUNEM)

Compression review v2 — Sections 25–50 — October 2026

This block continues the revised opening. Section and equation numbers retain their thesis sequence; references to §§1–24 refer to the approved 18-page opening.

25. From individual collisions to a fusion-power density

Consider a small volume dV containing $n_D dV$ deuterons and $n_T dV$ tritons. Choose one deuteron. During a short time dt, tritons with relative speed v sweep through the effective collision tube of area $\sigma(v)$ and length $v dt$. The expected number of its reactions is

$$n_T\, \sigma(v)\, v\, dt.$$

Particles in a thermal plasma do not all have the same relative speed. Average this expression over the normalized relative-speed distribution and multiply by the number $n_D dV$ of deuterons. Dividing by $dV\, dt$ gives the reaction rate per unit volume:

$$\mathcal{R}_{DT} = n_D n_T \langle \sigma v \rangle, \qquad \langle \sigma v \rangle = \int_0^\infty \sigma(v) v\, f_{\mathrm{rel}}(v)\, dv. \tag{53}$$

The measured cross-section $\sigma(v)$ supplies the nuclear reaction probability; $f_{\mathrm{rel}}(v)$ supplies the thermal statistics.

If each reaction releases total energy $E_f = 17.6\,\mathrm{MeV}$, then

$$p_{\mathrm{fus}} = E_f \mathcal{R}_{DT} = E_f n_D n_T \langle \sigma v \rangle, \qquad P_{\mathrm{fus}} = p_{\mathrm{fus}} V. \tag{54}$$

This gives the factors behind the scaling in §17: density, volume and the temperature-dependent reactivity jointly determine fusion output.

Model used here. Equation (53) assumes independent binary collisions and a thermal distribution. At reactor density, D–T fusion remains dilute enough for that description; collective plasma physics controls confinement, not the short-range nuclear reaction once two ions meet.

> **Design consequence.** Fusion output depends on the product $n_D n_T$. At fixed fuel density, the mixture must maximize that product.

26. Why the fuel mixture is nearly fifty–fifty

Let the total fuel-ion density be fixed:

$$n_i = n_D + n_T.$$

Write $n_D = x n_i$. Then $n_T = (1-x)n_i$, and the density product in Eq. (54) is

$$n_D n_T = x(1-x)n_i^2. \tag{55}$$

Complete the square:

$$x(1-x) = x - x^2 = \frac{1}{4} - \left(x - \frac{1}{2}\right)^2.$$

The squared term is nonnegative, so the largest possible value is $1/4$, achieved at $x = 1/2$. Thus

$$n_D = n_T = \frac{n_i}{2} \quad \Longrightarrow \quad p_{\text{fus}} = \frac{1}{4} n_i^2 \langle \sigma v \rangle E_f. \tag{56}$$

Equal fuel fractions maximize the reaction rate at fixed total fuel-ion density and temperature.

The pressure used in magnetic confinement also includes electrons. A quasi-neutral 50–50 D–T mixture has electron density $n_e = n_i$. We write the primary fuel density as $n \equiv n_i$. If the ions and electrons share a temperature T measured in energy units, their ideal-gas pressures add:

$$p = (n_i + n_e)T = 2 n_i T. \tag{57}$$

Combining Eqs. (56) and (57) explains the scaling chain on §17, but with its factors visible: at fixed T, $n_i = p/(2T)$ and hence $p_{\text{fus}} \propto p^2$.

> **Design consequence.** The density needed for fusion also creates pressure. The same fuel state must satisfy the magnetic-pressure and density constraints.

27. Where the fusion energy goes decides what the machine must add

The D–T products are an alpha particle and a neutron. In the center-of-mass frame they leave with equal and opposite momentum, so their momentum magnitudes are equal: $m_\alpha v_\alpha = m_n v_n$. Their kinetic energies are $E_\alpha = p^2/(2m_\alpha)$ and $E_n = p^2/(2m_n)$. Therefore

$$\frac{E_n}{E_\alpha} = \frac{m_\alpha}{m_n} \simeq 4, \qquad E_\alpha = \frac{1}{5} E_f \simeq 3.5\,\text{MeV}, \quad E_n = \frac{4}{5} E_f \simeq 14.1\,\text{MeV}.$$

The charged alpha particle can remain confined long enough to slow down by collisions and heat the plasma. The neutron has no electric charge, so magnetic fields do not bend it; it escapes the plasma and deposits its energy only after meeting material outside the vacuum region.

With complete alpha-energy deposition, $P_\alpha = (E_\alpha/E_f)P_{\text{fus}} \simeq P_{\text{fus}}/5$. Neutron power is $P_n \simeq 4P_{\text{fus}}/5$. The first term enters the plasma self-heating balance; the second must be captured outside the confined plasma. *Model used here.* The power balances below assume thermal fuel, common ion and electron temperature, and complete alpha-energy deposition. Alpha escape reduces the deposited fraction.

> **Design consequence.** Alpha energy can sustain the plasma; neutron energy must be collected in surrounding material. The two fractions enter different power balances.

28. Stored energy and the meaning of confinement time

Before asking whether alpha heating is enough, we need a careful measure of loss. For an ideal gas with three translational degrees of freedom, the internal energy density is $3nT/2$. In the equal-temperature D–T plasma there are n_i ions and $n_e = n_i$ electrons, so the total thermal energy is

$$W = \left(\frac{3}{2}n_iT + \frac{3}{2}n_eT\right)V = 3n_iTV. \tag{58}$$

The energy-confinement time is defined, rather than guessed, by

$$\tau_E \equiv \frac{W}{P_{\text{loss}}}, \qquad P_{\text{loss}} = \frac{W}{\tau_E}. \tag{59}$$

It answers a plain question: if all heating stopped and the loss rate stayed at its instantaneous value, how long would the present thermal energy take to leave?

Here P_{loss} includes transport and radiation, each counted once. The instantaneous ratio W/P_{loss} remains useful even when the energy decay is not exponential.

Using Eq. (57), Eq. (58) can also be written $W = (3/2)pV$. Thus a high-pressure plasma stores more energy at a given volume, but it also demands proportionally more heating power if τ_E is fixed.

> **Design consequence.** The loss power is W/τ_E. Ignition requires alpha heating to cover it at the proposed density and temperature.

29. The Lawson condition follows from one inequality

Ignition means that alpha heating alone can replace the plasma losses:

$$P_\alpha \geq P_{\text{loss}}. \tag{60}$$

Use the equal-mixture fusion result, Eq. (56), but retain only the alpha energy E_α:

$$P_\alpha = \frac{1}{4}n_i^2\langle\sigma v\rangle E_\alpha V.$$

Use Eq. (58) and Eq. (59) for the loss:

$$P_{\text{loss}} = \frac{3n_iTV}{\tau_E}.$$

Substitution into Eq. (60) gives

$$\frac{1}{4}n_i^2\langle\sigma v\rangle E_\alpha V \geq \frac{3n_iTV}{\tau_E}.$$

Cancel the positive common factor n_iV, then multiply by $4\tau_E$:

$$\boxed{n_i\tau_E \geq \frac{12T}{\langle\sigma v\rangle E_\alpha}.} \tag{61}$$

Every factor has a visible origin: $1/4$ from the equal fuel mix, 3 from ions plus electrons and three translational degrees of freedom, and E_α rather than E_f because neutrons do not heat the confined plasma.

Multiplying Eq. (61) by T gives the triple product $n_iT\tau_E$. Its threshold depends on the chosen temperature and D–T reactivity.

Design consequence. Ignition requires a compatible combination of density, temperature and confinement. External heating supplies any remaining power deficit.

30. Gain is a ratio, not a synonym for ignition

A plasma can make useful fusion power without being ignited if external heating supplies the missing energy. Define the plasma gain

$$Q_{\text{plasma}} \equiv \frac{P_{\text{fus}}}{P_{\text{aux}}}, \tag{62}$$

where P_{aux} is the heating delivered to the plasma by neutral beams, radio frequency systems, or other actuators. In steady state, a stripped-down energy balance is

$$P_\alpha + P_{\text{aux}} = P_{\text{loss}}. \tag{63}$$

As P_{aux} tends to zero while the equality remains possible, Q_{plasma} tends to infinity: that limiting case is ignition. At finite Q_{plasma}, the plant must supply auxiliary power and also pay the inefficiency of producing and delivering it.

Net electric power additionally depends on neutron-energy capture, thermal conversion and recirculating loads such as refrigeration, pumps and controls. Q_{plasma} measures the plasma contribution to that wider power balance.

Design consequence. Finite plasma gain requires auxiliary power. Its required level depends on transport through τ_E.

31. Why transport is the unresolved clock

A particle in a uniform magnetic field circles with a characteristic gyroradius ρ. A collision changes the direction of its velocity; the center of the next gyro-orbit is displaced by an amount of order ρ. If collisions occur at rate ν, then in time t there are approximately $N = \nu t$ independent steps. Random-walk addition gives

$$\langle \Delta r^2 \rangle \sim N\rho^2 = \nu t \rho^2.$$

By the definition $\langle \Delta r^2 \rangle \sim 2Dt$ in one radial direction, the classical cross-field diffusion estimate is

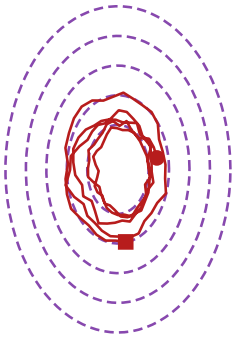
$$D_{\text{cl}} \sim \frac{\rho^2 \nu}{2}. \tag{64}$$

A container of minor radius a would then have a diffusion time of order

$$\tau_{\text{diff}} \sim \frac{a^2}{D_{\text{cl}}}. \tag{65}$$

This is a transparent reason strong field helps: $\rho \propto 1/B$, so the classical diffusion coefficient falls roughly as $1/B^2$ if the collision rate is held fixed.

Measured energy transport often exceeds this classical estimate. Fluctuations and turbulence correlate motion across flux surfaces. The random walk supplies a baseline; reactor prediction uses a confinement model calibrated to experiments.



one guiding centre, 150 collisions:
each kick moves it about one gyro-radius ρ

the estimate that follows

$N = \nu t$ collisions in time t

$\langle \Delta r^2 \rangle \sim N\rho^2 = \nu t\, \rho^2$

$D_{\mathrm{cl}} \sim \rho^2 \nu / 2$

a diffusion coefficient built from ONE step size and ONE step rate --- this is the CLASSICAL estimate

Real tokamaks lose heat faster than this. Toroidal geometry adds a neoclassical enhancement, and TURBULENCE adds a further channel that this picture does not model at all.

Explanatory sketch; schematic, not to scale.

Figure 17. Independent collisional steps provide a classical transport scale; correlated fluctuations add further transport.

Design consequence. Cross-field transport sets the loss time. Reactor calculations therefore need a confinement model with a stated applicability range.

32. Confinement is calibrated to experiments

For reactor studies, the working confinement time is often represented as

$$\tau_E = H\, \tau_{E,\mathrm{ref}}(I_p, B_0, n, P_{\mathrm{loss}}, R_0, a, \kappa, \ldots). \tag{66}$$

Here $\tau_{E,\mathrm{ref}}$ is a regression fitted to a specified database of experiments, and H is the ratio of the observed confinement to that reference. Its use requires the database, operating regime, units, heating conditions and shape range of the selected regression. Treat the regression as data with a stated applicability range.

The distinction between a derived equation and a fit matters especially here. Equation (64) followed from a stated random-walk model. Equation (66) is a bookkeeping identity once a reference scaling is chosen; its predictive content lives in the data and the applicability range of $\tau_{E,\mathrm{ref}}$.

A practical design calculation must therefore report at least three things: the selected reference law, the assumed H, and a sensitivity range. An assumed H above comparable experimental performance identifies the confinement improvement the design requires.

Scope. Confinement is represented here by an empirical reference law and multiplier H. The selected law, units and applicability range accompany the numerical reconstruction.

Design consequence. Confinement closes the energy balance alongside geometry and pressure. The assumed reference law and H must be tested together.

33. Pressure cannot rise freely: the origin of a beta limit

The local pressure-versus-line-bending balance gives a scaling estimate. Practical stability thresholds require a specified equilibrium and stability model.

The beta in §8 compares plasma pressure with magnetic pressure:

$$\beta = \frac{p}{B_0^2/(2\mu_0)}.$$

Why should there be an upper limit? Imagine a bulge of a pressure-carrying flux tube. The pressure force trying to enlarge it scales as p times its cross-sectional area. Bending the field lines that surround it costs magnetic energy; the restoring scale is set by B_θ^2/μ_0, where B_θ is the poloidal field that ties the plasma current to the geometry.

Ampere's-law scale at the edge is

$$B_\theta \sim \frac{\mu_0 I_p}{2\pi a}.$$

Equating the pressure and field-bending scales gives

$$p_{\max} \sim \frac{B_\theta^2}{\mu_0} \sim \frac{\mu_0 I_p^2}{4\pi^2 a^2}.$$

Divide by $B_0^2/(2\mu_0)$ to obtain the dimensionless structural form

$$\beta_{\max} \propto \left(\frac{I_p}{aB_0}\right)^2 \quad \text{in this crude local balance.} \tag{67}$$

The balance identifies pressure drive and magnetic restoring force. A global ideal-MHD calculation includes geometry and profiles when determining a stability boundary.

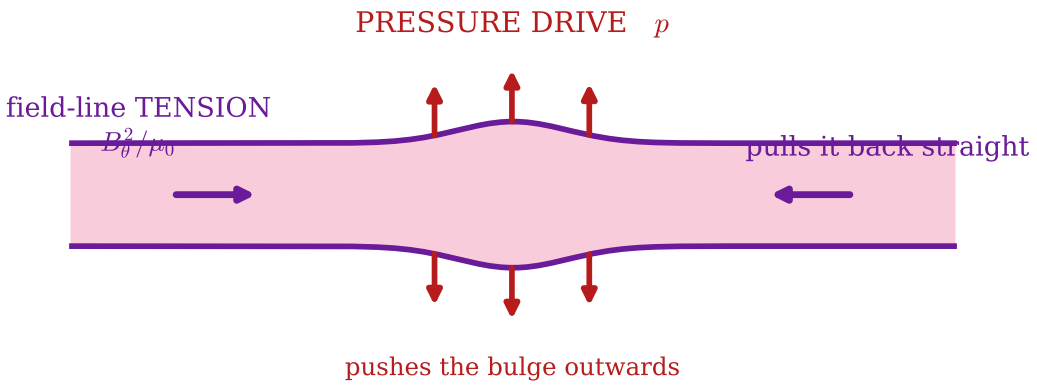


Figure 18. Pressure drives a bulge while magnetic line bending supplies a restoring force. The local balance gives a scaling estimate.

Design consequence. Pressure support depends on current, field and shape. A practical stability constraint must express that coupling.

34. The normalized beta used in reactor design

Detailed ideal-MHD calculations and experiments are commonly summarized by a normalized beta. With β written as a fraction rather than a percentage, define

$$\beta_N \equiv \frac{\beta\, a(\mathrm{m})\, B_0(\mathrm{T})}{I_p(\mathrm{MA})}. \tag{68}$$

This numerical convention uses a in metres, B_0 in tesla and I_p in MA. Using beta in percent instead would give a normalized-beta value 100 times larger. It is useful because comparable tokamak equilibria show an approximate operating boundary of the form

$$\beta \lesssim \beta_{N,\max} \frac{I_p(\mathrm{MA})}{a(\mathrm{m}) B_0(\mathrm{T})}. \tag{69}$$

The coefficient $\beta_{N,\max}$ is empirical/model-dependent; it must be imported from a specified stability study or design paper.

Equation (69) makes the current–pressure coupling explicit: larger allowable current raises allowable pressure, while a larger aB_0 at fixed current lowers this particular normalized beta. Combining it with the current ceiling on §15 makes the coupling visible: the kink limit restricts I_p, which then restricts the pressure that can be claimed for fusion power.

> **Design consequence.** The pressure allowance depends on the current permitted by kink stability. Both constraints must hold at the same operating point.

35. Density has a separate experimental limit

It would be tempting to use Eq. (57) and conclude that a chosen pressure can always be obtained by trading density against temperature. Tokamak operation does not permit an unlimited trade. A widely used experimental reference is the Greenwald density

$$n_G = \frac{I_p(\mathrm{MA})}{\pi a^2(\mathrm{m}^2)} \times 10^{20}\ \mathrm{m}^{-3}. \tag{70}$$

Operational scenarios usually report the fraction $f_G = n_e/n_G$, rather than merely a raw density.

Scope. The Greenwald relation is an empirical operating reference. The chosen scenario must state the density averaging convention and allowable fraction f_G; the numerical comparison uses a line-averaged electron density.

The formula still carries useful transparent geometry. At fixed current, a smaller cross-section raises the reference density as $1/a^2$. At fixed a, more current raises it. But that same current is limited by q_* and the same density contributes to pressure and radiation. The fraction f_G records the resulting density demand relative to the reference.

> **Design consequence.** Density has its own operating constraint. Increasing it must remain compatible with pressure, current and confinement.

36. Toroidal geometry traps some particles

The toroidal field is stronger on the inboard side because approximately $B_\varphi \propto 1/R$. A particle moving into stronger field converts parallel motion into perpendicular motion while conserving, to lowest adiabatic order,

$$\mu = \frac{m v_\perp^2}{2B} \quad \text{and} \quad \frac{1}{2} m (v_\parallel^2 + v_\perp^2).$$

Let the particle start where the field is $B_{\min}$ and let α be its pitch angle there, so $v_\perp = v \sin\alpha$. At the turning point $v_\parallel = 0$. Conservation of μ gives

$$\frac{v^2 \sin^2\alpha}{B_{\min}} = \frac{v^2}{B_{\max}}, \qquad \sin^2\alpha = \frac{B_{\min}}{B_{\max}}.$$

Particles with larger perpendicular fraction reach this condition before the high-field side and bounce back. For isotropically distributed velocity directions evaluated at $B_{\min}$, the local trapped fraction is therefore

$$f_t = \sqrt{1 - \frac{B_{\min}}{B_{\max}}}. \tag{71}$$

The same $1/R$ field that closes the magnetic corridor therefore creates trapped orbits. Neoclassical transport combines these toroidal guiding-centre orbits with Coulomb collisions; pressure gradients also produce a self-generated current.

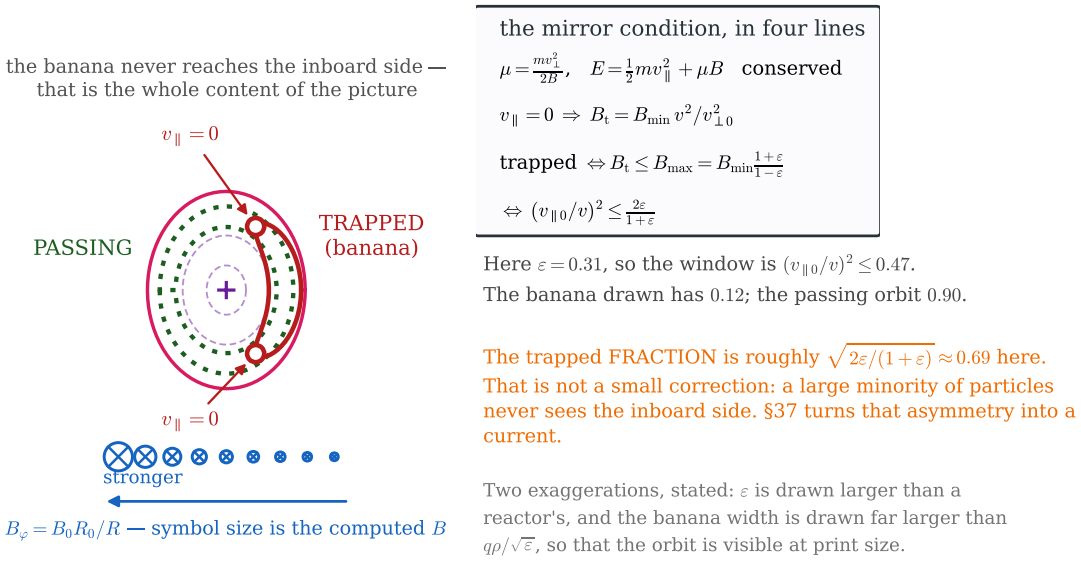


Figure 19. Passing and trapped guiding-centre orbits. Turning points follow from conservation of magnetic moment and energy; orbit widths are schematic.

Design consequence. Toroidal field variation creates trapped particles. Their response to a pressure gradient introduces a source of plasma current.

37. A pressure gradient can generate some of the required current

A trapped particle does not sample the two sides of the torus symmetrically. Its guiding-center drifts and collisions depend on energy and pitch angle. When pressure falls outward, there are more energetic particles on the inner side of a local radial interval than on the outer side. The resulting imbalance in trapped-particle motion is communicated by collisions to passing electrons. The outcome is a net toroidal current called the *bootstrap current*.

The direction and detailed coefficient require kinetic theory, because they depend on collisions, geometry, temperature gradients, and the current profile. What can be stated before importing such a coefficient is the structure:

$$j_{\mathrm{bs}} \propto -\frac{dp}{dr} \quad \text{and} \quad I_{\mathrm{bs}} = \int j_{\mathrm{bs}}\, dA. \tag{72}$$

No pressure gradient means no bootstrap drive in this simplified statement. A steeper gradient can produce more self-driven current, but it also moves the plasma closer to pressure-driven stability limits. Bootstrap current couples the pressure profile to the current needed for magnetic geometry.

Define the bootstrap fraction

$$f_{\mathrm{bs}} \equiv \frac{I_{\mathrm{bs}}}{I_p}. \tag{73}$$

The fraction f_{bs} is the primary parameter f_B. A kinetic model supplies its profile-dependent value, which sets the remaining transformer or external-drive demand.

Model used here. The proportionality describes the pressure-gradient drive in a reduced model. A quantitative prediction requires geometry, collisionality, species gradients and profiles.

> **Design consequence.** The pressure profile supplies part of the current. Its bootstrap contribution must remain compatible with stability and confinement.

38. The full current ledger

The toroidal plasma current can have several sources. Write the accounting identity

$$I_p = I_{\mathrm{ohm}} + I_{\mathrm{CD}} + I_{\mathrm{bs}}. \tag{74}$$

Equation (74) is an accounting identity; performance enters through the three terms. The first term is transformer-induced ohmic current; it is available while the central solenoid changes linked flux. The second is externally driven non-inductive current, for example from waves or beams transferring directed momentum. The third is the pressure-gradient-driven bootstrap current from §37.

For a stationary non-inductive state with negligible transformer contribution, $I_{\mathrm{ohm}} \simeq 0$, and Eq. (74) reduces to

$$I_{\mathrm{CD}} = (1 - f_{\mathrm{bs}}) I_p. \tag{75}$$

The larger the bootstrap fraction, the smaller the externally driven current requirement. But current drive consumes power and its efficiency depends on density, temperature, magnetic field, launch geometry, and the actuator itself. The bootstrap fraction must come from a compatible profile model.

> **Design consequence.** The required current must be supplied by transformer action, bootstrap current and affordable external drive. Their balance depends on operating mode.

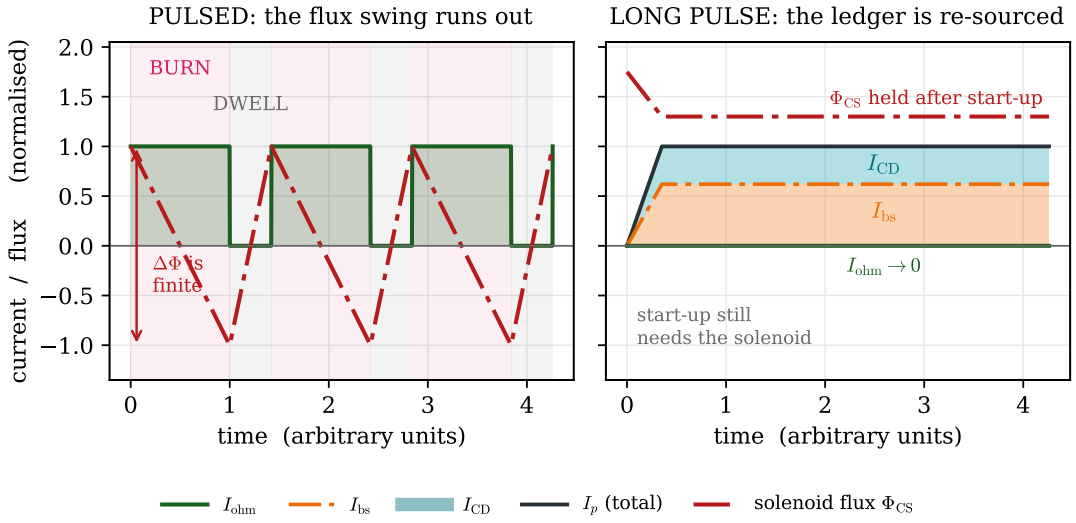
39. Pulsed operation is also a legitimate design choice

The central solenoid is a transformer primary. Faraday's law gives the loop voltage around the torus:

$$\oint \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d\Phi_{\mathrm{link}}}{dt}.$$

For a torus of major radius R_0, a representative toroidal electric field satisfies $2\pi R_0 E_\varphi = -d\Phi_{\mathrm{link}}/dt$. The solenoid has a finite usable flux swing $\Delta\Phi_{\mathrm{link}}$, so it can induce current only for a finite time. This is the physical origin of the transformer clock introduced on §11.

There are two coherent responses. One is a pulsed plant: use transformer current during a burn, then recharge the solenoid between pulses. The other is a long-pulse or steady-state plant: replace the lost ohmic contribution with the current-drive and bootstrap terms in Eq. (74). Neither option is free. Pulsing creates thermal cycling and dwell-time questions; non-inductive operation creates actuator power and accessibility questions. The operating choice must account for both the plant duty cycle and actuator demand.



Left, the only arithmetic on the figure: $V_{\mathrm{loop}} = -d\Phi/dt$ and $I_{\mathrm{ohm}} = V_{\mathrm{loop}}/R_p$ give $t_{\mathrm{burn}} = \Delta\Phi/(I_{\mathrm{ohm}} R_p)$ — a finite swing buys a finite burn. (Internal inductance is neglected; start-up consumes part of $\Delta\Phi$ before the burn begins.) Right, the $I_{\mathrm{bs}}/I_{\mathrm{CD}}$ split drawn is ILLUSTRATIVE: only the sum is constrained here. Neither panel claims its architecture is preferable— each states a different price.

Figure 20. Two current-supply strategies: transformer-supported burns with recharge intervals, or non-inductive current sustaining the burn. The illustrated current shares are schematic.

Design consequence. Transformer flux limits the inductive burn. Extending it requires more usable flux or a larger non-inductive current contribution.

40. The kink limit returns as a constraint on every current source

The current ledger does not allow arbitrary total current. A helical disturbance can be labelled by integers (m, n) through a displacement pattern $\xi \propto \cos(m\theta - n\varphi)$. Along an unperturbed field line, a poloidal turn changes θ by 2π while the toroidal angle changes by $2\pi q$. The phase change is therefore $2\pi(m - nq)$. When $q = m/n$, the disturbance remains in phase after a circuit; field-line bending has the least geometric leverage to average it away.

This is the accessible reason safety factor is a stability variable rather than only a geometric ratio. The full ideal-MHD energy principle determines the precise stability boundary. Reactor studies commonly use a specified global constraint

$$q_* > q_K, \tag{76}$$

with q_K supplied by the chosen stability model. Substituting the shaped safety-factor formula, Eq. (39), reproduces the current ceiling already derived in Eq. (40). Every term in the current ledger must fit under that same ceiling; bootstrap current helps meet a required I_p but does not abolish the stability constraint on I_p itself.

Model used here. The resonance argument motivates the role of field-line pitch. The global threshold q_K is supplied separately by the selected stability model.

> **Design consequence.** Kink stability constrains the total current, regardless of its source. Bootstrap current reduces external drive demand within that same ceiling.

41. The operating constraints must overlap

For an ignited candidate state, the coupled operating requirements include

$$q_* > q_K \qquad \text{current and kink margin,} \tag{77}$$

$$\beta \lesssim \beta_{N,\max}\frac{I_p}{aB_0} \qquad \text{pressure and MHD margin,} \tag{78}$$

$$n_e \lesssim n_G \qquad \text{density reference,} \tag{79}$$

$$n_i\tau_E \geq \frac{12T}{\langle\sigma v\rangle E_\alpha} \qquad \text{self-heating requirement,} \tag{80}$$

$$I_p = I_{\text{ohm}} + I_{\text{CD}} + I_{\text{bs}} \qquad \text{current accounting.} \tag{81}$$

The beta relation uses the metre–tesla–MA convention of §34. The notation $\lesssim$ in the second and third lines is deliberate. Those limits are modelled or empirical margins, not exact equalities. The last line is an identity, but its terms depend on operating mode and profiles. The first and fourth lines are inequalities derived from explicit physical balances, with a stability threshold or reactivity curve still to be specified.

Increasing density might help Eq. (80) but challenge Eq. (79); increasing current can improve one normalized beta bound but challenge Eq. (77); improving confinement changes needed heating and current drive. The correct question is whether the inequalities overlap with usable margin.

> **Design consequence.** An operating point must satisfy the coupled constraints with margin. The remaining geometric input is the radial space required around the plasma.

42. From plasma requirements to the radial build

The plasma-side roles of Freidberg's Table-1 parameters: $a, R_0, \kappa, T, n, p, \tau_E, B_0, \beta, I, q_*, f_B$ have now been introduced. Quantities such as $A, \delta, q(r), \beta_N, H, I_{\text{CD}}, Q_{\text{plasma}}$ remain supporting quantities, not independent primary knobs. The missing hardware parameters are b and c; they must be derived before numerical closure begins. Neutron interactions, magnetic forces and current-carrying area determine their roles.

Design consequence. Neutron capture and magnet forces require radial space. Deriving b and c completes the reactor geometry.

43. A magnetic bottle does not catch the neutron

The D–T reaction in Eq. (1) has two products with very different fates. The charged alpha particle remains magnetically guided and can transfer energy back to the plasma. The neutron has no electric charge. Its Lorentz force is exactly zero:

$$\mathbf{F}_n = 0 \cdot (\mathbf{E} + \mathbf{v}_n \times \mathbf{B}) = 0.$$

It crosses the magnetic surfaces in a nearly straight line until it interacts with surrounding material.

The neutron carries 14.1 MeV, approximately four fifths of the fusion energy. A power-producing machine must therefore do four jobs outside the plasma: survive the first impact, slow and capture neutrons, breed replacement tritium, and protect the magnets and vacuum boundary from the remaining radiation. These jobs force a radial region between plasma and TF coil. We call its total inboard/outboard thickness b.

Design consequence. Neutron energy requires a surrounding nuclear region. Its thickness competes with the radius available to the inboard magnet.

Scope. The following one-dimensional model relates thickness to interaction lengths. Material-specific neutron transport, heat removal and tritium recovery determine the engineered build.

44. Why the nuclear island has a thickness

First consider the slowing-down part of the problem. In a homogeneous material with target number density N and an effective interaction cross section σ_s, the mean distance between relevant collisions is

$$\lambda_s = \frac{1}{N\sigma_s}. \tag{82}$$

This follows directly from probability. In a thin distance dx, a neutron has collision probability $N\sigma_s dx = dx/\lambda_s$. Thus the chance of no collision decreases exponentially with distance.

For a transparent first model, suppose each collision removes a fixed fraction ξ of the neutron energy on average. Over many collisions, the average energy loss per distance can be represented by

$$\frac{dE}{dx} = -\frac{\xi E}{\lambda_s}. \tag{83}$$

Separating variables gives

$$\frac{dE}{E} = -\frac{\xi\, dx}{\lambda_s} \quad \Longrightarrow \quad E(x) = E_0 \exp\!\left(-\frac{\xi x}{\lambda_s}\right).$$

To reduce a 14.1 MeV neutron from E_0 to a chosen capture-friendly energy E_c, the associated slowing-down distance is therefore

$$b_{\text{slow}} = \frac{\lambda_s}{\xi} \ln\!\left(\frac{E_0}{E_c}\right). \tag{84}$$

The logarithm matters: a large energy ratio does not require a proportional thickness, but it does require several interaction lengths. Equation (84) already shows why a thin wall cannot be a reactor solution.

Now consider breeding. One useful reaction is

$$n + {}^{6}\mathrm{Li} \longrightarrow {}^{4}\mathrm{He} + {}^{3}\mathrm{H} + 4.8\,\mathrm{MeV}.$$

The emitted tritium replaces fuel consumed by D–T fusion. If slow neutrons are absorbed with macroscopic probability per length $\Sigma_B = N_6\sigma_B$, their surviving flux Γ obeys

$$\frac{d\Gamma}{dx} = -\Sigma_B\Gamma, \qquad \Gamma(x) = \Gamma(0)e^{-x/\lambda_B}, \qquad \lambda_B = \frac{1}{\Sigma_B}.$$

Capturing a chosen fraction f_{cap} then requires

$$b_{\mathrm{breed}} = -\lambda_B \ln(1 - f_{\mathrm{cap}}). \tag{85}$$

Both λ_s and λ_B depend on material and neutron energy. Thus Eqs. (84)–(85) retain those material and energy dependencies.

45. From microscopic collisions to the reactor dimension b

The nuclear region performs several functions. At the level needed for a first reactor layout, write

$$b = b_{\mathrm{FW}} + b_{\mathrm{mult}} + b_{\mathrm{slow}} + b_{\mathrm{breed}} + b_{\mathrm{shield}} + b_{\mathrm{vac}}. \tag{86}$$

Here b_{FW} is the first-wall allowance; b_{mult} represents any neutron-multiplying region; b_{slow} and b_{breed} are motivated by the preceding transport calculation; b_{shield} protects coils and surrounding structures; and b_{vac} is the vacuum-boundary allowance. Some concepts combine or reorder these functions. The terms in Eq. (86) are non-overlapping radial allowances; combined functions are counted once.

The first wall also sees a power-density demand. In the large-aspect-ratio circular approximation, a torus with major radius R_0 and minor radius a has surface area

$$S_{\mathrm{wall}} \simeq (2\pi R_0)(2\pi a) = 4\pi^2 R_0 a.$$

If P_n is the neutron power and W_n is an allowed average neutron wall loading,

$$\frac{P_n}{4\pi^2 R_0 a} \le W_n \qquad \Longrightarrow \qquad R_0 a \ge \frac{P_n}{4\pi^2 W_n}. \tag{87}$$

Elongation and detailed surface shape modify this estimate; they do not remove its message. At fixed fusion-neutron power, a smaller first-wall area means a harsher wall loading. Nuclear performance therefore couples the two plasma dimensions R_0 and a even before we make a plasma-transport calculation.

The more immediate geometric consequence of b is radial crowding. On the inboard side, the blanket lies between the plasma edge and the TF coil. It displaces the coil toward the symmetry axis, where a toroidal field is stronger. The next section turns that qualitative statement into an equation.

Design consequence. The nuclear build b separates plasma from magnet. That separation changes the attainable field through the inboard geometry.

46. The field gets stronger as the coil moves inward

The continuous-coil model already gave

$$B_\varphi(R) = \frac{\mu_0 N I_c}{2\pi R}.$$

At the plasma centre $R = R_0$, write this as B_0. The inboard edge of the plasma is at $R_0 - a$. After the nuclear island and an inboard TF-coil build have been added, the innermost high-field location lies approximately at

$$R_{\text{in}} = R_0 - a - b - c. \tag{88}$$

The sign is important: moving inward means decreasing R, which increases the toroidal field for fixed coil current.

Using $B_\varphi(R)R = B_0 R_0$, the maximum field in this simplified radial build is

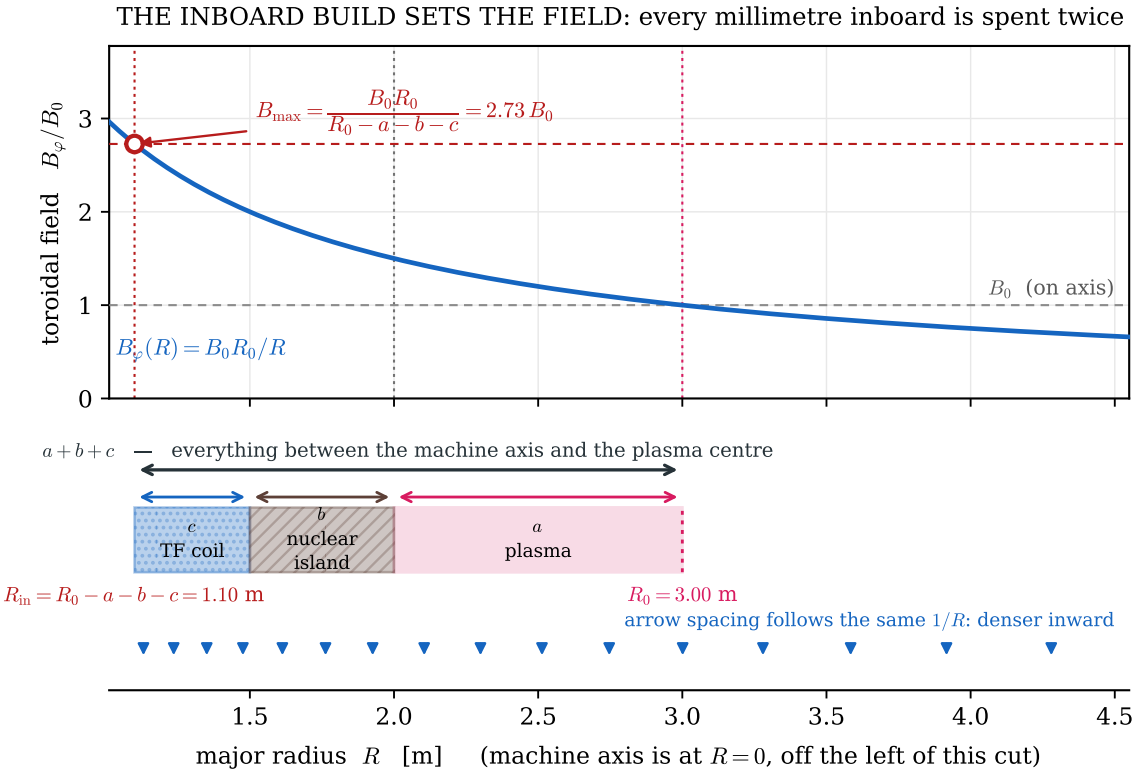
$$B_{\max} = \frac{B_0 R_0}{R_0 - a - b - c}. \tag{89}$$

Equivalently,

$$B_0 = B_{\max}\left(1 - \frac{a+b+c}{R_0}\right). \tag{90}$$

This is the first compact expression of the radial-build problem. A larger plasma minor radius, a thicker nuclear island, or a thicker TF coil leaves less inboard radius. At fixed material limit $B_{\max}$, each reduces the attainable central field B_0.

The equation also explains why a tokamak cannot be designed by choosing a desirable B_0 and adding the blanket and magnets afterwards. Those supposedly external components feed back directly into the field available to the plasma.



Rearranged, Eq. (90): $B_0 = B_{\max}(1 - (a+b+c)/R_0) = 2.73 \times (1 - 1.90/3.00) = 1.00\,B_0$ — the identity, evaluated. The consequence is the one the primer draws out: at a FIXED material limit $B_{\max}$, a larger plasma, a thicker nuclear island or a thicker coil each reduce the attainable B_0. The three compete for the same inboard radius, and none of them is free. a, b, c here are ILLUSTRATIVE: b is not derived until the nuclear-island section and c not until the coil-stress section.

Figure 21. The inboard radial ledger connects nuclear thickness b and magnet thickness c to the field available at the plasma. Dimensions shown are illustrative.

47. A field coil must carry current and survive its own force

A magnetic field stores energy. In vacuum its energy density is

$$u_B = \frac{B^2}{2\mu_0}. \tag{91}$$

One way to obtain this expression is to build a solenoid gradually. Its magnetic energy is $U = \frac{1}{2}LI^2$; dividing by volume and substituting the ideal-solenoid relations $L = \mu_0 N^2 A/\ell$ and $B = \mu_0 NI/\ell$ gives $U/(A\ell) = B^2/(2\mu_0)$.

The same scale appears as a magnetic pressure. If a field-filled region expands by a small volume dV at fixed flux, the released magnetic energy supplies mechanical work $p_B dV$. The resulting stress scale is

$$p_B = \frac{B^2}{2\mu_0}. \tag{92}$$

At high field this is enormous: the dependence is quadratic, so doubling field quadruples this stress scale. A TF coil is therefore not just superconducting cable. It needs a structural load path that prevents the winding pack from being crushed, pulled apart, or driven beyond allowable strain.

For a transparent thickness estimate, let a representative magnetic force be $F \sim p_B A_{\rm load}$ and let structural material of cross-sectional area $A_{\rm str}$ be limited to stress $\sigma_{\rm allow}$. Force balance requires

$$\sigma_{\rm allow} A_{\rm str} \gtrsim p_B A_{\rm load}.$$

If the geometry converts these areas into a representative support length ℓ_s, then the structural contribution obeys the scaling

$$c_{\rm str} \sim \frac{B_{\rm max}^2}{2\mu_0 \sigma_{\rm allow}} \ell_s \times (\text{geometry factor}). \tag{93}$$

The geometry factor incorporates tensile, centering, bending and wedging loads. It connects the field-pressure scale to the required support thickness.

Scope. The support-thickness estimate uses a representative load path. Detailed sizing requires the coil geometry and material allowables at the operating temperature and fluence.

48. The other part of coil thickness: enough current-carrying area

Stress is not the only reason a coil occupies space. Ampère's law requires a total enclosed current approximately

$$NI \simeq \frac{2\pi R_{\rm in} B_{\rm max}}{\mu_0}. \tag{94}$$

The product NI, called ampere-turns, cannot be carried through zero area. If $J_{\rm wp}$ is the allowed current density averaged over the complete winding pack (including conductor, stabilizer and internal pack allowances), the required current-carrying area scales as

$$A_{\rm wp} \gtrsim \frac{NI}{J_{\rm wp}}. \tag{95}$$

For a representative coil height $h_{\rm wp}$, this yields a winding-pack thickness scale

$$c_{\rm wp} \sim \frac{2\pi R_{\rm in} B_{\rm max}}{\mu_0 J_{\rm wp} h_{\rm wp}}. \tag{96}$$

The area-to-thickness conversion uses the representative height $h_{\rm wp}$. Its geometric factor depends on how the total ampere-turns are distributed among coils.

The full primary parameter is therefore a radial ledger:

$$c = c_{\rm str} + c_{\rm wp} + c_{\rm ins} + c_{\rm cool} + c_{\rm margin}. \tag{97}$$

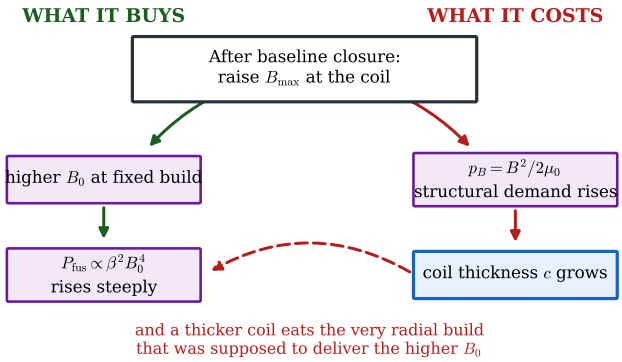
Here $c_{\rm str}$ denotes support outside the winding pack. The remaining allowances cover insulation, cooling and margins outside that pack; internal allowances already included in $J_{\rm wp}$ are counted only once. The first two terms, Eqs. (93) and (96), are the two irreducible physical demands: survive the field and produce the field.

49. Higher field changes both performance and magnet build

We can now say precisely what a higher-field magnet changes. At fixed radial geometry, Eq. (90) makes a higher allowable $B_{\rm max}$ available as a higher central field B_0. But Eqs. (93) and (96) also show the price: field increases magnetic stress quadratically and imposes a technology-dependent current-density requirement. Both effects enter the magnet-build calculation.

High-temperature superconductors (HTS) enter the argument after the baseline closure identifies a field requirement beyond the conventional magnet model. They enter this trade through the field and current density their conductors can sustain. Their value to a reactor depends on the resulting plasma performance, magnet build and engineering cost. The reference calculation will compare field choices within the same coupled constraints.

Design consequence. A higher allowable coil field can raise B_0, while stress and current-density requirements consume radial space. The benefit requires a coupled calculation.



Explanatory sketch. No magnet technology, vendor or numerical performance claim is implied; this is the shape of the trade-off only.

Figure 22. Higher coil field raises the available plasma field and the structural demand. The radial build couples those two effects.

50. The primary design vocabulary is complete

The primary parameter set is now complete, with $n = n_i$, $I = I_p$ and $f_B = f_{\mathrm{bs}}$:

$$a,\ R_0,\ \kappa,\ b,\ c,\ T,\ n,\ p,\ \tau_E,\ B_0,\ \beta,\ I,\ q_*,\ f_B.$$

The set combines geometry, plasma state, magnetic balance and the radial builds forced by neutron transport and magnet engineering. Its variables are coupled:

$$p = 2nT, \qquad q_* \propto \frac{a^2 B_0}{R_0 I}, \qquad B_0 = B_{\max}\left(1 - \frac{a+b+c}{R_0}\right).$$

These relations show how a component change propagates through the machine.

The next part applies Freidberg's 2015 reference calculation. It will select explicit engineering and nuclear assumptions, express the primary parameters as coupled functions rather than isolated knobs, and then compare the demanded plasma state with four operational tests: Greenwald density, Troyon beta, kink safety factor, and achievable bootstrap fraction. The core question will be whether all four inequalities can hold simultaneously for a conventional baseline.

Design consequence. The parameter set now supports a numerical closure. The reference calculation will test whether its operating constraints overlap.

Scope. The reference is Freidberg, Mangiarotti and Minervini (2015), Sections 3–6. Its reduced blanket and analytic TF-coil models supply the numerical design assumptions.

51. Entering the 2015 argument: a model, not a verdict

We now change tasks. The first fifty pages asked, "Why must the machine contain these parts?" The next task is, "If a particular conventional-reactor model is required to make a specified amount of power, what plasma state does that model demand?" This is the calculation made analytically by Freidberg, Mangiarotti, and Minervini in 2015.

Their reference design is intentionally simplified. It specifies an electric output, an allowed neutron wall loading, a thermal conversion efficiency, a maximum inboard coil field, magnet stress and winding-pack limits, and a cap on recirculating power. It then asks whether a conventional plasma can simultaneously meet the resulting density, beta, kink-safety-factor, and bootstrap-current requirements.

> **Design consequence.** The following conclusion, if reproduced, will be conditional: there is no self-consistent reactor within the stated baseline assumptions. It will not say that no tokamak can ever work. That distinction is where honest engineering begins.

Model and limits. Every numerical input in this reconstruction will be visibly tagged as either an economic choice, an engineering limit, a nuclear-data/model choice, or an empirical plasma-physics fit. Changing one of them defines a different reactor case; it is not a mathematical error.

52. The reference design asks the machine to do a specific job

For the 2015 baseline, the primary imposed mission choices are

$$P_E = 1000\,\mathrm{MW_e}, \qquad W_n = 4\,\mathrm{MW\,m^{-2}}, \qquad \eta_T = 0.40.$$

P_E is desired electric output, W_n is the chosen upper bound on average neutron wall loading, and η_T converts fusion heat into electric power:

$$P_E = \eta_T P_F. \tag{98}$$

The subscript F denotes the total thermal power credited to fusion and blanket reactions in the paper's accounting model. The point of displaying these numbers is not to bless them. A $1000\,\mathrm{MW_e}$ plant, a $4\,\mathrm{MW\,m^{-2}}$ wall limit, and 40% conversion efficiency are assumptions to be audited.

The reference magnet model also takes

$$B_{\max} = 13\,\mathrm{T}, \qquad \sigma_{\mathrm{allow}} = 600\,\mathrm{MPa}, \qquad J_{\mathrm{wp}} = 20\,\mathrm{MA\,m^{-2}}.$$

These are a baseline conventional-superconductor and structural model, not claims about the best magnets available today. They are precisely the assumptions later challenged by the high-field option.

Supporting quantity — reference inputs. The quantities $P_E, W_n, \eta_T, B_{\max}, \sigma_{\mathrm{allow}}$, and J_{wp} are inputs to a chosen design case. They guide the determination of the primary parameter set; they are not additional Table-1 parameters.

53. Power and wall loading turn two dimensions into one family

For the shaped boundary, the source uses the standard mean-square ellipse-perimeter approximation. Revolving it around the major radius gives the first-wall area

$$S_{\mathrm{wall}} \simeq 4\pi^2 R_0 a \left(\frac{1+\kappa^2}{2}\right)^{1/2}. \tag{99}$$

This agrees with the circular result $4\pi^2 R_0 a$ when $\kappa = 1$. It is an analytic shape approximation, not a replacement for an actual first-wall surface.

The D–T neutron carries $E_n = 14.1\,\mathrm{MeV}$ of the fusion energy $E_{\mathrm{DT}} = 17.6\,\mathrm{MeV}$. In the reference energy accounting,

$$P_n = \frac{E_n}{E_F} P_F = \frac{E_n}{E_F}\frac{P_E}{\eta_T}.$$

Equating neutron power to the allowed wall loading times the wall area, $P_n = W_n S_{\mathrm{wall}}$, gives

$$R_0 a = \frac{E_n P_E}{4\pi^2 E_F W_n \eta_T}\left(\frac{2}{1+\kappa^2}\right)^{1/2}. \tag{100}$$

All quantities on the right are chosen inputs or previously earned parameters. Thus the design does not choose R_0 and a independently:

$$R_0(a) = \frac{C_{\mathrm{wall}}}{a}, \qquad C_{\mathrm{wall}} \equiv \frac{E_n P_E}{4\pi^2 E_F W_n \eta_T}\left(\frac{2}{1+\kappa^2}\right)^{1/2}. \tag{101}$$

This is why Freidberg can make a one-parameter scan. After fixing the mission and the blanket/magnet model, each trial minor radius implies a corresponding major radius.

Model and limits. Equation (99) uses a simple perimeter approximation and an average wall loading. A final reactor analysis needs local neutron transport and the actual first-wall surface. The one-parameter logic survives those refinements.

54. The radial build makes the field a coupled result

The preceding page makes R_0 a function of a. For its particular analytic TF-coil calculation, Freidberg defines the inboard field-build fraction

$$\epsilon_B = \frac{a+b}{R_0}$$

and writes

$$B_0 = B_{\max}(1-\epsilon_B).$$

This is the source model's field convention: c is then found from its stress and winding-pack equations, rather than inserted into ϵ_B. The physical radial ledger on pages 46–48 remains useful for understanding why c exists; reproducing the source's numerical scan requires us not to mix that fuller ledger with its analytic convention. Its logical structure is more important than its algebra:

$$\begin{gathered}\text{mission and wall loading} \implies R_0(a),\\ \text{neutron transport} \implies b,\\ \text{field, stress, and ampere-turns} \implies c(a, B_0),\\ \text{inboard radial build} \implies B_0(a).\end{gathered}$$

The central field is therefore neither a free wish nor merely a property of the superconductor. It is the outcome of a coupled geometric and mechanical problem.

This is also the first place to be alert to a future HTS comparison. If a different conductor technology changes the credible $B_{\max}$ or J_{wp}, it changes the solution for c and therefore the accessible B_0. We will not assume that this improves the reactor until the baseline plasma constraints have been tested.

PLACE IMAGE HERE — Causal ledger diagram: how the reference design reduces to a scan in a. Make a compact flow diagram of the four arrows displayed above. Use a distinct style for chosen inputs, primary parameters, and derived supporting quantities. End at "test plasma constraints," not at a success claim. This is a reset explanatory figure, not a cumulative machine image.

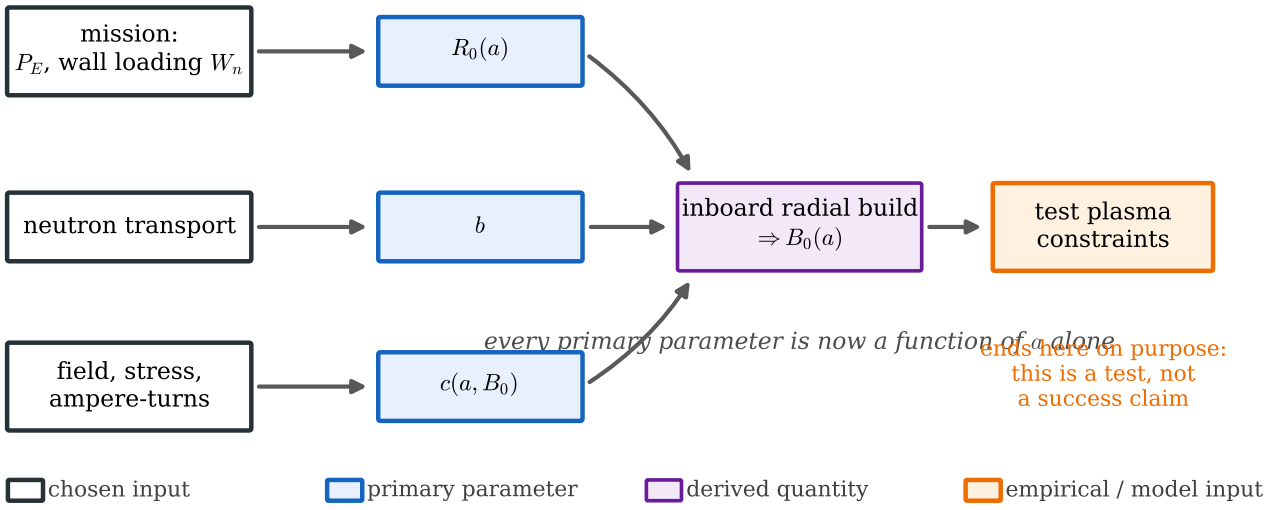


Figure 30: Causal ledger diagram: how the reference design reduces to a scan in a. *Idealized schematic; not a fabrication drawing.*

55. Fusion power fixes the required pressure

The D–T reaction rate in an equal fuel mixture is

$$\mathcal{R} = \frac{n^2}{4}\langle\sigma v\rangle(T),$$

because $n_D = n_T = n/2$. Multiplying by the energy released per reaction and integrating over plasma volume gives

$$P_{\rm fus} = \frac{\langle\sigma v\rangle(T) E_{\rm DT}}{4} \int n^2 \, dV. \tag{102}$$

For a first transparent estimate, use volume-averaged n and T:

$$P_{\rm fus} \simeq \frac{\langle\sigma v\rangle(T) E_{\rm DT}}{4} n^2 V.$$

The equal-temperature quasineutral model gave $p = 2nT$, hence

$$P_{\rm fus} \simeq \frac{\langle\sigma v\rangle(T) E_{\rm DT}}{16 T^2} p^2 V. \tag{103}$$

Power balance therefore determines a required pressure:

$$p_{\rm req} \simeq \left[\frac{16 T^2 P_{\rm fus,req}}{\langle\sigma v\rangle(T) E_{\rm DT} V}\right]^{1/2}. \tag{104}$$

This is a crucial conceptual reversal. The plasma physicist does not simply choose a pleasantly low pressure; the requested electric output and available volume demand one.

The 2015 reference model selects $T \simeq 14\,\mathrm{keV}$ because the ratio $\langle\sigma v\rangle/T^2$ is near its favorable broad maximum for D–T fuel. Later we will reproduce that statement from a cited reactivity fit and replace the uniform approximation with the paper's declared density and temperature profiles.

56. Ignition turns pressure into a required confinement time

Fusion power does not remain as useful plasma heat. The neutron escapes; for D–T, the alpha fraction is

$$\frac{E_\alpha}{E_{\rm DT}} = \frac{3.5}{17.6} \simeq \frac{1}{5}.$$

For an ignited plasma, alpha heating must replace the energy conducted and transported out of the core. Let W be thermal energy stored in the plasma and define

$$\tau_E = \frac{W}{P_{\rm loss}}.$$

The ignition balance is $P_\alpha = P_{\rm loss} = W/\tau_E$, or

$$\tau_{E,\rm req} = \frac{W}{P_\alpha}. \tag{105}$$

In the equal-temperature model, the thermal energy density is $w = \frac{3}{2}(n_e + n_i)T = 3nT = \frac{3}{2}p$. Thus $W \simeq \frac{3}{2}pV$. With $P_\alpha \simeq P_{\rm fus}/5$,

$$\tau_{E,\rm req} \simeq \frac{15pV}{2P_{\rm fus}}. \tag{106}$$

Substituting the pressure demanded by Eq. (104) gives the next causal chain:

$$P_E,\ W_n,\ \eta_T,\ a \implies R_0,\ V,\ p \implies \tau_{E,\rm req}.$$

The remaining, decisive question is whether an empirical tokamak confinement law can deliver this τ_E without demanding an unacceptable plasma current. That is where the one indispensable plasma-physics input enters the 2015 design.

57. Before testing a limit: what kind of statement is it?

The four tests ahead do not all have the same epistemic status. It would be a serious error to place them in one box labelled "first principles." We will use three labels.

Derived relation	follows from stated equations and stated approximations; for example, $p = 2nT$ and $q_* \propto a^2 B_0/(R_0 I)$.
Physics model	follows after adopting a specified equilibrium, profile, or kinetic model; it may be calculated but is not universal.
Empirical boundary	summarizes experimental operation or a regression; its coefficient and range are evidence, not a theorem.

Greenwald density is an empirical boundary. The practical Troyon beta coefficient is an experiment- and model-informed stability boundary. The kink threshold is a result of a specified MHD stability model, with profile and wall dependence. The bootstrap fraction requires a neoclassical kinetic model plus a current profile. Their physical motivation can be derived; their reactor-grade numerical coefficients cannot honestly be manufactured from the motivation alone.

The test itself is nevertheless elementary. For every constraint define a demand-to- allowance ratio,

$$\mathcal{R} \equiv \frac{\text{quantity demanded by the reference design}}{\text{quantity allowed by the stated constraint}}. \tag{107}$$

A baseline is admissible only if each relevant $\mathcal{R} \leq 1$. This notation forces us to name both numerator and denominator before announcing pass or failure.

> **Design consequence.** The goal is not to make empirical plasma physics look more certain than it is. The goal is to expose exactly which evidence a conventional reactor would need to satisfy, and whether the same design point can satisfy all of it at once.

58. The one empirical input that determines the required current

The 2015 reference case uses an ELMy H-mode confinement regression. H-mode is a high-confinement tokamak regime with a strong edge transport barrier. ELMy means that the edge undergoes recurring edge-localized-mode (ELM) relaxations: short bursts that expel a small amount of edge energy and particles. The word identifies the experiment class behind the regression; it does not add an independently derived law.[2] In the declared units, the regression is

$$\tau_E = 0.145H\,\frac{I^{0.93}R_0^{1.39}a^{0.58}\kappa^{0.78}\bar{n}^{0.41}B_0^{0.15}A^{0.19}}{P_\alpha^{0.69}} \quad \text{s.} \tag{108}$$

Here I is in MA, R_0 and a in m, $\bar{n}$ in $10^{20}\,\mathrm{m}^{-3}$, B_0 in T, P_α in MW, $A = 2.5$ is the mean ion mass used by the fit, and H is the confinement multiplier. The coefficient is meaningless if these units or the database/regime are changed.

Equation (108) is not deduced from the Lorentz force. It is the particular empirical closure selected by the paper. Its physical direction is intelligible: larger current and machine size correlate with better confinement; greater heating power correlates with more difficult power exhaust. But the exponents are measured regression exponents, not universal laws of nature.

Supporting quantity — reference confinement fit. Equation (108) is the explicit version of $\tau_E = H\tau_{E,\mathrm{ref}}$ used in this reconstruction. It is the one direct plasma-performance input that turns the ignition requirement into a required current. The freely available source is Freidberg, Mangiarotti, and Minervini (2015), Eq. (28), linked on page 1 of this thesis.

Model and limits. The reference paper assumes $H = 1$ for its conventional baseline. We will retain that assumption only while reproducing that baseline, then vary it openly when testing the advanced-confinement option.

[2] A freely accessible ITER explanation describes H-mode's steep edge pressure region and the associated edge-localized modes: https://www.iter.org/node/20687/progress-elm-physics-and-elm-control.

59. Solving the confinement fit for the current

The ignition calculation has already supplied $\tau_{E,\mathrm{req}}$. Set it equal to the right side of Eq. (108) and isolate the only unknown in this step, I:

$$I^{0.93} = \frac{\tau_{E,\mathrm{req}} P_\alpha^{0.69}}{0.145 H R_0^{1.39} a^{0.58} \kappa^{0.78} \bar{n}^{0.41} B_0^{0.15} A^{0.19}}.$$

Raising both sides to the power $1/0.93$ gives

$$I_{\mathrm{req}} = \left[\frac{\tau_{E,\mathrm{req}} P_\alpha^{0.69}}{0.145 H R_0^{1.39} a^{0.58} \kappa^{0.78} \bar{n}^{0.41} B_0^{0.15} A^{0.19}} \right]^{1/0.93} \mathrm{MA}. \tag{109}$$

There is no mystery step here: the current is whatever value makes the selected empirical confinement law equal the confinement time required by ignition.

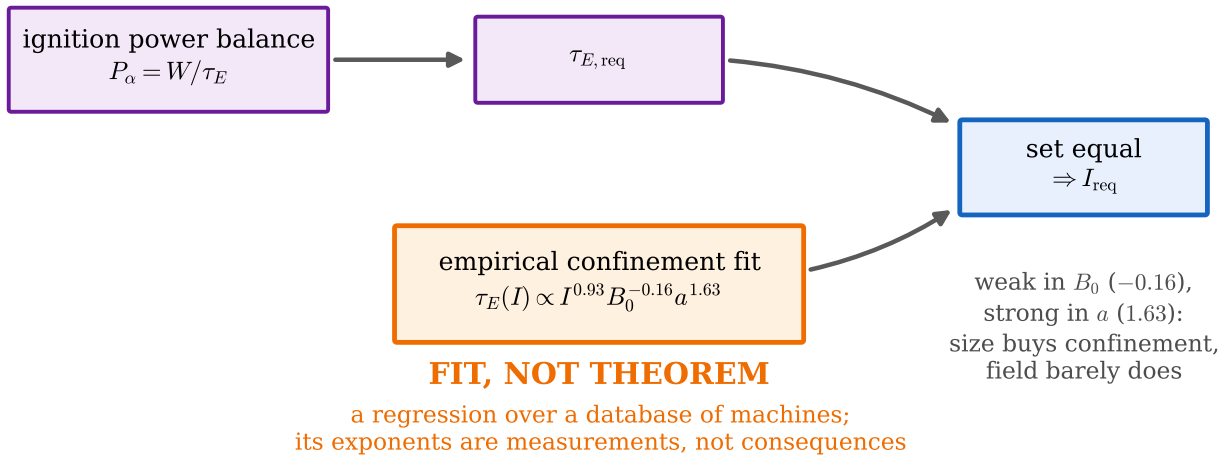
After substituting the reference design's previously determined relations, Freidberg obtains the compact scaling

$$I_{\mathrm{req}} \simeq 12.1 \, \frac{a^{1.63}}{B_0^{0.16}} \ \mathrm{MA}, \tag{110}$$

where the remaining fixed baseline assumptions and the stated units are absorbed into the coefficient. The striking feature is not the exact decimal exponents. It is the strong approximately $a^{5/3}$ current demand and the weak $B_0^{-0.16}$ dependence.

For the reference point $a = 1.34\,\mathrm{m}$ and $B_0 = 6.83\,\mathrm{T}$, the paper finds $I_{\mathrm{req}} = 14.3\,\mathrm{MA}$. This is a demanded current, not a performance target selected because it is easy to sustain.

PLACE IMAGE HERE — Reset derivation sketch: confinement requirement becomes current demand. Show a left-to-right chain: ignition power balance gives $\tau_{E,\mathrm{req}}$; the explicit empirical fit gives $\tau_E(I)$; equating them produces I_{req}. Put the words "fit, not theorem" directly beneath the confinement-law block. Include the weak $B_0^{-0.16}$ and strong $a^{1.63}$ reference scaling.



Explanatory sketch. Exponents are those of the reference regression quoted in the text; they are empirical.

Figure 31: Reset derivation sketch: confinement requirement becomes current demand. *Idealized schematic; not a fabrication drawing.*

60. High required current immediately tests field-line twist

The safety factor was introduced as the number of toroidal circuits made by a field line per poloidal circuit. At the plasma edge, more plasma current produces more poloidal field, so it reduces this twist count. For the elongated reference geometry, the paper uses

$$q_* = \frac{2\pi a^2 B_0}{\mu_0 R_0 I}\left(\frac{1+\kappa^2}{2}\right). \tag{111}$$

The first factor is the Ampère-law estimate already derived from $B_\theta \sim \mu_0 I/(2\pi a)$; the shape factor records the paper's chosen elongation correction. Thus substituting I_{req}, not an independently chosen current, produces the required q_*.

Why does a lower q_* matter? A current-carrying plasma column contains free energy. A helical displacement can bend and reconnect the field in a way that releases some of that current-associated magnetic energy. The exact stability boundary depends on current profile, pressure profile, wall response, rotation, and mode number. The reference model uses the practical external-kink/disruption condition

$$q_* \geq q_K, \qquad q_K \simeq 2. \tag{112}$$

Here a disruption means an abrupt loss of controlled plasma equilibrium and confinement, commonly accompanied by a rapid current termination and large transient loads. The mechanism is MHD current-driven instability; the numerical threshold is a stated model boundary, not a number obtained by setting two elementary equations equal.

Supporting quantity — kink ratio. The transparent test is $\mathcal{R}_K = q_K/q_*$. It is below one only when the demanded current leaves enough field-line twist to meet the adopted kink-safety criterion.

61. The four tests are now motivated, not imported

At the same reference design point, evaluate four distinct ratios:

$$\mathcal{R}_G = \frac{n}{n_G}, \qquad n_G = \frac{I}{\pi a^2} \times 10^{20}\,\mathrm{m}^{-3}, \tag{113}$$

$$\mathcal{R}_T = \frac{\beta}{\beta_T}, \qquad \beta_T = \beta_N \frac{I}{aB_0}, \tag{114}$$

$$\mathcal{R}_K = \frac{q_K}{q_*}, \qquad q_K \simeq 2, \tag{115}$$

$$\mathcal{R}_B = \frac{f_B}{f_{\mathrm{NC}}}. \tag{116}$$

Units in the Greenwald and normalized-beta forms are the conventional MA–m–T units stated earlier. Each ratio has a different physical origin:

- $\mathcal{R}_G$ checks the experimentally observed density boundary introduced on page 35;
- $\mathcal{R}_T$ checks pressure against the practical MHD beta allowance motivated on pages 33–34;
- $\mathcal{R}_K$ checks the current-driven kink boundary just explained;
- $\mathcal{R}_B$ checks whether non-inductive current sources can replace the required current in steady state.

The notation makes the decision rule uniform but does not erase the different kinds of evidence behind the four denominators. A claim that the reactor "passes plasma physics" will be permitted only after every one of these ratios is shown, with its assumptions, at the same design point.

Design consequence. The current required by confinement appears in every test: directly in n_G, β_T, and q_*, and indirectly in the current balance. This is why the conventional-tokamak difficulty has a common root rather than four unrelated problems.

62. First audit: the density requirement

The fusion-power calculation demands the reference average density

$$n = 1.43 \times 10^{20}\,\mathrm{m}^{-3}.$$

The Greenwald reference density, using the confinement-demanded $I = 14.3\,\mathrm{MA}$ and $a = 1.34\,\mathrm{m}$, is

$$n_G = \frac{14.3}{\pi(1.34)^2} \times 10^{20}\,\mathrm{m}^{-3} = 2.54 \times 10^{20}\,\mathrm{m}^{-3}.$$

Therefore

$$\mathcal{R}_G = \frac{1.43}{2.54} = 0.56 < 1. \tag{117}$$

The baseline passes this particular test with margin. That result is valuable: it shows the later failure is not created by declaring every well-known limit violated.

Model and limits. The use of the Greenwald relation at reactor scale remains an extrapolation. The conclusion here is only that the 2015 baseline lies below the boundary assumed by that model. It is not a first-principles proof that such a density is attainable in a burning reactor.

63. Second audit: pressure is barely allowed

The same reference design requires

$$\beta = 4.13\%.$$

Its adopted normalized-beta coefficient is $\beta_N = 2.8$. With $I = 14.3\,\mathrm{MA}$, $a = 1.34\,\mathrm{m}$, and $B_0 = 6.83\,\mathrm{T}$,

$$\beta_T = 2.8\,\frac{14.3}{(1.34)(6.83)}\% = 4.39\%.$$

Thus

$$\mathcal{R}_T = \frac{4.13}{4.39} = 0.94 < 1. \tag{118}$$

The comparison passes, but only narrowly. It is not a reserve that can be spent silently on profile changes, control margin, or modelling error.

Model and limits. The 2.8 coefficient is the reference model's Troyon-normalized-beta assumption. A different equilibrium, wall, profile, or stability-control model can change it. The algebra above is exact once that assumption and its conventional units are stated; the coefficient itself is not derived from the algebra.

64. Third audit: the confinement-driven current fails the kink test

Insert the same demanded current into Eq. (111). The reference calculation gives

$$q_* = 1.56.$$

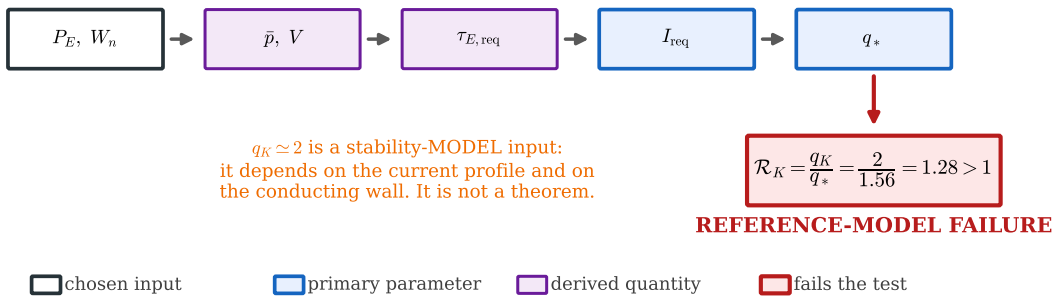
The adopted disruption-avoidance threshold is $q_K \simeq 2$, so

$$\mathcal{R}_K = \frac{2}{1.56} = 1.28 > 1. \tag{119}$$

This is a failure of the stated baseline. The required current produces insufficient edge field-line twist for the adopted kink-safety margin.

Notice the causal chain. The kink problem was not inserted after the reactor was chosen. Electric output and wall loading demanded pressure and volume; ignition demanded τ_E; the confinement fit demanded I; that I forced q_* below the adopted stability boundary. Reducing current would improve q_*, but would then fail to provide the empirical confinement time used in the baseline.

PLACE IMAGE HERE — Causal failure sketch: why the kink ratio exceeds one.
Draw the chain $P_E, W_n \to p, V \to \tau_{E,\mathrm{req}} \to I_{\mathrm{req}} \to q_* \to \mathcal{R}_K$. Mark the last comparison $1.28 > 1$ as a reference-model failure. Include a note that $q_K \simeq 2$ is a profile/wall-dependent stability-model input.



Explanatory sketch. The failure is a failure OF THE SELECTED MODEL SET at the reference point, not a statement about tokamaks in general.

Figure 32: Causal failure sketch: why the kink ratio exceeds one. *Idealized schematic; not a fabrication drawing.*

65. Fourth audit begins with a current ledger, not a magic bootstrap number

A steady-state tokamak cannot rely on a transformer indefinitely. Its current ledger must be

$$I = I_{\mathrm{CD}} + I_{\mathrm{bs}},$$

where I_{CD} is externally driven current and I_{bs} is bootstrap current. Divide by the required total current:

$$1 = \frac{I_{\mathrm{CD}}}{I} + f_B, \qquad f_B = 1 - \frac{I_{\mathrm{CD}}}{I}. \tag{120}$$

This equation is only charge accounting. It does not claim that the plasma can supply the required f_B.

For the reference recirculating-power and lower-hybrid current-drive model, the paper obtains $I_{\mathrm{CD}} = 2.25\,\mathrm{MA}$. With $I = 14.3\,\mathrm{MA}$,

$$f_B = 1 - \frac{2.25}{14.3} = 0.84. \tag{121}$$

The required bootstrap fraction is therefore very large because the demanded current is large while economically permitted external drive is limited.

The next task is not to announce a bootstrap limit from nowhere. We must derive the physical source of bootstrap current from trapped-particle collisional asymmetry, state the selected neoclassical current-density model, and integrate it over the declared profiles to obtain the achievable fraction f_{NC}. Only then can we evaluate $\mathcal{R}_B = f_B/f_{\mathrm{NC}}$.

Design consequence. Three audited comparisons now exist: density passes, beta is marginal, and kink safety fails. The remaining steady-state test will ask whether the pressure gradients that help make fusion power can naturally supply enough current to replace what the transformer cannot.

66. Where bootstrap current actually comes from

Page 36 established the first ingredient: a torus has trapped as well as passing particles. Page 37 stated the second: the pressure falls from the hot, dense core toward the edge. Put those facts together, but do not yet call the result a formula.

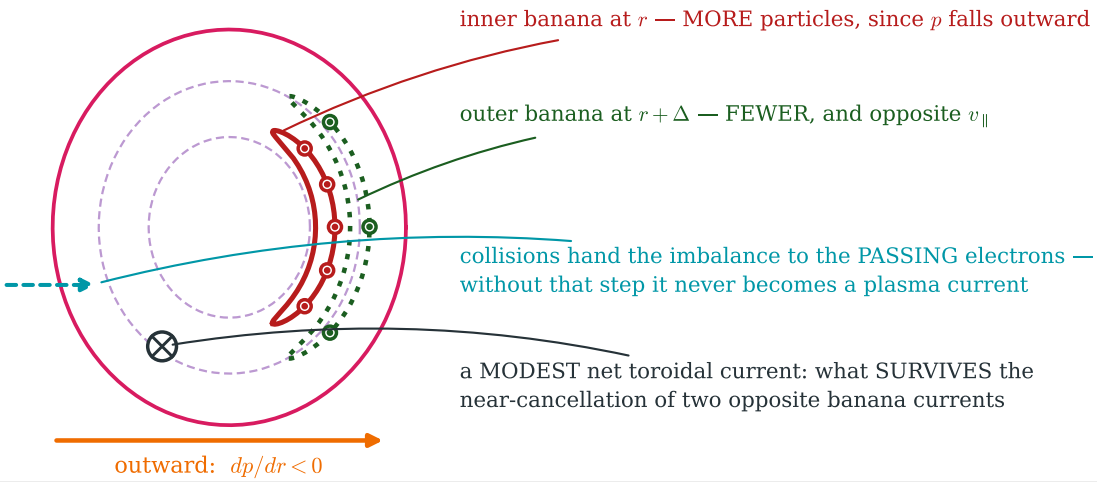
In a uniform straight field, particles moving parallel to **B** in the two directions can cancel one another's current. In a torus, trapped particles bounce between mirror points and do not complete a full parallel circuit. A pressure gradient means that the population arriving from the core side of a narrow radial shell is not identical to the population arriving from the edge side. Grad-B and curvature drifts then separate the two populations slightly. Collisions transfer part of that imbalance between trapped and passing particles. The cancellation of parallel electron motion is therefore incomplete: a toroidal current remains.

The energy source is not mysterious. It is the pressure profile itself. Flatten the profile so that $dp/dr = 0$, and the drive disappears. Make the profile steeper, and the available drive grows, while the pressure-gradient stability burden grows with it. This is precisely why bootstrap current belongs in the same closure problem as beta and kink safety.

PLACE IMAGE HERE — Physics sketch: from trapped particles to bootstrap current. Draw a small annular plasma shell. On the inboard side show stronger field and trapped electron bounce points; on the outboard side show passing electrons. Add an outward pressure-gradient arrow and a collision arrow transferring the trapped-particle asymmetry to passing electrons. End with a modest toroidal-current arrow labelled "incomplete cancellation," not "free current."

Design consequence. The pressure gradient that is required to make fusion power also creates a possible self-current. Whether it is large enough is not an independent choice: it must be calculated from a kinetic model, the same profiles, and the magnetic geometry already demanded by the reactor.

FROM TRAPPED PARTICLES TO A CURRENT — and why it is not free



1. At a given point the two adjacent bananas carry OPPOSITE $v_\parallel$, so their currents nearly cancel.
2. Because $dp/dr<0$, slightly more particles sit on the inner orbit — the cancellation is INCOMPLETE.
3. Collisions transfer that residue to the passing electrons. That step is what makes it a plasma current.
4. Scaling: $j_{\rm bs}\sim -C\frac{\sqrt{\varepsilon}}{B_\theta}\frac{dp}{dr}$, the $\sqrt{\varepsilon}$ inherited from the trapped fraction $\sim\sqrt{2\varepsilon/(1+\varepsilon)}$.

The coefficient C is left unwritten ON PURPOSE. It is where the kinetic model enters — collisionality regime, profile shape, impurity content — and the primer's own audit of the 2015 baseline turns on exactly that number. Quoting one here would hide the modelling choice the audit later argues about.

This is an INCOMPLETE CANCELLATION, not free current. It exists only while the pressure gradient is maintained, and maintaining that gradient is what the heating system is for.

Explanatory sketch; schematic, not to scale. The banana width and the orbit separation are exaggerated for legibility.

Figure 33: Physics sketch: from trapped particles to bootstrap current. *Idealized schematic; not a fabrication drawing.*

67. The first irreducibly kinetic input is declared, not smuggled in

The mathematical object behind the preceding paragraph is a drift-kinetic equation. In schematic form for a distribution function $f(\mathbf{x}, \mathbf{v})$ it is

$$\left(v_{\parallel}\mathbf{b} + \mathbf{v}_d\right)\cdot\nabla f + \frac{qE_{\parallel}}{m}\frac{\partial f}{\partial v_{\parallel}} = C[f]. \tag{122}$$

Here $\mathbf{b} = \mathbf{B}/B$, $\mathbf{v}_d$ represents the guiding-center drifts, and $C[f]$ is the Coulomb collision operator. Solving Eq. (122) with trapped orbits is beyond a first course in differential equations; writing a number for it without acknowledgement would be worse than leaving a difficult step visible.

Freidberg, Mangiarotti, and Minervini therefore choose a particular large-aspect-ratio, low-collisionality neoclassical closure. They assume equal ion and electron temperatures, $T_i = T_e = T$, one effective ion charge, $Z = 1$, and a circular area-equivalent radius

$$\hat{a} = \kappa^{1/2}a, \qquad \rho = \frac{r}{\hat{a}}. \tag{123}$$

Their local bootstrap-current model is their Eq. (41):

$$J_B(\rho) = -2.44\left(\frac{r}{R_0}\right)^{1/2}\frac{p}{B_\theta}\left(\frac{1}{n}\frac{dn}{dr} + 0.055\,\frac{1}{T}\frac{dT}{dr}\right). \tag{124}$$

The coefficient 2.44 and the relative temperature-gradient weight 0.055 are the kinetic closure, expressed in the paper's conventional working units; they are not deducible from Maxwell's equations and elementary calculus alone. Everything from this point onward—the gradients, the poloidal field, the area integral, and the comparison with the required fraction—can and will be shown explicitly.

Model and limits. Equation (124) is a declared neoclassical model, not a universal bootstrap law. Its regime assumptions, effective charge, collisionality treatment, and chosen profiles are part of the result. The accessible source is Freidberg, Mangiarotti, and Minervini (2015), Eq. (41).

68. The selected profiles make the pressure-gradient term calculable

The same paper does not leave $n(r)$ and $T(r)$ implicit. Its standard-profile choice (its Eq. (20)) is

$$T = 2\bar{T}(1-\rho^2), \qquad p = 2.5\bar{p}(1-\rho^2)^{3/2}, \qquad n = 1.5\bar{n}(1-\rho^2)^{1/2}. \tag{125}$$

The numerical factors make the bars cross-sectional averages. For example, $\int_0^1 2\rho\,[2\bar{T}(1-\rho^2)]\,d\rho = \bar{T}$. Because $d\rho/dr = 1/\hat{a}$, ordinary differentiation gives

$$\frac{1}{n}\frac{dn}{dr} = \frac{1}{1.5\bar{n}(1-\rho^2)^{1/2}}\left[1.5\bar{n}\,\frac{1}{2}(1-\rho^2)^{-1/2}\,(-2\rho)\,\frac{1}{\hat{a}}\right] = -\frac{\rho}{\hat{a}(1-\rho^2)}, \tag{126}$$

$$\frac{1}{T}\frac{dT}{dr} = \frac{1}{2\bar{T}(1-\rho^2)}\left[2\bar{T}\,(-2\rho)\,\frac{1}{\hat{a}}\right] = -\frac{2\rho}{\hat{a}(1-\rho^2)}. \tag{127}$$

Thus the bracket in Eq. (124) is

$$\frac{1}{n}\frac{dn}{dr} + 0.055\frac{1}{T}\frac{dT}{dr} = -\frac{(1+0.110)\rho}{\hat{a}(1-\rho^2)}. \tag{128}$$

Substitution of Eqs. (125) and (128) into Eq. (124) then gives, with $2.44(1.110)(2.5) = 6.77 \simeq 6.8$,

$$J_B(\rho) = 6.8\,\frac{\bar{p}}{\hat{a}^{1/2}R_0^{1/2}}\,\frac{\rho^{3/2}(1-\rho^2)^{1/2}}{B_\theta(\rho)}. \tag{129}$$

The sign convention is now carried by the direction chosen for B_θ and plasma current. The useful content is transparent: this model predicts a current density which vanishes at the axis and edge and is driven in the finite-gradient interior.

69. The poloidal field is obtained from the assumed total-current profile

Equation (129) still contains $B_\theta(\rho)$. It cannot be chosen after seeing the desired answer: it is produced by the total plasma-current profile. The paper selects

$$J_T(\rho) = \frac{I}{\pi \hat{a}^2} \frac{9\rho^{1/4}}{8} \frac{\alpha^2(1-x)e^{\alpha x}}{e^\alpha - 1 - \alpha}, \qquad x = \rho^{9/4}, \quad \alpha = 2.53. \tag{130}$$

This is another profile assumption, chosen to put appreciable current away from the axis. It is normalized correctly: using $\rho = x^{4/9}$ gives

$$\rho^{1/4}\rho\, d\rho = \frac{4}{9}\, dx, \qquad \frac{I_{\text{enc}}}{I} = \int_0^x \frac{\alpha^2(1-u)e^{\alpha u}}{e^\alpha - 1 - \alpha}\, du. \tag{131}$$

The antiderivative is

$$\int \alpha^2(1-u)e^{\alpha u}\, du = (1 + \alpha - \alpha u)\, e^{\alpha u}, \tag{132}$$

so evaluation between 0 and x yields

$$\frac{I_{\text{enc}}}{I} = \frac{(1+\alpha-\alpha x)e^{\alpha x} - 1 - \alpha}{e^\alpha - 1 - \alpha}. \tag{133}$$

Ampere's law $2\pi r B_\theta = \mu_0 I_{\text{enc}}$ now gives

$$b_\theta(\rho) \equiv \frac{B_\theta}{\mu_0 I/(2\pi\hat{a})} = \frac{1}{\rho}\left[\frac{(1+\alpha-\alpha x)e^{\alpha x} - 1 - \alpha}{e^\alpha - 1 - \alpha}\right]. \tag{134}$$

This is the missing magnetic denominator in the bootstrap calculation. No additional machine component has appeared; we have only made the previously required current profile explicit.

70. Integrate the local current, then test the steady-state requirement

In the area-equivalent circular cross-section, $dA = 2\pi\hat{a}^2\rho\,d\rho$ after integrating over the polar angle. Thus

$$I_B = 2\pi\hat{a}^2 \int_0^1 J_B(\rho)\rho\,d\rho \tag{135}$$

$$= 2\pi\hat{a}^2 \int_0^1 \left[6.8\frac{\bar{p}}{\hat{a}^{1/2}R_0^{1/2}}\frac{\rho^{3/2}(1-\rho^2)^{1/2}}{B_\theta}\right]\rho\,d\rho. \tag{136}$$

Replace B_θ using Eq. (134), divide by I, and collect constants. Since $4\pi^2(6.8) = 268.5$, the result is the paper's Eq. (44):

$$f_{\rm NC} \equiv \frac{I_B}{I} = 268\frac{\hat{a}^{5/2}\bar{p}}{\mu_0 R_0^{1/2} I^2}\int_0^1 \frac{\rho^{5/2}(1-\rho^2)^{1/2}}{b_\theta(\rho)}\,d\rho. \tag{137}$$

This integral has no elementary antiderivative. That is not a gap in the argument: it is a fully specified one-dimensional numerical quadrature. With $\alpha = 2.53$, its dimensionless value is 0.318 (three significant figures). For the reference point,

$$\kappa = 1.70, \quad \hat{a} = \sqrt{1.70}(1.34\,\mathrm{m}) = 1.747\,\mathrm{m}, \quad \bar{p} = 7.57\,\mathrm{atm}, \quad R_0 = 5.34\,\mathrm{m}, \quad I = 14.3\,\mathrm{MA}, \tag{138}$$

where $1\,\mathrm{atm} = 101325\,\mathrm{Pa}$ and $1\,\mathrm{MA} = 10^6\,\mathrm{A}$. Equation (137) then gives

$$f_{\rm NC} = 0.44. \tag{139}$$

The required value from Eq. (121) was $f_B = 0.84$. The audited comparison is therefore

$$\mathcal{R}_B = \frac{f_B}{f_{\rm NC}} = \frac{0.84}{0.44} = 1.91 > 1. \tag{140}$$

This is a failure of the selected standard-profile neoclassical model: its calculated self-current is about half the required self-current. It is not a proof that no tokamak can ever operate steadily.

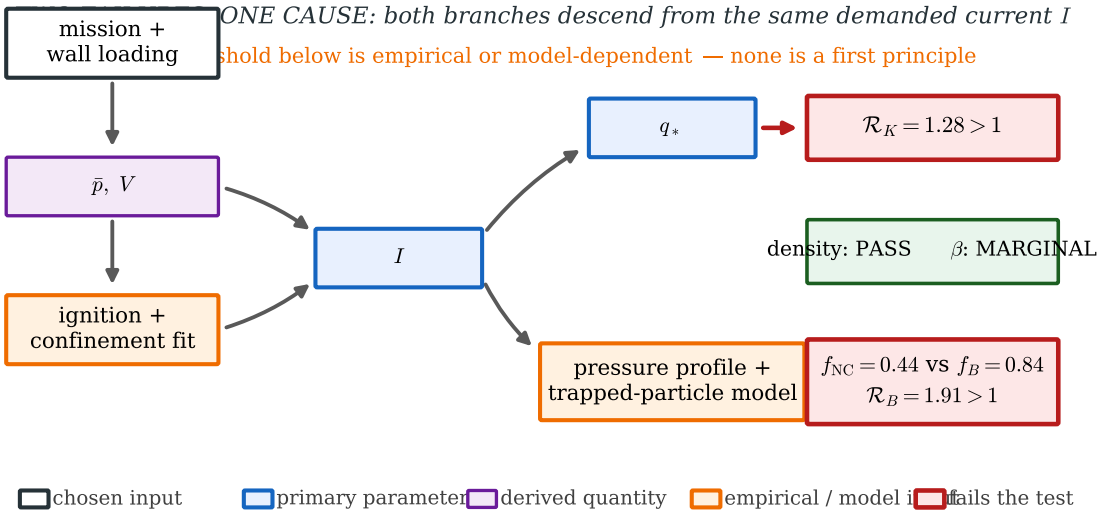
71. What the four tests have, and have not, established

The baseline can now be read as one causal argument rather than a bag of named limits:

Test	ratio	status	meaning
Greenwald density	$\mathcal{R}_G = 0.56$	passes	density is below the adopted empirical boundary
Troyon beta	$\mathcal{R}_T = 0.94$	marginal	pressure is near the adopted MHD boundary
Kink safety	$\mathcal{R}_K = 1.28$	fails	required current gives too little q_* margin
Bootstrap closure	$\mathcal{R}_B = 1.91$	fails	calculated self-current is below required current share

The first two boundaries are empirical or model-dependent operating margins. The last two failures are coupled: fusion mission and wall loading demand p and volume; ignition and the confinement fit demand I; that current lowers q_*, while the same pressure profile can supply only a limited bootstrap share under the declared kinetic model.

PLACE IMAGE HERE — Reset sketch: the conventional baseline's two coupled closure failures. Draw one concise logic map. Mission and wall loading lead to p, V; ignition and the confinement fit lead to I. Split into two branches: $I \to q_* \to \mathcal{R}_K = 1.28 > 1$, and pressure profile plus trapped-particle kinetic model $\to f_{\rm NC} = 0.44$ compared with $f_B = 0.84$, hence $\mathcal{R}_B = 1.91 > 1$. Keep density and beta as side checks marked pass and marginal. Label every threshold or regression as model/empirical rather than first principle.



Explanatory sketch. Both ratios are properties of the selected 2015 model set at its reference point, not universal tokamak limits.

Figure 34: Reset sketch: the conventional baseline's two coupled closure failures. *Idealized schematic; not a fabrication drawing.*

Design consequence. The conventional reference point is not rejected because a named limit appeared from the left field. It fails two explicit closure tests after their physical origin, profile assumptions, and calculation have been exposed. The next question is which previously declared demands can be changed together without simply moving the failure elsewhere.

72. A reference point is not a design-space conclusion

The reference point $a = 1.34\,\mathrm{m}$ was chosen in the source by imposing $R_0/a = 4$. It is a useful check, but it cannot establish that the conventional baseline fails at every size. The wall-loading calculation fixes instead

$$R_0(a) = \frac{7.16}{a}\ \mathrm{m} \qquad (\kappa = 1.70,\ 2015\ \text{baseline inputs}), \tag{141}$$

so a is the remaining geometric trial variable.

For every trial a, the source's analytic radial-build convention supplies

$$\epsilon_B(a) = \frac{a+b}{R_0(a)}, \qquad B_0(a) = B_{\max}[1-\epsilon_B(a)], \quad b = 1.20\,\mathrm{m}, \quad B_{\max} = 13\,\mathrm{T}. \tag{142}$$

Its TF-coil stress and winding-pack model then determines $c(a)$; the plasma calculation does not get to prescribe c afterward. This is an important bookkeeping point. The physical build developed earlier explains why the magnet has finite radial thickness. Equation (142) is the particular reduced convention required to reproduce this paper's numerical scan.

The already-derived power and ignition relations become compact functions of a:

$$\bar{p}(a) = \frac{8.76}{a^{1/2}}\ \mathrm{atm}, \qquad \bar{n}(a) = \frac{1.66}{a^{1/2}} \times 10^{20}\,\mathrm{m}^{-3}, \qquad \tau_E(a) = 0.81 a^{1/2}\ \mathrm{s}. \tag{143}$$

These are not new laws of nature. They are the stated mission, profile, and wall-loading model expressed as a one-parameter family.

73. The current-drive allowance must move with the trial design

In the reference case, $I_{\mathrm{CD}} = 2.25\,\mathrm{MA}$ was obtained from a fixed recirculating-power allowance. A scan must not hold that number fixed when density, field, and size change. The power allocated to waves is

$$P_{\mathrm{CD}} = \eta_{\mathrm{RF}}\eta_{\mathrm{kly}} f_{\mathrm{RP}} P_E = (0.80)(0.50)(0.10)P_E = 40\,\mathrm{MW}. \tag{144}$$

The source's lower-hybrid model writes the local efficiency in its conventional units as

$$\eta_{\mathrm{CD}} \equiv \frac{R_0 \bar{n} I_{\mathrm{CD}}}{P_{\mathrm{CD}}} \simeq \frac{1.2}{n_{\parallel}^2}. \tag{145}$$

Here $\bar{n}$ is measured in $10^{20}\,\mathrm{m}^{-3}$, I_{CD} in MA, and P_{CD} in MW. The refractive index is calculated at the intended deposition surface, $\rho_m \simeq 0.8$, from

$$n_{\parallel} \simeq \frac{\omega_{pe}}{\Omega_e} + \left(1 + \frac{\omega_{pe}^2}{\Omega_e^2}\right)^{1/2} \left(1 - \frac{\omega_{LH}^2}{\omega^2}\right)^{1/2}, \qquad \omega \simeq 2\omega_{LH}(\rho_m). \tag{146}$$

The frequencies follow from the local density and field:

$$\omega_{pe}^2 = \frac{n e^2}{\epsilon_0 m_e}, \qquad \Omega_e = \frac{eB}{m_e}, \qquad \omega_{LH}^2 = \frac{\omega_{pi}^2}{1 + \omega_{pe}^2/\Omega_e^2}. \tag{147}$$

Thus each trial design gives $n_{\parallel}(a)$, then $\eta_{\mathrm{CD}}(a)$, then $I_{\mathrm{CD}}(a) = P_{\mathrm{CD}}\eta_{\mathrm{CD}}/[R_0(a)\bar{n}(a)]$, and finally

$$f_B(a) = 1 - \frac{I_{\mathrm{CD}}(a)}{I(a)}. \tag{148}$$

This is why the bootstrap-demand curve cannot be drawn from the single reference fraction alone.

Model and limits. Equations (145)– (147) are a specified lower-hybrid current-drive model, not a universal efficiency theorem. They are included because the scan needs its assumptions exposed as clearly as the bootstrap model did.

74. Four ratio functions, one admissibility question

For each permitted a, use the confinement regression and the relations above to obtain $I(a)$, then calculate

$$\mathcal{R}_G(a) = \frac{\bar{n}(a)}{I(a)/[\pi a^2]}, \tag{149}$$

$$\mathcal{R}_T(a) = \frac{\beta(a)}{2.8\, I(a)/[aB_0(a)]}, \tag{150}$$

$$\mathcal{R}_K(a) = \frac{2}{q_*(a)}, \tag{151}$$

$$\mathcal{R}_B(a) = \frac{f_B(a)}{f_{\mathrm{NC}}(a)}. \tag{152}$$

The denominator in the first line is in the usual Greenwald units. The 2.8 in the second line is the stated Troyon-normalized-beta model input, in percent convention. The 2 in the third line is the adopted kink-safety threshold. The final denominator is the profile-integrated neoclassical result derived on pages 66–70.

The calculation has a clean pass rule:

$$\mathcal{R}_{\max}(a) = \max\{\mathcal{R}_G, \mathcal{R}_T, \mathcal{R}_K, \mathcal{R}_B\}, \qquad \text{admissible only if} \qquad \mathcal{R}_{\max}(a) \le 1. \tag{153}$$

The scan used here is implemented in the accompanying transparent arithmetic file:

`freidberg_baseline_scan.py`

It evaluates the published analytic coil, confinement, lower-hybrid, and bootstrap closures directly; it does not fit a curve to the source's published figure.

> **Design consequence.** A claimed "operating window" is an intersection, not four separate checks. Changing a is acceptable only if it reduces the largest violation without raising another ratio above one.

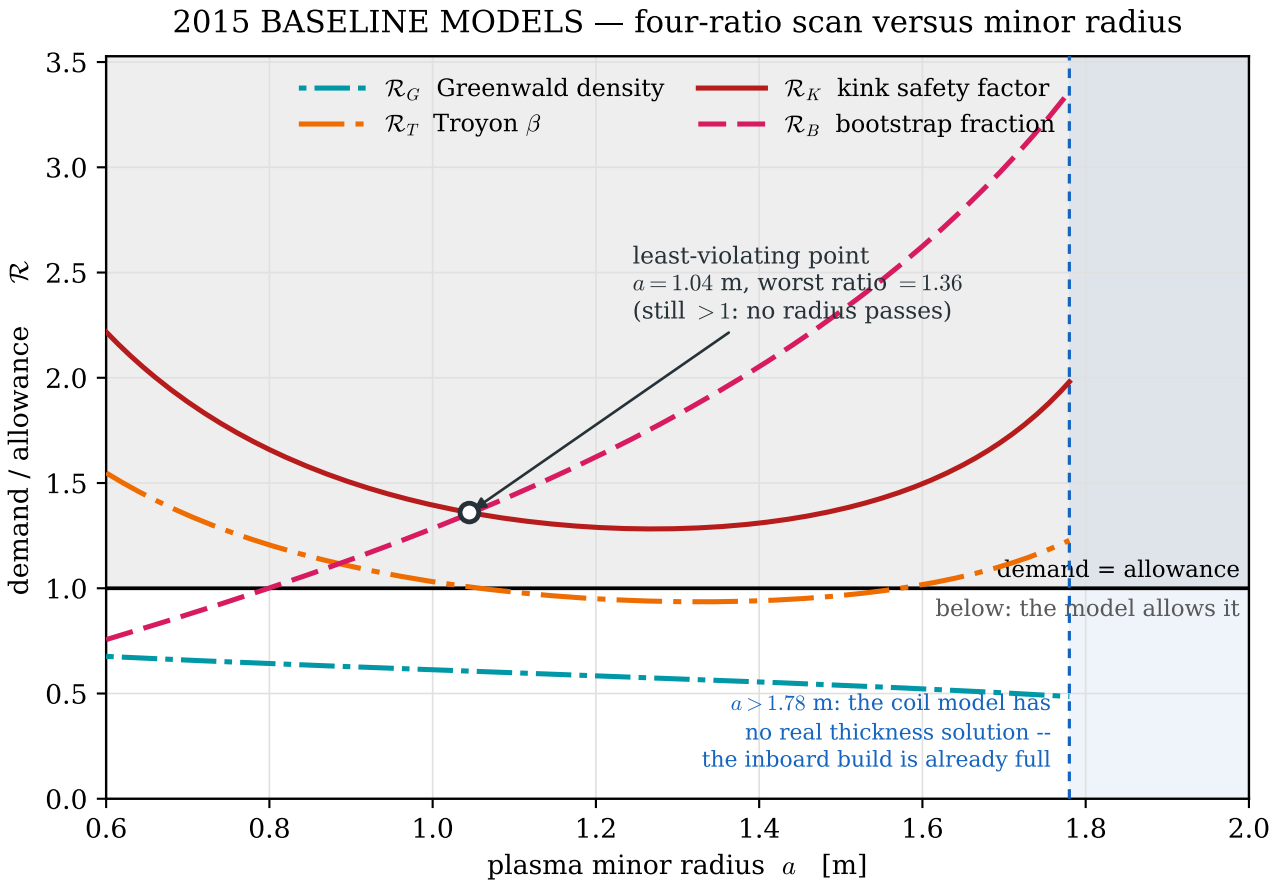
75. The conventional baseline has no simultaneous window in the scan

The table below gives representative outputs of the fully disclosed calculation. The valid high-a end is limited by the source's TF-coil stress/current-density model: with the stated inputs, its last real radial-build solution is $a \simeq 1.782\,\mathrm{m}$. Beyond that point the model has no real coil thickness, so the inboard build has failed before a plasma ratio can be evaluated. The small-a end is already excluded by the rapidly worsening kink and beta ratios.

a (m)	$\mathcal{R}_G$	$\mathcal{R}_T$	$\mathcal{R}_K$	$\mathcal{R}_B$	$\mathcal{R}_{\max}$
0.40	0.72	2.31	3.56	0.53	3.56
0.80	0.64	1.21	1.66	1.00	1.66
1.00	0.61	1.03	1.40	1.28	1.40
1.05	0.61	1.00	1.36	1.36	1.36
1.20	0.58	0.95	1.29	1.62	1.62
1.34	0.56	0.94	1.28	1.91	1.91
1.60	0.52	1.02	1.50	2.62	2.62

The least-violating sampled point is near $a = 1.05\,\mathrm{m}$, but it still has $\mathcal{R}_K \simeq 1.36$ and $\mathcal{R}_B \simeq 1.36$. No row makes all four ratios at most one. This independently reproduces the logical content of the source's Fig. 5: shrinking the machine makes kink and beta worse; enlarging it eventually makes the required bootstrap share worse. Greenwald density is not the bottleneck in this baseline family.

PLACE IMAGE HERE — Reset plot: reproducible four-ratio scan versus minor radius. Plot the four calculated ratios from `freidberg_baseline_scan.py` against a. Add a horizontal pass line at one, shade only the region above one, and mark the least-violating point near $a = 1.05\,\mathrm{m}$. Label the plot "2015 baseline models," not a universal tokamak limit. Do not add ST or HTS hardware.



Reproduced from primer/freidberg_baseline_scan.py. These are the ratios implied by ONE declared model set — an analytic coil build, one empirical confinement fit, one stability threshold, one kinetic bootstrap coefficient. This is NOT a universal tokamak limit: it is the behaviour of that model set, and each ingredient is a modelling choice that can be argued with.

Figure 35: Reset plot: reproducible four-ratio scan versus minor radius. *Idealized schematic; not a fabrication drawing.*

76. The first conventional conclusion is conditional and specific

We can now state precisely what has been demonstrated. Within the chosen mission, wall-loading, blanket, TF-coil, confinement, lower-hybrid-current-drive, standard-profile, and neoclassical-bootstrap models, changing conventional tokamak size alone does not close both the kink and steady-state current requirements.

This is a strong result, but it is not a slogan about fusion. It does *not* prove that conventional tokamaks cannot operate, that advanced scenarios cannot improve profiles, or that a modern magnet system cannot alter the trade. It identifies the two demands that any proposed remedy must address: lower the confinement-driven total current, and/or increase the allowed current and field-line-twist margin without merely transferring the burden to beta, wall loading, or recirculating power.

The next section of the 2015 paper tests remedies in a disciplined order. First it asks what improved confinement ($H > 1$) buys and what beta margin it costs. Second it raises the engineering field ceiling $B_{\max}$—the point at which a modern HTS-enabled magnet discussion becomes relevant. Only after those conventional trade-offs are visible will the thesis introduce alternative aspect-ratio geometry.

Design consequence. The conventional baseline has now earned the question "what changes the current closure?" High field and HTS enter next as a response to a calculated failure, not as an unmotivated technology preference.

77. Remedy one: what an H factor does and does not mean

The first remedy in the 2015 paper does not add a new piece of hardware. It asks whether the plasma can lose energy more slowly than the reference H-mode regression predicts. Write that possibility honestly as

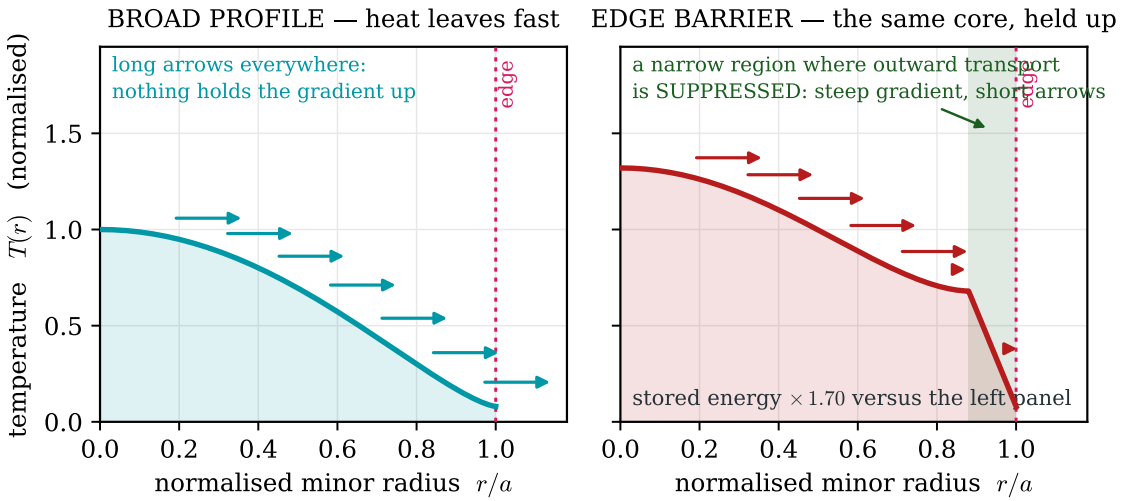
$$\tau_E = H\tau_{E,\mathrm{ref}}, \qquad H > 1. \tag{154}$$

The symbol H is a measured-performance multiplier relative to the particular regression on page 58. It is not a new conservation law, a material property, or a guarantee that a reactor can be run at the stated value.

There is nevertheless a physical picture behind trying it. Turbulent eddies carry heat and particles down their gradients. Carefully shaped pressure and current profiles can, in favorable regimes, change the magnetic shear and flow environment felt by those eddies; an internal or edge transport barrier can then reduce the effective outward transport. Maintaining such a state requires profile control and must be demonstrated in the relevant operating regime. In an ignited plasma, alpha heating also makes that control problem less independent than it is in an externally heated experiment.

PLACE IMAGE HERE — Reset physics sketch: why an H factor is not a knob on the machine. Draw a toroidal plasma cross-section only, with a broad radial temperature profile on the left and a steeper, protected-gradient region on the right. Show small outward turbulent heat arrows in the first case and shorter suppressed arrows in the second. Label the second as a transport-barrier and profile-control empirical performance change, not a new coil or reactor component. Do not add ST, HTS, divertor, or a spherical plasma.

Model and limits. Raising H means assuming a better-than-reference confinement state and then testing its consequences. It is a scenario assumption with experimental burden of proof, not an available design margin.



$H \equiv \tau_E/\tau_E^{\text{scaling}}$ is a RATIO TO A FIT, and therefore an OUTPUT. There is no dial marked H on the machine: nothing here is a new coil, a new supply or a new component. What changed is a TRANSPORT STATE, reached empirically and held only while its conditions hold — an input-power threshold, a shape, a divertor configuration — and paid for with edge-localised relaxations and impurity retention. The $\times 1.70$ above is the integral of the two curves drawn here, so even that number is a property of these two shapes, not a design input. Assuming a value of H and then quoting the performance it implies is assuming the conclusion.

Figure 36: Reset physics sketch: why an H factor is not a knob on the machine. *Idealized schematic; not a fabrication drawing.*

78. Better confinement lowers current—and moves every coupled test

Equation (109) contains H only in the denominator inside a power $1/0.93$. Holding the other inputs of that equation fixed gives

$$I_{\mathrm{req}}(H) = I_{\mathrm{req}}(1)\, H^{-1/0.93}. \tag{155}$$

Thus a larger H lowers the current that the empirical fit demands. Since $q_* \propto 1/I$ at fixed geometry,

$$q_*(H) = q_*(1)\, H^{1/0.93}. \tag{156}$$

This is the intended benefit: less current produces more field-line twist and hence more margin in the adopted kink test.

But the same reduced current tightens two allowance ratios. The Greenwald allowance is proportional to I, and the adopted Troyon pressure allowance is also proportional to I. At fixed geometry and fusion pressure,

$$\mathcal{R}_G(H) = \mathcal{R}_G(1)H^{1/0.93}, \qquad \mathcal{R}_T(H) = \mathcal{R}_T(1)H^{1/0.93}. \tag{157}$$

In words: lowering current helps kink safety but makes a fixed density and pressure look larger relative to current-based operating boundaries.

Bootstrap closure shifts in the helpful direction at fixed geometry. From Eq. (137), $f_{\mathrm{NC}} \propto p/I^2$, while $f_B = 1 - I_{\mathrm{CD}}/I$. Lower I both raises the calculated bootstrap fraction and reduces the required bootstrap share if the current-drive allowance is held fixed. That is why increasing H can remove the two original current-closure failures. It cannot be called a solution until the density and beta checks are repeated.

> **Design consequence.** Improved confinement is not a free improvement. It trades a current-and-kink problem for reduced current-based density and beta headroom. The only fair test is a simultaneous recalculation.

79. The H scan: close bootstrap first, then inspect the remaining margins

For each trial H, the 2015 procedure chooses the radius that closes the steady-state current ledger:

$$f_B\big(a, H\big) = f_{\mathrm{NC}}\big(a, H\big). \tag{158}$$

This does not make the design acceptable; it simply removes one degree of freedom before the other ratios are tested. The same transparent calculation file used for the size scan now evaluates this condition, using the changed current in the defining safety-factor relation rather than reusing a baseline q_*.

H	a (m)	$\mathcal{R}_G$	$\mathcal{R}_T$	$\mathcal{R}_K$	$\mathcal{R}_B$
1.00	0.799	0.64	1.21	1.66	1.00
1.10	0.990	0.68	1.15	1.26	1.00
1.20	1.167	0.72	1.17	1.06	1.00
1.26	1.263	0.74	1.20	1.00	1.00
1.30	1.323	0.75	1.24	0.97	1.00
1.40	1.459	0.78	1.37	0.93	1.00

The table exhibits the trade directly. Near $H = 1.26$, bootstrap and kink closure are both obtained and the Greenwald ratio remains below one, but the Troyon ratio is about 1.20: the selected 2.8% normalized-beta allowance is exceeded. Larger H only increases that beta demand in this family.

Audit note. The paper reports the same qualitative conclusion and quotes a still larger required normalized beta, 3.6% rather than 2.8%, at its threshold case. The printed analytic ledger reproduced here gives approximately 3.37% on a 10^{-3} m radius grid. Both exceed 2.8%. We retain the discrepancy rather than silently tune a coefficient to reproduce a prose number: the published paper does not supply the full numerical implementation behind its figure.

80. What the advanced-confinement remedy has established

The H exercise has sharpened the diagnosis. If an advanced transport state were reliably available, it could lower the confinement-driven current enough to close the baseline kink and bootstrap ledgers. In the declared conventional family, it does so by asking the plasma to operate above the adopted normalized-beta allowance. The missing quantity is not an unexplained new component; it is MHD pressure margin.

This is a useful result even if future experiments improve that margin. It tells us exactly what must be supplied together: reproducible confinement enhancement, profile control, and an MHD-stability basis for the higher normalized beta. Calling only the first item the solution would conceal the coupled demand.

The next remedy changes a different primitive constraint: the maximum field that the inboard TF system can bear. Higher field changes the required current, the current drive efficiency, and the MHD ratios without assuming $H > 1$. That is the logical place to introduce high-temperature superconductors: first establish what a higher $B_{\max}$ buys in this model, then ask whether an HTS magnet can provide it while surviving its structural, thermal, and manufacturing constraints.

> **Design consequence.** The advanced-confinement route is a conditional plasma-physics remedy, not a completed reactor design. The high-field route will be tested next as a distinct engineering remedy, with its cost and stress penalties kept visible.

81. Freidberg's second remedy: raise the field ceiling, not the confinement claim

The preceding remedy changed the empirical confinement multiplier H. Freidberg's second remedy holds $H = 1$ and changes an engineering input instead: the maximum magnetic field allowed at the inboard TF coil, $B_{\max}$. This is a clean counterfactual. It asks what the already-derived conventional machine would demand if its coil system could credibly tolerate a higher peak field.

At a fixed trial geometry, the immediate result is already known from Eq. (90), or from the source convention Eq. (142):

$$B_0 = B_{\max}(1 - \epsilon_B). \tag{159}$$

The central field consequently rises in direct proportion to the inboard field ceiling if the fractional radial build is held fixed. The plasma pressure required by the mission does not disappear. By the definition of beta,

$$\beta = \frac{2\mu_0 p}{B_0^2}. \tag{160}$$

Thus at fixed pressure and geometry, doubling B_0 quarters β. This is the first reason high field can relieve a pressure-stability problem: it increases the magnetic pressure available to balance the same plasma pressure.

The reference compact expression for beta had 13 T absorbed into its numerical coefficient. To vary $B_{\max}$ honestly, restore that dependence:

$$\beta(\%) = \frac{1.31}{a^{1/2}(1 - \epsilon_B)^2} \left(\frac{13\,\mathrm{T}}{B_{\max}} \right)^2. \tag{161}$$

Equation (161) is not a new plasma law; it is Eq. (160) after the 2015 power-and-size relations have been substituted. Writing the factor explicitly prevents a common audit error: one must not use a coefficient calibrated at 13 T unchanged in a high-field scan.

Supporting quantity — raised field ceiling $B_{\max}$. This is an engineering input, not a new primary Freidberg parameter. It changes the primary central field B_0 and the coil build c, then indirectly changes the current, stability, and current-drive calculations.

82. What field can help—and what it cannot help by itself

First hold geometry fixed as a thought experiment. The selected confinement regression gave the compact current requirement

$$I_{\mathrm{req}} \propto B_0^{-0.16}.$$

The exponent is weak: a field increase alone reduces the regression's current demand, but not dramatically. The more consequential changes occur after the current, safety factor, current-drive, bootstrap, and radial-build relations are recalculated together.

The safety-factor relation makes one helpful direction visible without any empirical claim:

$$q_* \propto \frac{a^2 B_0}{R_0 I}. \tag{162}$$

At unchanged geometry and current, a higher toroidal field increases field-line twist. If the confinement relation also permits a lower current, the two effects both raise q_*. Thus high field can address the same kink failure that improved confinement addressed, but it does so through a different physical lever.

There is no free magnetic pressure. The field energy density and representative mechanical stress scale are

$$u_B = p_B = \frac{B_{\mathrm{max}}^2}{2\mu_0}. \tag{163}$$

This is the same result derived on page 47, now applied to the proposed remedy. Higher field means larger Lorentz loads and, for a fixed allowable stress, more structure. The source's analytic coil model calculates this response as a larger c, rather than assuming that a better superconductor can make the support material vanish.

Design consequence. Raising B_{max} is physically promising because it lowers beta and tends to raise q_*. It is physically expensive because the very field that helps the plasma makes the coil's magnetic pressure grow as B_{max}^2. The scan must retain both statements.

PLACE IMAGE HERE — Reset physics sketch: a stronger inboard field changes two ledgers. Show a conventional inboard midplane radial cut, not a new machine. Left branch: $B_{\mathrm{max}} \uparrow \Rightarrow B_0 \uparrow \Rightarrow \beta = 2\mu_0 p/B_0^2 \downarrow$ and $q_* \propto B_0/I \uparrow$. Right branch: $B_{\mathrm{max}} \uparrow \Rightarrow p_B = B_{\mathrm{max}}^2/(2\mu_0) \uparrow \Rightarrow$ greater support demand and coil build c. Mark the right branch as a mechanical consequence, not a drawback that superconductivity erases. No ST or divertor.

This brief asks for the two-sided field diagram that Figure 29 already carries; its right branch is identical and its left differs only by two further nodes. Under the standing rule that redundancy is fixed rather than expanded, no second figure is drawn — the extra nodes belong on that figure, and this brief should be retired.

83. Why field also changes the selected current-drive calculation

The reference current ledger limits externally driven current because wave power is limited. Its lower-hybrid model does not treat current-drive efficiency as a constant:

$$\eta_{\mathrm{CD}} \simeq \frac{1.2}{n_{\parallel}^{2}}, \qquad I_{\mathrm{CD}} = \frac{P_{\mathrm{CD}}\eta_{\mathrm{CD}}}{R_0 \bar{n}}. \tag{164}$$

Here $n_{\parallel} \equiv ck_{\parallel}/\omega$ is the parallel refractive index of the launched wave. In the selected outside-launch approximation it contains the ratio

$$\frac{\omega_{pe}}{\Omega_e}, \qquad \omega_{pe}^{2} = \frac{ne^{2}}{\epsilon_0 m_e}, \qquad \Omega_e = \frac{eB}{m_e}. \tag{165}$$

The first expression measures the electron-plasma oscillation rate; the second is the electron cyclotron rate. At fixed density, increasing local B increases Ω_e and decreases ω_{pe}/Ω_e. In this particular lower-hybrid launch model, the resulting change in $n_{\parallel}$ improves the available current-drive efficiency. That improvement reduces the bootstrap share required by the ledger

$$f_B = 1 - \frac{I_{\mathrm{CD}}}{I}.$$

This is a model-dependent bridge between magnet field and steady-state current. It must not be generalized into the claim that every wave system becomes more efficient at higher field. Launch geometry, accessibility, damping, parasitic absorption, current profile, and the power plant's recirculating-power budget all require a separate calculation.

Model and limits. For Section 6.2 we keep Freidberg's 40 MW wave-power allocation, its lower-hybrid accessibility model, its profiles, and $H = 1$. We change only $B_{\max}$ and then repeat the whole coupled calculation. This isolates what the high-field proposal actually claims.

84. The high-field scan: one closure condition, then four independent tests

For every trial $B_{\max}$, retain the mission and wall-loading relations, compute $R_0(a)$, $B_0(a, B_{\max})$, the analytic TF build c, pressure, density, confinement-demanded current, lower-hybrid drive, and bootstrap fraction. As in the H scan, choose the remaining radius by the steady-state closure condition

$$f_B(a, B_{\max}) = f_{\mathrm{NC}}(a, B_{\max}). \tag{166}$$

Only after that bookkeeping condition is met do we inspect the Greenwald, Troyon, kink, and bootstrap ratios. The procedure is implemented directly in `code/calculations/freidberg_baseline_scan.py`; its radius grid is 0.001 m, so the last displayed digit should not be read as a precision claim.

$B_{\max}$ (T)	a (m)	B_0 (T)	c (m)	$\mathcal{R}_G$	$\mathcal{R}_T$	$\mathcal{R}_K$	$\mathcal{R}_B$
13.0	0.799	10.10	1.02	0.64	1.21	1.66	1.00
15.0	0.875	11.20	1.27	0.65	1.00	1.30	1.00
17.0	0.944	12.19	1.55	0.65	0.85	1.06	1.00
17.6	0.964	12.47	1.64	0.65	0.82	1.00	1.00
18.0	0.976	12.66	1.70	0.65	0.80	0.97	1.00
20.0	1.036	13.53	2.05	0.65	0.70	0.83	1.00

The table has a simple reading. Greenwald changes little along this bootstrap-matched family. Troyon becomes comfortable before the threshold. The kink ratio is the last ratio to fall to one. On this transparent grid the first all-pass point is $B_{\max} = 17.65\,\mathrm{T}$; rounding to the precision of the underlying model gives about 17.6 T, the threshold reported in the 2015 paper.

Audit note. This is a conditional result, not a prediction of a built reactor. It says that the selected 2015 confinement regression, current-drive model, bootstrap model, and four boundaries overlap when the inboard field ceiling is raised to roughly 17.6 T. Changing those models may move the number; it does not remove the need to test their overlap.

85. The price is measurable: more engineered volume and more exhaust burden

The increasing c column is already a direct mechanical cost of high field. The source also records two simple engineering figures of merit. Its engineered-volume proxy is

$$\frac{V_I}{P_E} = \frac{V_B + V_{\mathrm{TF}}}{P_E}, \tag{167}$$

where V_B is blanket-region volume and V_{TF} is TF-coil volume in its specified geometry. It is not a literal construction-price quotation. It is a transparent proxy for how much highly engineered blanket and magnet material must be provided per electric megawatt.

The source's parallel-heat-flux proxy follows the alpha power along the field:

$$Q_{\parallel} = \frac{P_\alpha B_0}{R_0}. \tag{168}$$

Its units are $\mathrm{MW\,T\,m^{-1}}$. The relation is a scaling measure, not a divertor-surface heat-flux calculation; a later chapter must introduce scrape-off layer width, magnetic expansion, radiation, and material limits before asking what a plate can survive.

For the source's compared cases, moving from its advanced-confinement case at 13 T to the high-field all-constraint case near 17.6 T increases V_I/P_E from 1.07 to $1.87\,\mathrm{m^3\,MW^{-1}}$ (about a factor 1.75) and reports $Q_{\parallel} = 657\,\mathrm{MW\,T\,m^{-1}}$, about 1.30 times its advanced-confinement comparison case. These are source-reported model outputs; we do not reverse-engineer extra precision from a plotted calculation whose full implementation was not published.

> **Design consequence.** The high-field branch closes the selected plasma ratios only by moving difficulty into magnet structure, engineered volume, and exhaust. That is not a refutation of high field. It is the reason magnet technology and heat exhaust must be developed as part of the same reactor, not treated as a footnote to plasma performance.

86. HTS is earned here: an enabler with a mechanical bill

A superconductor is useful because below its operating critical surface—set by temperature, magnetic field, current density, field angle, strain, and material state—it can carry a large current with negligible DC electrical resistance. The important engineering quantity is therefore not the slogan zero resistance. It is a measured, margin-bearing critical-current surface

$$I_c = I_c(B, T, \theta, \varepsilon, \text{conductor architecture}).$$

The conductor must remain below that surface throughout normal operation and credible transients, while its stabilizer, insulation, joints where present, cryogenic system, and protection system handle disturbances.

The 13 T reference model was deliberately a conventional-magnet assumption. The preceding scan has now made a more precise statement: *within this conventional $H = 1$ route and these four declared plasma tests*, its $B_{\max} = 13\,\text{T}$ ceiling is insufficient, whereas a ceiling near 17.6 T makes the selected constraints overlap. A credible high-field conductor is therefore necessary for this particular remedy. It is not a theorem that HTS is necessary for every fusion strategy.

Rare-earth barium copper oxide (REBCO), commonly called an HTS conductor, is a serious candidate because it can retain useful current capability in fields and temperatures where a conventional low-temperature-superconductor design is much more constrained. This is no longer purely aspirational: a representative-scale REBCO TF model coil was built and tested to 20.1 T peak field on conductor at 20 K. [3]

That last fact is the corrective to magical thinking. HTS can open a higher $B_{\max}$ window; it does not repeal Eq. (163). A fusion magnet program must still close stress and strain, quench detection and energy extraction, radiation tolerance, joint resistance where demountability is desired, manufacturing yield, cryogenic power, inspection, and replacement. The next sections remain in Freidberg's conventional design space; only after its closure problem is fully mapped will the thesis ask whether a changed aspect ratio can solve a different geometric problem.

PLACE IMAGE HERE — Cumulative high-field design state: the inboard TF system now has a technology requirement. Use cumulative state 13, preserving its conventional tokamak geometry. In the inboard TF-coil bracket *c*, distinguish winding pack, structural case, insulation/cooling, and the nuclear island *b*. Add a callout: high-field branch requires a conductor technology whose operating critical-current margin supports $B_{\max} \gtrsim 17.6\,\text{T}$. Put force arrows on the structural case and state explicitly that HTS does not remove them. Do not introduce ST geometry or a divertor.

[3] Z. S. Hartwig *et al.*, The SPARC Toroidal Field Model Coil Program, open preprint, 2023: https://arxiv.org/abs/2308.12301. The report also records 40.5 kA terminal current, large Lorentz loading, and nearly 1 GPa structural-case stress.

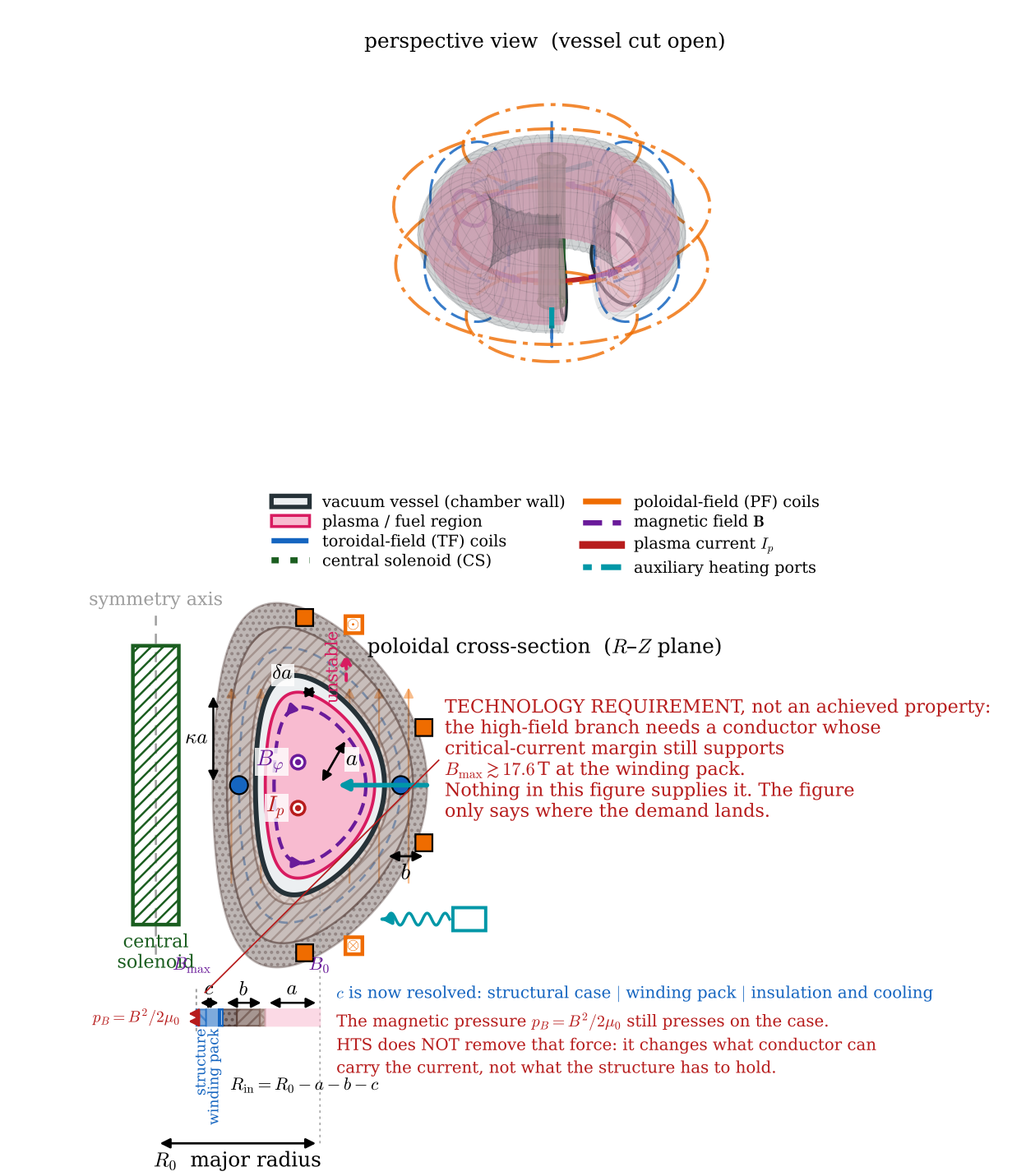


Figure 37: The same machine as the previous state; only the coil bracket c is resolved further, into structural case, winding pack and insulation/cooling. The field threshold is drawn as a REQUIREMENT on a conductor, not as a property the machine now has, and the force arrows stay: HTS changes what carries the current, not what the case must hold.

87. Freidberg's third remedy: change the plant scale

The high-field route changed a magnet limit. The next route changes a mission input: the requested electric output P_E. The reference case chose $P_E = 1000\,\mathrm{MW_e}$. Raising that number may sound like making the problem harder, because a larger plant must make more fusion power. At fixed allowed neutron wall loading, however, it first forces a larger wall area:

$$R_0 a = 7.16 \left(\frac{P_E}{1000\,\mathrm{MW}} \right) \mathrm{m}^2, \tag{169}$$

where the coefficient retains the 2015 blanket, shape, wall-loading, and thermal conversion assumptions.

The equation has an immediate geometrical meaning. For a chosen a, raising output raises R_0 in proportion. The machine gains toroidal length and plasma volume rather than being asked to generate arbitrarily more power per square metre of first wall. This is the physical content behind the familiar phrase economy of scale. It is not an accounting trick and it is not free: the absolute amount of plant, inventory, construction, and grid-connected power also grows.

For this test Freidberg holds the original plasma and magnet assumptions:

$$H = 1, \qquad B_{\max} = 13\,\mathrm{T}, \qquad W_n = 4\,\mathrm{MW\,m^{-2}},$$

and again uses $f_B = f_{\mathrm{NC}}$ to select the compatible a for each trial output. The purpose is narrow: discover whether scale alone can make the four chosen plasma-model constraints overlap without asserting better confinement or a higher-field magnet.

88. More output at fixed wall loading means more volume, not higher pressure

At the fixed temperature and profiles of the reference model, fusion power has the form

$$P_{\text{fus}} \propto p^2 V. \tag{170}$$

For a torus of fixed elongation, $V \propto R_0 a^2$. Insert Eq. (169):

$$V \propto (R_0 a) a \propto P_E a.$$

The thermal fusion power is proportional to P_E, so

$$P_E \propto p^2 (P_E a) \quad \Longrightarrow \quad p \propto a^{-1/2}.$$

The factor P_E cancels. At fixed a, the mission asks for a proportionally larger volume at proportionally larger power, leaving the required average pressure unchanged. With the fixed $T \simeq 14\,\text{keV}$ model and $p = 2nT$, the same is true of the average density:

$$n \propto a^{-1/2}.$$

The confinement time makes the cancellation visible a second way. The stored thermal energy scales as $W \sim pV \propto P_E a^{1/2}$, while alpha heating scales as $P_\alpha \propto P_E$. Thus

$$\tau_{E,\text{req}} = \frac{W}{P_\alpha} \propto a^{1/2}, \tag{171}$$

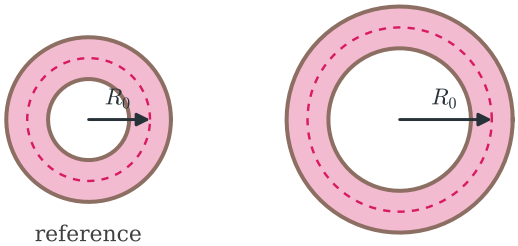
again with no direct P_E dependence at fixed a.

Design consequence. Higher output at fixed wall loading does not buy lower pressure by itself. It buys a larger volume in which to produce the larger total power. The benefit, if any, must enter through the way that larger R_0 changes the confinement-driven current and the current ledger.

PLACE IMAGE HERE — Reset physics sketch: scale at fixed neutron wall loading. Draw two conventional torus silhouettes at the same labelled wall loading W_n. The right machine has larger R_0, larger wall area, and proportionally larger P_E; keep the plasma cross-section and pressure shading comparable. Beside it show the short chain $P_E \uparrow \Rightarrow R_0 a \uparrow \Rightarrow V \uparrow$ and $P_E \propto p^2 V \Rightarrow p$ unchanged at fixed a. This is a reset scaling sketch, not a claim that large plants are automatically economical. No ST, HTS, or divertor.

TWO WAYS TO GROW THE SAME MACHINE — and what each one exports

A. fixed wall loading W_n — the plant gets bigger because P_E did



reference

$R_0 \times 1.50$, wall area $\times 1.50$, $P_E \times 1.50$

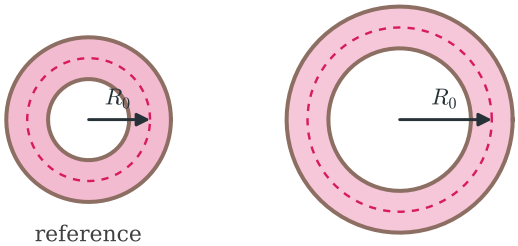
$P_E \uparrow \Rightarrow R_0 a \uparrow \Rightarrow V \uparrow$

$P_E \propto p^2 V$, and V grew by the same factor, so at fixed a:

$p \times 1.00$ — UNCHANGED

Exported: capital scale — nothing about the plasma itself got any easier.

B. fixed P_E, lower W_n — the plant gets bigger to spread the SAME neutron power



reference

$R_0 \times 1.50$, $W_n \times 0.67$, $p, n \times 0.82$

$W_n \downarrow \Rightarrow V \uparrow \Rightarrow p, n \downarrow$

same P_E over $\times 1.50$ the volume:

$p \times 0.82$, $W_n \times 0.67$

Exported, and not small: MORE blanket, MORE shield, MORE TF-coil volume — all competing for the inboard radius §46 already showed is full.

Both rows are the SAME drawing with a different quantity held fixed — which is why they are one figure, not two. Every ratio is computed from $V = 2\pi^2 R_0 a^2$ and $A_w = 4\pi^2 R_0 a$ at fixed a; none is estimated. Row A's "p unchanged" is an identity, not an approximation. A scaling sketch only: it says nothing about economics.

Explanatory sketch; schematic, not to scale. Plan view: the annulus is the first wall, its shading the pressure. Not a cumulative machine state.

Figure 38: Two ways to grow the same machine. Row A is this brief (fixed W_n); row B is the brief on lower wall loading. They are one figure because they are the same drawing with a different quantity held fixed — and only side by side can the two chains be compared.

89. The enlarged machine changes the current requirement

The selected confinement relation, solved for current on page 59, has the dependence

$$I^{0.93} \propto \frac{\tau_{E,\mathrm{req}} P_\alpha^{0.69}}{R_0^{1.39} a^{0.58} n^{0.41} B_0^{0.15}}.$$

At fixed a, the preceding page showed that $\tau_{E,\mathrm{req}}$ and n do not carry a direct power-output factor. In contrast,

$$P_\alpha^{0.69} \propto P_E^{0.69}, \qquad R_0^{1.39} \propto P_E^{1.39}.$$

Ignoring for one line the smaller change in B_0 caused by the changed radial geometry, their ratio is $P_E^{-0.70}$. Raising both sides to $1/0.93$ gives

$$I_{\mathrm{req}} \propto P_E^{-0.70/0.93} \simeq P_E^{-0.75}. \tag{172}$$

This is the central mechanism of the power-output remedy. The larger major radius helps the empirical confinement scaling more strongly than the increased alpha power hurts it, so the regression asks for less current at a fixed cross-section.

The current-drive budget must be updated at the same time. If the selected recirculating-power fraction remains fixed, $P_{\mathrm{CD}} \propto P_E$. But

$$I_{\mathrm{CD}} = \frac{P_{\mathrm{CD}} \eta_{\mathrm{CD}}}{R_0 n}$$

contains the same factor $R_0 \propto P_E$ in its denominator. At fixed a its direct power dependence approximately cancels, leaving field and wave-accessibility changes to determine the remainder. The required total current, however, falls according to Eq. (172). Therefore the bootstrap fraction required by the ledger falls:

$$f_B = 1 - \frac{I_{\mathrm{CD}}}{I}.$$

This is why a larger plant can ease both the kink test and steady-state closure in the stated model. It is not because a large reactor creates bootstrap current from nothing.

Model and limits. The exponent -0.75 is a consequence of the selected empirical confinement fit, fixed wall loading, fixed temperature/profiles, and a fixed recirculating-power fraction. It is a transparent model implication, not a size-scaling theorem for all tokamaks.

90. The power-output scan: the model closes near $1.55\,\mathrm{GW_e}$

For each P_E, the calculation holds $H = 1$ and $B_{\max} = 13\,\mathrm{T}$, imposes $f_B = f_{\mathrm{NC}}$, and then tests all four ratios. The output and radius grids are 1 MW and 0.001 m respectively.

P_E (MW)	a (m)	R_0 (m)	B_0 (T)	$\mathcal{R}_G$	$\mathcal{R}_T$	$\mathcal{R}_K$	$\mathcal{R}_B$
1000	0.799	8.96	10.10	0.64	1.21	1.66	1.00
1200	1.030	8.34	9.52	0.71	1.09	1.31	1.00
1400	1.263	7.94	8.97	0.77	1.02	1.10	1.00
1550	1.436	7.73	8.57	0.81	0.99	1.00	1.00
1600	1.493	7.67	8.44	0.82	0.99	0.97	1.00
1800	1.717	7.51	7.95	0.87	0.97	0.89	1.00

The kink and Troyon ratios are the last two to close. The script's first all-pass point is $P_E = 1556\,\mathrm{MW}$ on its displayed grids; Freidberg reports 1550 MW, an agreement appropriate to a simplified model and discrete scan. Greenwald remains below one with appreciable margin.

Audit note. This result does not say that a $1.55\,\mathrm{GW_e}$ reactor is preferred, financeable, grid-compatible, licensable, or buildable. It says only that raising the output is a third way to make the stated four plasma-model inequalities overlap while retaining $H = 1$ and the paper's 13 T magnet assumption.

91. A larger plant trades plasma closure for absolute scale and heat exhaust

The source's engineered-volume proxy at the $1550\,\mathrm{MW_e}$ case is

$$\frac{V_I}{P_E} = 1.17\,\mathrm{m^3\,MW^{-1}},$$

close to its $1.07\,\mathrm{m^3\,MW^{-1}}$ advanced-confinement comparison. That is the limited sense in which the source identifies an economy of scale: the engineered volume *per megawatt* need not rise sharply. It does not mean the total engineering volume or capital required for a $1550\,\mathrm{MW_e}$ plant is small; both are necessarily larger in absolute terms.

The same source reports its parallel-heat-flux proxy rising to

$$Q_{\parallel} = 673\,\mathrm{MW\,T\,m^{-1}}$$

at the high-output point, compared with $507\,\mathrm{MW\,T\,m^{-1}}$ in its advanced-confinement comparison case. The proxy is a warning that core plasma-constraint closure does not settle power exhaust. It is not a divertor design calculation, and it must not be translated directly into a material surface limit.

Design consequence. The third remedy establishes a genuine alternative within the 2015 model: conventional field and ordinary confinement can meet the four selected plasma tests at a sufficiently large output. It also leaves us with a larger, more-concentrated generating station and a more demanding exhaust proxy. A reactor is not complete merely because one family of plasma ratios reaches one.

92. Freidberg's fourth remedy: lower the permitted wall loading

The last single-knob remedy changes the allowed average neutron wall loading W_n. The reference value was $4\,\mathrm{MW\,m^{-2}}$. Lowering it means demanding a larger first-wall area for the same neutron power:

$$R_0 a = 7.16 \left(\frac{4\,\mathrm{MW\,m^{-2}}}{W_n} \right) \mathrm{m}^2 \qquad (P_E = 1000\,\mathrm{MW_e}). \tag{173}$$

This is not an arbitrary way to make the plasma easier. It represents a conservative nuclear-material and first-wall design choice: less neutron power is assigned to each square metre, so more square metres must be built.

The immediate trade is visible before any plasma fit is used. At fixed a, $R_0 \propto W_n^{-1}$ and $V \propto R_0 a^2 \propto W_n^{-1}$. The same fusion output is produced in a larger volume. That lowers the required power density, pressure, and density, but it makes the reactor physically larger. Freidberg holds

$$P_E = 1000\,\mathrm{MW_e}, \qquad H = 1, \qquad B_{\max} = 13\,\mathrm{T},$$

and again imposes bootstrap closure to select $a(W_n)$ before testing the four ratios.

Design consequence. A lower wall-loading limit is not a plasma improvement supplied for free. It is a decision to purchase lower power density with more wall, blanket, shielding, magnets, and building-scale volume. The next pages quantify how that engineering choice reaches the current problem.

93. Lower wall loading reduces the required pressure

At fixed output and temperature, use again

$$P_{\text{fus}} \propto p^2 V.$$

From Eq. (173), $V \propto W_n^{-1} a$ at fixed elongation. Holding total fusion power fixed therefore gives

$$p \propto \left(\frac{W_n}{a}\right)^{1/2}, \qquad n \propto \left(\frac{W_n}{a}\right)^{1/2}. \tag{174}$$

Unlike the higher-output case, there is no cancellation: reducing W_n genuinely reduces the average pressure and density required to make the same total power.

The stored energy scales as

$$W \sim pV \propto \left(\frac{W_n}{a}\right)^{1/2} \left(\frac{a}{W_n}\right) = \left(\frac{a}{W_n}\right)^{1/2}.$$

The alpha power is unchanged because the total fusion output is unchanged. Hence

$$\tau_{E,\text{req}} \propto \left(\frac{a}{W_n}\right)^{1/2}. \tag{175}$$

This longer required confinement time may look like a penalty, but it must be placed beside the larger R_0, lower density, and changed field geometry in the empirical confinement law before judging the net current demand.

PLACE IMAGE HERE — Reset physics sketch: lower wall loading trades density for machine scale. Draw two conventional torus silhouettes at the same P_E and temperature. The right machine has a larger R_0, more first-wall area, lower labelled W_n, and lighter pressure/density shading. Show $W_n \downarrow \Rightarrow V \uparrow \Rightarrow p, n \downarrow$, then the explicit warning: more blanket, shield, and TF-coil volume. This is a reset scaling sketch, not a cumulative machine state. No ST, HTS, or divertor.

This brief and the previous one ask for the same drawing with a different quantity held fixed, so they share one figure: see row B of Figure 38.

94. The net current direction follows from all terms together

Use the same selected confinement dependence:

$$I^{0.93} \propto \frac{\tau_{E,\mathrm{req}} P_\alpha^{0.69}}{R_0^{1.39} a^{0.58} n^{0.41} B_0^{0.15}}.$$

At fixed a, Eqs. (174) and (175) give

$$\tau_{E,\mathrm{req}} \propto W_n^{-1/2}, \qquad R_0^{-1.39} \propto W_n^{1.39}, \qquad n^{-0.41} \propto W_n^{-0.205}.$$

Multiplying the three explicit wall-loading powers gives

$$W_n^{-0.500+1.390-0.205} = W_n^{0.685}.$$

Consequently,

$$I_{\mathrm{req}} \propto W_n^{0.685/0.93} \simeq W_n^{0.74}, \tag{176}$$

apart from the smaller self-consistent change in B_0. Lowering wall loading lowers the current demanded by the selected regression. It also lowers beta directly through Eq. (174), while the larger R_0 and lower current improve the kink ratio in the full calculation.

External current drive deserves the same caution as before. With fixed output, P_{CD} is fixed. As W_n falls, $R_0 n \propto W_n^{-1/2}$ grows, which tends to reduce the externally driven current in the selected efficiency model. Yet I_{req} falls more rapidly in this family; the required bootstrap share still decreases after all quantities are recalculated. The conclusion is a coupled result, not a consequence of holding I_{CD} fixed by hand.

95. The wall-loading scan closes near $2.1\,\mathrm{MW\,m^{-2}}$

The calculation now scans W_n while retaining $P_E = 1000\,\mathrm{MW_e}$, $H = 1$, and $B_{\max} = 13\,\mathrm{T}$. At each value it first sets $f_B = f_{\mathrm{NC}}$ and then tests the same four ratios.

W_n (MW m^{-2})	a (m)	R_0 (m)	B_0 (T)	$\mathcal{R}_G$	$\mathcal{R}_T$	$\mathcal{R}_K$	$\mathcal{R}_B$
4.00	0.799	8.96	10.10	0.64	1.21	1.66	1.00
3.50	0.890	9.19	10.04	0.65	1.11	1.49	1.00
3.00	1.009	9.46	9.96	0.67	1.00	1.32	1.00
2.50	1.171	9.78	9.85	0.68	0.89	1.14	1.00
2.10	1.349	10.11	9.72	0.69	0.80	1.00	1.00
2.00	1.403	10.21	9.68	0.70	0.78	0.97	1.00

The kink criterion is again the final one to close. The transparent script finds the largest all-pass wall loading as $2.09\,\mathrm{MW\,m^{-2}}$ on its displayed grids; the source reports $W_n \lesssim 2.1\,\mathrm{MW\,m^{-2}}$. This agreement is a useful check on the algebra, not validation of every physical assumption beneath the 2015 model.

Audit note. Lower wall loading can close the selected plasma constraints without advanced confinement or a higher-field magnet. It does so by selecting a larger, lower-power-density machine. Any claim that this is the best reactor strategy must also compare its blanket, shield, coil, construction, maintenance, and financing burden.

96. The reward is lower exhaust proxy; the price is a much larger plant

The source reports the low-wall-loading case's engineered-volume proxy as

$$\frac{V_I}{P_E} = 2.59\,\mathrm{m^3\,MW^{-1}}$$

near $W_n = 2.1\,\mathrm{MW\,m^{-2}}$, compared with $1.07\,\mathrm{m^3\,MW^{-1}}$ in its advanced-confinement comparison: about a factor 2.42 increase. This is qualitatively expected from the enlarged radial and toroidal dimensions, though the exact number belongs to its specified blanket and TF-coil geometry.

The reported parallel-heat-flux proxy moves in the favorable direction, from 507 to $374\,\mathrm{MW\,T\,m^{-1}}$ in those compared cases. Lower average power density makes an exhaust problem less severe, but it does not solve it. Divertor and first-wall design still require a separate calculation of power sharing, scrape-off-layer transport, radiation, magnetic expansion, detachment, and material response.

Design consequence. The four single-variable remedies have now revealed four distinct currencies: empirical confinement margin, magnet field and structure, plant output, and neutron-wall area. Each can close the selected plasma ledger under stated assumptions; each exports difficulty into another part of the reactor. The next task is to compare the remedies on one honest ledger before moving beyond Freidberg's conventional baseline.

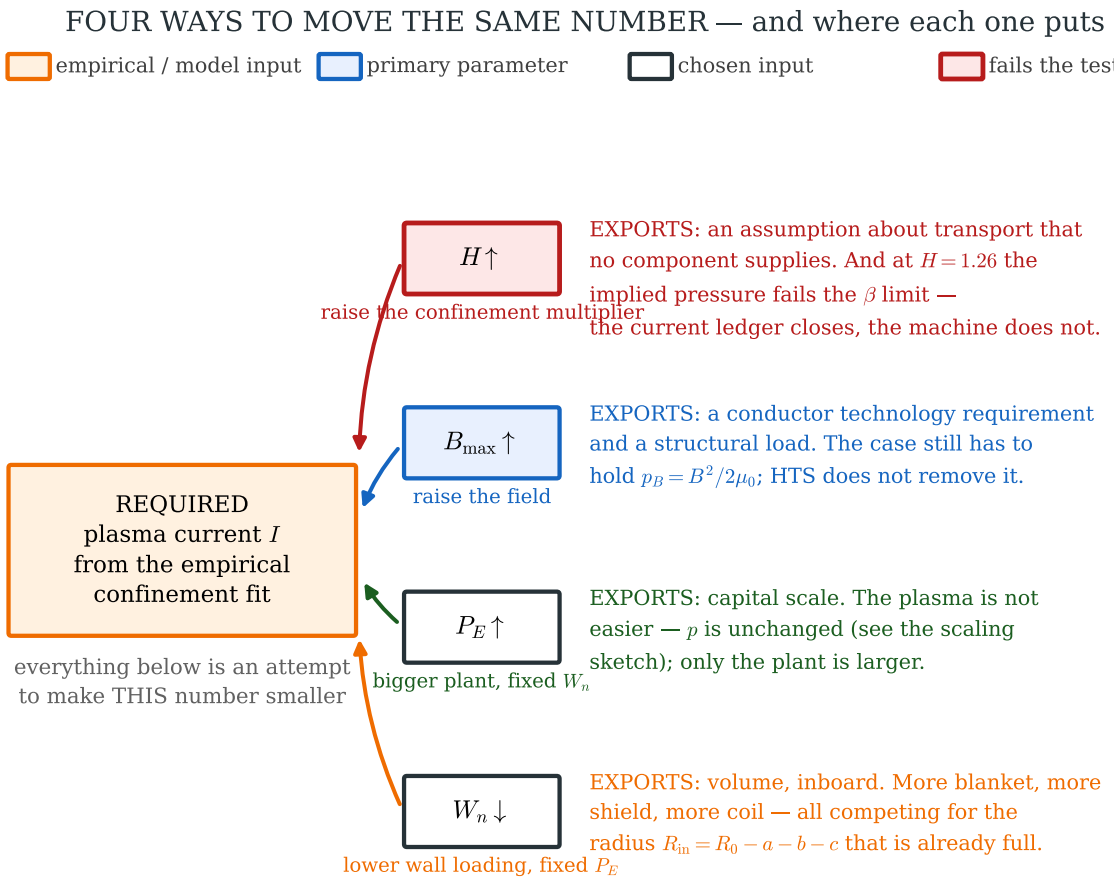
97. One comparison ledger: which changes actually close the four tests?

The four options are not four versions of the same physical claim. Each changes one declared input while retaining the others as closely as the coupled calculation permits:

Option	Changed demand	Result in the stated model
Improved confinement	$H : 1 \to 1.26$	Kink and bootstrap close, but the Troyon ratio is about $1.20 > 1$; not a full closure.
High field	$B_{\max} : 13 \to 17.6\,\mathrm{T}$	All four ratios close; magnet stress, coil build, and engineered-volume proxy rise.
Higher output	$P_E : 1000 \to 1550\,\mathrm{MW_e}$	All four ratios close; absolute plant size and heat-exhaust proxy rise.
Lower wall loading	$W_n : 4 \to 2.1\,\mathrm{MW\,m^{-2}}$	All four ratios close; engineered volume per megawatt rises strongly.

The first row is deliberately retained. It teaches the method: finding one simultaneous equality, such as $f_B = f_{\mathrm{NC}}$, is not enough. Every inequality must be retested at the same state. The remaining three routes pass this particular four-ratio screen only because the changed input alters the confinement-driven current, the current ledger, and the pressure/stability ledger together.

PLACE IMAGE HERE — Reset comparison sketch: four ways to move the same current closure. Draw one compact causal map with a central block labelled required plasma current from the empirical confinement fit. Four incoming branches: $H \uparrow$ (empirical transport assumption); $B_{\max} \uparrow$ (field and structural demand); $P_E \uparrow$ at fixed W_n (larger plant); and $W_n \downarrow$ at fixed P_E (larger lower-power-density plant). Each branch must show a distinct exported burden. Mark $H = 1.26$ as beta failure, and the other three as four-ratio closure only within the stated 2015 models. This is a reset causal graphic, not a new machine drawing.



Only the top branch is marked as a FAILURE. $H = 1.26$ closes the current ledger and then violates the β limit — it is not a fourth option, it is the same difficulty moved where the ledger cannot see it. The other three DO close the four ratios, but only inside the stated 2015 model set, whose confinement fit, stability threshold, coil build and bootstrap coefficient are each a modelling choice. None of the four is a result about tokamaks — all four are results about that model set, and each hands its cost to a different engineer.

Explanatory sketch; a causal map, not a machine drawing. Branch positions carry no ordering.

Figure 39: Reset comparison sketch: four ways to move the same current closure. *Idealized schematic; not a fabrication drawing.*

98. The source's final cases, stated without pretending to measure reality

For orientation, the 2015 paper reports the following rounded states. The values are not independent targets; they are outputs of its coupled equations after the listed input change. The reference row is shown because it makes the original mismatch visible.

	reference	$H = 1.26$	high field	high output	low wall load
$B_{\max}$ (T)	13	13	17.6	13	13
P_E (MW)	1000	1000	1000	1554	1000
W_n (MW m^{-2})	4.0	4.0	4.0	4.0	2.1
a (m)	1.34	1.26	0.97	1.44	1.35
R_0 (m)	5.34	5.57	7.39	7.70	10.2
I (MA)	14.3	10.0	7.64	11.2	8.52
q_K/q_*	1.28	1.00	1.00	1.00	1.00
$f_B/f_{\rm NC}$	1.90	1.00	1.00	1.00	1.00
β/β_T	0.94	1.21	0.82	1.00	0.80

The table confirms three conceptual lessons. First, the reference failure is current-centered: its q_K/q_* and bootstrap ratio exceed one. Second, reducing current is not sufficient if the way it is reduced also tightens the beta allowance, as the $H = 1.26$ row shows. Third, high field, plant scale, and low wall loading are not interchangeable technologies; they reach current closure through different geometric and engineering commitments.

Model and limits. This table reproduces rounded values reported in the freely available 2015 source. Our companion script independently reproduces the three closure thresholds to the resolution stated in the relevant scans. Neither result turns the source's profile choices, empirical boundaries, or coil model into universally valid reactor predictions.

99. What Freidberg's calculation has rigorously established

The word rigorously must be used carefully. The following implication has been established by transparent substitution and numerical quadrature:

$$\begin{array}{c}\text{2015 mission, geometry, coil, profile, confinement, current-drive,}\\ \text{bootstrap, and four-limit assumptions}\end{array} \quad \Longrightarrow \quad \begin{array}{l}\text{the reference conventional state fails ki}\\ \text{steady-state bootstrap closure simultan}\end{array}$$

The size scan shows that merely choosing another minor radius within that baseline does not make all four tests pass. The remedy scans then identify which changed assumptions can restore overlap, and what new burden each imports.

What has *not* been established is equally important. The calculation is not a proof that every conventional tokamak fails, that any one high-field plant can be built, or that a ratio below one makes a reactor safe. The Greenwald relation and the chosen normalized-beta allowance are operational boundaries, not theorems. The confinement and current-drive laws are empirical/model closures. The bootstrap result depends on the chosen equilibrium and profiles. The analytic coil model is not a detailed magnet design.

Design consequence. The real achievement of the reconstruction is not a slogan about tokamaks. It is a map of conditional causal failures: which physics assumption or engineering choice causes each failure, what intervention can remove it in the model, and what new design obligation is thereby created.

100. Four ratios are not a complete reactor acceptance test

The 2015 paper intentionally keeps several major reactor questions outside its four-ratio screen. The most immediate is power exhaust. $Q_{\parallel}$ is only a parallel-transport proxy; no scrape-off-layer width, divertor geometry, impurity radiation, detachment window, transient load, or material lifetime has yet been calculated. A passing value of q_K/q_* is likewise not a complete disruption-risk assessment. Vertical stability, control authority, error fields, energetic particles, resistive walls, and nonlinear evolution are not exhausted by one edge safety factor.

On the engineering side, the radial-build calculation has identified why blanket, shield, winding pack, and structure consume space, but it has not produced a maintainable nuclear island. Neutron damage, tritium inventory, remote handling, cryoplant electricity, joint reliability where relevant, fatigue, and component replacement remain design problems. On the plant side, net electric output and recirculating power must eventually be joined to availability, construction schedule, licensing, fuel cycle, and grid integration.

None of these omissions invalidates the four-ratio argument. They prevent us from misrepresenting it as a completed reactor design. The correct next move is to retain the conventional baseline as a reference and ask whether a different aspect-ratio geometry can improve the current-and-field trade *without* hiding the radial build and central-stack obligations it creates.

101. The conventional baseline is now complete enough to earn the next question

We began without a tokamak and let the conventional machine reveal itself: a hot charged fusion plasma required magnetic confinement; the escape direction required a torus and TF coils; toroidal drifts required current, poloidal field, and shaping; steady state required a current ledger; fusion neutrons required the nuclear island; and the field required a finite inboard magnet build. The primary parameter set

$$a,\ R_0,\ \kappa,\ b,\ c,\ T,\ n,\ p,\ \tau_E,\ B_0,\ \beta,\ I,\ q_*,\ f_B$$

has acquired physical meaning before it was used as engineering input.

Freidberg's reference calculation then exposed the conventional current closure: under its declared assumptions, an ordinary reference machine needs too much current for both its adopted kink margin and its available steady-state bootstrap share. The high-field result gives HTS its correct causal position: it is a technology that can open a needed field window in one closure route, while intensifying structural and protection demands.

Only now is it legitimate to consider low-aspect-ratio, spherical-tokamak geometry. It will not be introduced as a compact machine-shaped answer. It must first be asked what aspect ratio changes in the same equations, which conventional failures it can relieve, and why its reduced central space creates the central-stack problem. The next chapter begins that comparison from the already-earned radial-build ledger.

102. A spherical tokamak changes one geometric ratio, not the laws of plasma

Define the plasma aspect ratio by

$$A \equiv \frac{R_0}{a}. \tag{177}$$

A conventional tokamak has A of order 3–4 or larger. A low-aspect-ratio tokamak, commonly called a spherical tokamak (ST), has A of order 1–2. The word spherical describes the compact toroidal shape; the plasma remains a torus, not a filled sphere. It still needs TF field, poloidal field, equilibrium control, heating, exhaust, a vacuum boundary, and a nuclear island.

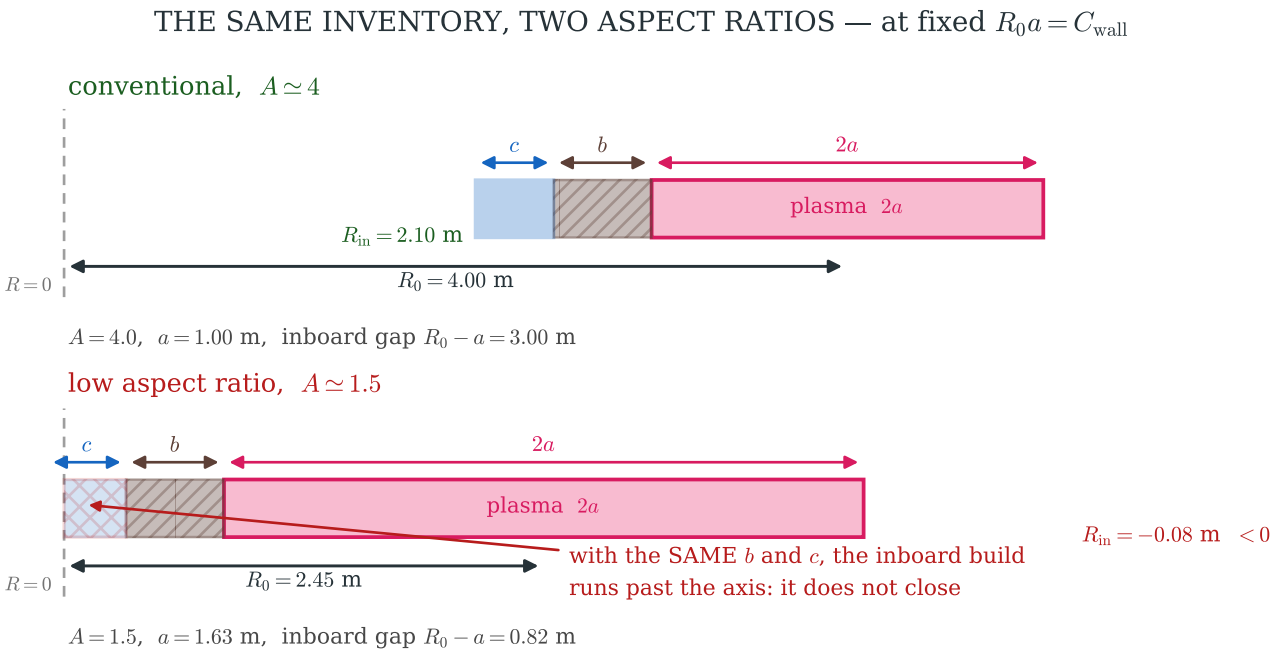
At a fixed fusion mission and wall-loading product $R_0 a = C_{\mathrm{wall}}$, the two geometric variables can instead be written in terms of A:

$$R_0 = (AC_{\mathrm{wall}})^{1/2}, \qquad a = \left(\frac{C_{\mathrm{wall}}}{A}\right)^{1/2}. \tag{178}$$

Thus lowering A makes the cross-section larger while bringing the plasma centre closer to the symmetry axis. It is not simply a uniformly shrunken conventional machine.

Design consequence. Aspect ratio is a causal geometric variable. Lowering it can alter field-line twist, trapped-particle physics, volume, and radial space at once. We must derive each direction separately before calling low aspect ratio an advantage.

PLACE IMAGE HERE — Reset geometry sketch: aspect ratio changes the torus, not the physics inventory. Show two midplane cuts with the same labelled $R_0 a = C_{\mathrm{wall}}$: a conventional $A \simeq 4$ plasma and a low-$A \simeq 1.5$ plasma. Mark R_0, a, and the symmetry axis. Retain only the already-earned plasma, nuclear-island bracket, and TF-system location as simplified outlines. Make the low-A inboard gap visibly smaller. Do not label either design as automatically superior and do not yet add an HTS brand or an ST-specific divertor.



Both rows hold $R_0 a = 4.00\,\mathrm{m}^2$, so the first-wall area is the same. $b = 0.50$ m and $c = 0.40$ m are fixed IN METRES: they are set by neutron attenuation and by conductor and structure, not by how round the plasma is — scaling them with a would assume the answer. Everything above is then arithmetic: $a = \sqrt{C_{\mathrm{wall}}/A}$, $R_0 = Aa$.

NEITHER design is superior here, and the figure does not say one is. It says something narrower: low A cannot be adopted while holding b and c fixed — the inboard allocations have to be renegotiated. That is exactly what the following sections do, and why they quote their own shielding allocation and centre-column envelope instead of inheriting the conventional ones. The physics inventory has not changed; the room it fits into has.

Explanatory sketch; midplane cuts at a common radial scale. No HTS brand and no ST divertor: neither is derived yet.

Figure 40: Reset geometry sketch: aspect ratio changes the torus, not the physics inventory. *Idealized schematic; not a fabrication drawing.*

103. Low aspect ratio strengthens toroidal field variation and trapping

The toroidal-field relation remains

$$B_\varphi(R)R = B_0R_0.$$

At the inboard and outboard plasma edges, $R = R_0 - a$ and $R = R_0 + a$. Hence

$$\frac{B_{\text{out}}}{B_{\text{in}}} = \frac{R_0 - a}{R_0 + a} = \frac{A - 1}{A + 1}. \tag{179}$$

When A falls, this ratio falls: the inboard field becomes much stronger than the outboard field. The mirror argument from page 36 then says that a larger fraction of particles can be trapped. In the simple low-collisionality estimate used earlier,

$$f_t \simeq \left(1 - \frac{B_{\text{min}}}{B_{\text{max}}}\right)^{1/2} \simeq \left(\frac{2}{A + 1}\right)^{1/2}. \tag{180}$$

For orientation, this expression gives $f_t \simeq 0.63$ at $A = 4$ and $f_t \simeq 0.89$ at $A = 1.5$.

More trapped particles make a larger bootstrap response physically plausible because bootstrap current begins with trapped-particle/passing-particle collisional asymmetry. But Eq. (180) is not a bootstrap fraction. The large-aspect-ratio local closure used on pages 67–70 is no longer controlled at $A \sim 1$–2. A finite-aspect-ratio kinetic calculation must replace it before claiming a numerical f_{NC} for an ST.

Model and limits. The direction of the trapping effect follows from the toroidal-field geometry and magnetic-moment argument. The resulting bootstrap fraction, profile, and collisionality dependence are finite-aspect-ratio neoclassical physics, not a number that can be extrapolated from Freidberg's large-aspect-ratio coefficient.

104. Aspect ratio also changes the kink–beta coupling

The shaped edge safety-factor estimate was

$$q_* = \frac{2\pi a^2 B_0}{\mu_0 R_0 I}\left(\frac{1+\kappa^2}{2}\right) = \frac{2\pi}{\mu_0}\left(\frac{1+\kappa^2}{2}\right)\frac{aB_0}{AI}. \tag{181}$$

Normalized beta contains the same dimensional group aB_0/I:

$$\beta_N \equiv \beta(\%)\frac{aB_0}{I_{\mathrm{MA}}}. \tag{182}$$

Here $I_{\mathrm{MA}} = I/(10^6\,\mathrm{A})$ and $\beta(\%) = 100\beta$. Substitute

$$\frac{aB_0}{I} = \frac{\beta_N}{10^8\beta}$$

into Eq. (181) to obtain

$$q_* = \frac{0.05}{A}\left(\frac{1+\kappa^2}{2}\right)\frac{\beta_N}{\beta}. \tag{183}$$

The numerical 0.05 comes only from $\mu_0 = 4\pi \times 10^{-7}$ and the conventional percent–MA units in β_N.

Equation (183) is a geometric insight, not a stability limit. At fixed β, β_N, and shape, lower A raises q_*. Equivalently, at a required q_* it permits a larger normalized beta within this relation. This is the first-principles reason low aspect ratio is a plausible response to the Freidberg kink–beta coupling: it changes the geometry of the current that produces the stabilizing poloidal field.

Audit note. The equation does *not* prove that an ST is stable at a chosen β_N. The allowable β_N remains an equilibrium, wall, profile, rotation, and kinetic-MHD question. Low aspect ratio changes the relevant stability physics, so the conventional Troyon coefficient 2.8 cannot simply be reused as an ST theorem.

105. What an ST can plausibly relieve in the Freidberg closure

The conventional reference failed because its required current simultaneously made q_* too small and made steady-state replacement too demanding. Low aspect ratio attacks both mechanisms in direction, though not yet in certified magnitude:

- Equation (183) shows that smaller A increases q_* for the same normalized-beta and shape inputs. This can relax the kink-versus-pressure coupling that was restrictive in the conventional scan.
- Equation (180) shows stronger toroidal trapping. Combined with the pressure gradients already required by fusion, this creates a physical route to larger bootstrap current and a smaller externally driven current share.
- At fixed $R_0 a$, Eq. (178) gives a larger minor radius. That changes volume, current profile, wall area distribution, shaping, and power-exhaust geometry; none may be held fixed without calculation.

This is enough to formulate a responsible test, not to announce its result. The question immediately next is narrow and mathematical: does a finite-aspect-ratio ST calculation satisfy the two ratios that failed in Freidberg's conventional baseline, namely the kink ratio and the non-inductive current ratio? We will use a published low-A equilibrium and current-drive model, state every replacement assumption, and evaluate those two ratios. The centre-stack problem is *not* a premise of that test; it is the separate engineering problem revealed only after the two Freidberg constraints have been answered.

Integrated ST studies make the same methodological point. Menard *et al.* report low-aspect-ratio pilot-plant configurations only after free-boundary equilibrium, detailed device layout, and three-dimensional neutronics are coupled; their configurations must also meet tritium breeding, divertor, and maintenance requirements.[4]

[4] J. E. Menard *et al.*, Fusion nuclear science facilities and pilot plants based on the spherical tokamak, *Nuclear Fusion* 56 (2016) 106023, freely accessible at https://iopscience.iop.org/article/10.1088/0029-5515/56/10/106023/pdf.

106. The ST question is exactly Freidberg's two failed ratios

Freidberg's conventional reference failed two current-centred tests:

$$\mathcal{R}_K = \frac{q_K}{q_*} > 1, \qquad \mathcal{R}_B = \frac{1 - I_{\mathrm{CD}}/I}{f_{\mathrm{NC}}} > 1.$$

The first says that the confinement-demanded current leaves too little safety factor; the second says that the calculated self-current is smaller than the self-current share required after permitted external current drive is subtracted. For the conventional reference, pages 64 and 70 obtained 1.28 and 1.91, respectively.

The ST test uses the same accounting, but it must replace the two large-aspect-ratio inputs rather than reuse them carelessly:

$$\mathcal{R}_{K,\mathrm{ST}} = \frac{q_{K,\mathrm{ST}}}{q_{*,\mathrm{ST}}}, \qquad \mathcal{R}_{B,\mathrm{ST}} = \frac{1 - I_{\mathrm{CD,allow}}/I_{\mathrm{ST}}}{f_{\mathrm{BS,ST}}}.$$

Here $q_{K,\mathrm{ST}}$ is an ST-specific kink boundary, $f_{\mathrm{BS,ST}}$ is obtained from an ST finite-aspect-ratio bootstrap model, and $I_{\mathrm{CD,allow}}$ is the external current that the plant can afford to supply under an independently stated power and efficiency constraint. The logic has not changed. A candidate closes Freidberg's bootstrap obstruction only when those two inputs are independently obtained and the ratio is at most one.

Design consequence. The centre-stack question is not included in Eq. (). It must not be used to weaken or postpone the answer. The task first is to calculate the ST kink ratio and to compare its finite-A bootstrap prediction with an independently justified current-drive allowance.

107. The finite-aspect-ratio bootstrap model is declared

Menard *et al.* use a low-aspect-ratio equilibrium-informed bootstrap closure of the form

$$f_{\mathrm{BS}} = C_{\mathrm{BS}}\sqrt{\epsilon}\,\beta_{\mathrm{pol,th}}, \qquad \epsilon \equiv \frac{a}{R_0} = \frac{1}{A}, \tag{184}$$

where $\beta_{\mathrm{pol,th}}$ is the thermal poloidal beta and C_{BS} is a profile/safety-factor coefficient. Their stated parameterization is

$$C_{\mathrm{BS}} = \max\left(1.2 - 0.2\frac{q_*}{q_{\min}},\, 0.6\right). \tag{185}$$

Here $q_{\min}$ is the minimum safety factor of the assumed equilibrium/current profile. It is an input to this finite-A closure, not a new universal constant. [5]

The definition of poloidal beta makes the physical content visible:

$$\beta_{\mathrm{pol,th}} \equiv \frac{2\mu_0\langle p_{\mathrm{th}}\rangle}{B_{\mathrm{pol}}^2}.$$

It increases when thermal pressure rises relative to the poloidal magnetic field. Equation (184) then combines the pressure-gradient source of bootstrap current with the low-A orbit geometry through $\sqrt{\epsilon}$. This is the finite-A replacement for Freidberg's large-aspect-ratio local integral on pages 67–70.

Model and limits. The new coefficient and the equilibrium-derived poloidal beta make Eq. (184) a model closure. That is acceptable and necessary: Freidberg's own numerical bootstrap result also required a declared neoclassical closure. What would be unacceptable is to use the conventional large-A coefficient at $A \simeq 1.7$ and call the extrapolation a proof.

[5] J. E. Menard *et al.*, cited on page 105, Sec. 2.1.5, Eqs. and surrounding text on bootstrap/current-drive modeling. They state that this form is consistent with low-aspect-ratio equilibrium calculations; it is not deduced from Maxwell's equations alone.

108. A published finite-A candidate supplies every quantity the two tests need

The following is not a generic ST slogan. It is a specified low-aspect-ratio systems calculation from Menard *et al.*: $A = 1.7$, $\kappa = 2.75$, $B_{T0} = 3\,\mathrm{T}$, ITER-$H_{98} = 1.25$, and 80 MW of neutral-beam heating/current-drive power. Their density scan at $R_0 = 1.6\,\mathrm{m}$ reports

$$\beta_N = 4.0\text{–}4.7, \qquad q_* \geq 3, \qquad I = 10.8\text{–}12\,\mathrm{MA}.$$

At normalized density near 0.8, their current model gives a bootstrap fraction of about $f_{\mathrm{BS}} = 0.76$ and a beam-driven share of about 0.24. [6]

These inputs give an ST safety factor and an ST bootstrap/current-drive distribution at the same operating point. They demonstrate that the authors can construct a fully non-inductive scenario in their model. They do *not*, by themselves, state the independently derived current-drive allowance required in the numerator of Eq. (). That distinction is the difference between constructing a current balance and testing Freidberg's economic current-drive constraint.

Audit note. The displayed $\beta_N = 4.0$–4.7 range characterizes this candidate, but it is not silently used as a passed stability test in the present two-constraint comparison. Menard *et al.* provide a separate normalized-beta/stability assessment for their ST equilibria. That analysis must be evaluated explicitly before this thesis makes any broader high-β_N stability claim.

PLACE IMAGE HERE — Freidberg two-constraint closure: published low-A ST operating point. Use a two-column, deliberately non-engineering graphic. Left: the conventional Freidberg reference with $q_* = 1.56 < 2$ and required/calculated bootstrap 0.84/0.44. Right: the specified $A = 1.7$ ST model with $q_* \geq 3$ against $q_{K,\mathrm{ST}} = 2.8$, and current fractions $f_{\mathrm{BS}} \simeq 0.76$ plus $f_{\mathrm{NBI}} \simeq 0.24$. Label the right-hand current sum as an identity imposed by a fully non-inductive scenario, not as an independently passed Freidberg ratio. State the resulting required external current explicitly. Do not draw magnets, centre stacks, HTS, or reactor hardware; this image separates the resolved kink result from the still-conditional bootstrap question.

[6] Menard *et al.*, cited on page 105, Secs. 2.1.3–2.1.6 and Figs. 8–9. Values read from the published scan are shown to two significant figures. The paper calculates total non-inductive current as the sum of bootstrap and neutral-beam current.

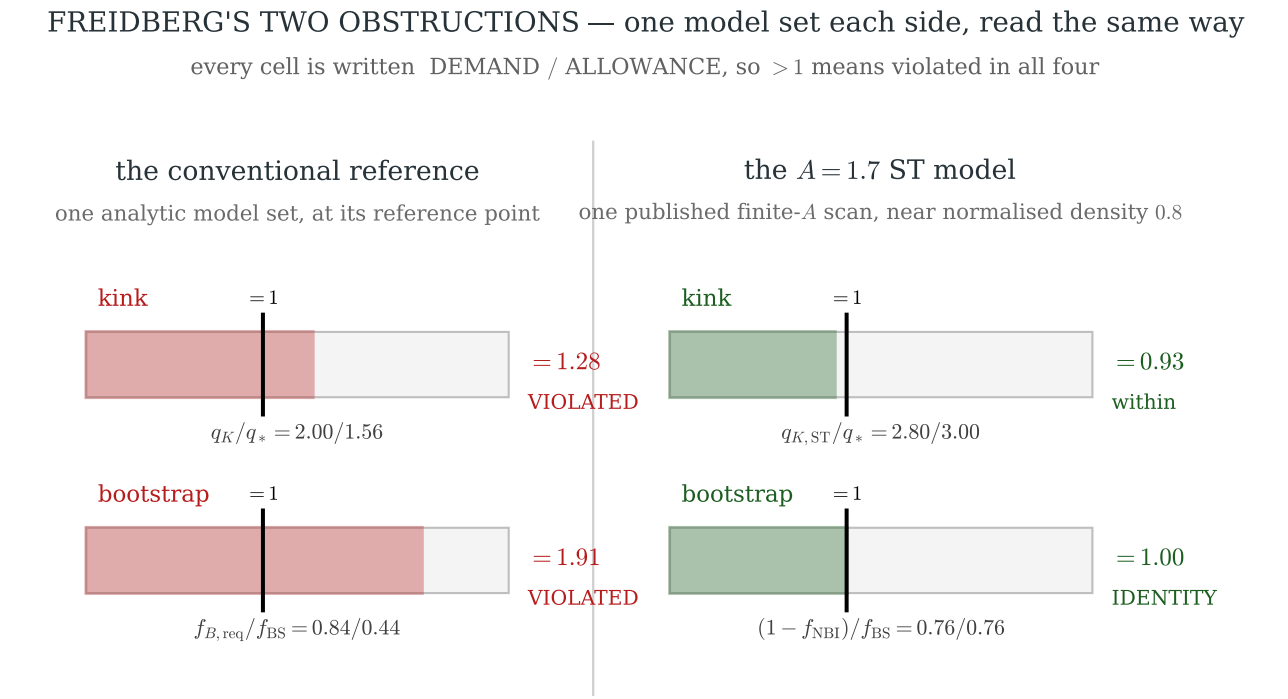


Figure 41: Freidberg's two obstructions, read with their required logical inputs visible. The ST kink comparison is an actual threshold test. The ST 0.76/0.24 current split is an identity of Menard's fully non-inductive scenario, not an independent bootstrap-ratio pass: it identifies the external current the scenario must provide. Both columns are declared model results, not properties of tokamaks or of STs.

109. First answer: the ST candidate closes the kink obstruction

The relevant ST disruption evidence used by Menard *et al.* is more demanding than Freidberg's conventional reference threshold: NSTX disruption data show a marked increase in disruptivity below $q_* \simeq 2.8$. Their stated ST design requirement is therefore

$$q_{K,\mathrm{ST}} = 2.8.$$

[7] The same candidate scan gives $q_* \geq 3$. Substitute these published values:

$$\mathcal{R}_{K,\mathrm{ST}} = \frac{2.8}{q_*} \leq \frac{2.8}{3.0} = 0.93 < 1. \tag{186}$$

This is an affirmative answer to Freidberg's kink obstruction within the declared finite-A model. It does not claim that every ST is kink stable; it demonstrates that the low-A candidate can satisfy a stricter ST-specific safety-factor criterion than the $q_K = 2$ criterion that the conventional reference failed. The geometrical reason anticipated on page 104 is now joined to an explicit operating-point test.

[7] Menard *et al.*, cited on page 105, Sec. 2.1.3. The numerical boundary comes from an NSTX data-based stability/disruption criterion and remains profile, wall, and operating-regime dependent.

110. The bootstrap result identifies the price; it does not yet test Freidberg's constraint

The current ledger does not change with aspect ratio:

$$I_{\mathrm{ST}} = I_{\mathrm{BS}} + I_{\mathrm{CD}}.$$

Divide by I_{ST}. The Menard candidate gives, to the precision supported by the published scan,

$$\frac{I_{\mathrm{BS}}}{I_{\mathrm{ST}}} \simeq 0.76, \qquad \frac{I_{\mathrm{CD}}}{I_{\mathrm{ST}}} \simeq 0.24,$$

and the two fractions sum to one. This equality is not an independent test. The Menard scenario is constructed to be fully non-inductive, so

$$\frac{I_{\mathrm{BS}}}{I} + \frac{I_{\mathrm{CD}}}{I} = 1$$

by its current-accounting definition. Dividing its own complement by its own bootstrap fraction produces 1 identically; it could not reveal a failure.

What the published point does give us is the physical and engineering burden that an independent Freidberg test must meet:

$$I_{\mathrm{CD,needed}} = (1 - f_{\mathrm{BS}})I \simeq 0.24I = 2.6\text{–}2.9\,\mathrm{MA}. \tag{187}$$

The scenario assumes 80 MW of 0.5 MeV neutral-beam heating/current-drive power to obtain this fully non-inductive balance. It therefore demonstrates a substantial finite-A bootstrap fraction and states the associated external-drive burden; it does not independently establish that a plant can afford that burden.

The proper Freidberg test remains conditional until an economic/recirculating-power analysis supplies $I_{\mathrm{CD,allow}}$ independently:

$$\mathcal{R}_{B,\mathrm{ST}} = \frac{1 - I_{\mathrm{CD,allow}}/I}{f_{\mathrm{BS}}} \leq 1 \quad \Longleftrightarrow \quad I_{\mathrm{CD,allow}} \geq I_{\mathrm{CD,needed}}. \tag{188}$$

This is the question the next research pass must answer. It is more rigorous than calling the identity $0.76/0.76 = 1$ a measured closure.

The condition can be priced with published scalars alone. Freidberg's allowance was 100 MW of wall electricity for current drive, of which 40 MW reaches the plasma (a 50% wall-to-klystron and 80% absorption chain); Menard's pilot-plant analysis likewise assumes a neutral-beam wall-plug efficiency of 0.4.[8] Keeping the published 80 MW beam state would therefore draw $80/0.4 = 200\,\mathrm{MW_e}$, twice that allowance; fitting under it would need a wall-plug efficiency of 0.8, or, if only the allowance's 40 MW were injected, a drive efficacy of about $2.6\,\mathrm{MA}/40\,\mathrm{MW} \simeq 65\,\mathrm{kA/MW}$ at that state. These are bounds on the published state, not on a redesigned ST: lowering the beam power changes heating, fast-ion pressure, fusion power, current, and bootstrap fraction together, and the FNSF point is a $Q_{\mathrm{DT}} \simeq 2$ neutron-source mission rather than a $1\,\mathrm{GW_e}$ plant. The affordability question can only be settled on a power-plant-mission ST scan carrying its own recirculating-power budget.

[8] Freidberg *et al.*, Sec. 4.6; Menard *et al.*, Sec. 2.1.6, where $\eta_{\mathrm{aux}} = 0.4$ is described as higher than presently achievable, and Sec. 5.3. The arithmetic is reproduced in `thesis/code/calculations/st_drive_affordability.py`.

111. Verdict: ST closes the kink obstruction; bootstrap closure has a stated condition

The published finite-aspect-ratio ST model answers one of Freidberg's two current-centred obstructions affirmatively:

$$\mathcal{R}_{K,\mathrm{ST}} \leq 0.93 < 1.$$

It also demonstrates a large bootstrap fraction, $f_{\mathrm{BS}} \simeq 0.76$, and makes the missing condition quantitative: the operating point needs approximately 2.6–2.9 MA of independently affordable current drive.

We must not call that second result closure until its current-drive allowance has been obtained independently. Freidberg's conventional comparison was meaningful because the permitted drive came from a separate recirculating-power calculation while the bootstrap fraction came from a neoclassical calculation. Reusing the complement of a fully non-inductive ST current balance would erase that distinction.

Design consequence. Low aspect ratio has now been shown to close the kink obstruction in a published finite-A model and to raise the bootstrap fraction to a quantified, promising level. Whether it closes Freidberg's bootstrap obstruction depends on the separate question $I_{\mathrm{CD,allow}} \geq 2.6$–$2.9\,\mathrm{MA}$.

112. Only after that answer does the separate centre-stack problem begin

The conclusion on page 111 does not say that the ST has no further problems. It says the kink question is answered affirmatively and the bootstrap question has been reduced to an explicit independent current-drive requirement. Lowering A simultaneously brings the plasma closer to the symmetry axis and strengthens the inboard toroidal field. That creates a new engineering question: can a D–T reactor fit shielding, breeding, TF conductor, support, cooling, protection, and perhaps current-initiation function inside the remaining central radius?

That question is logically separate. It cannot retroactively turn the resolved kink ratio into a failure, any more than it can replace the independent current-drive analysis still required for bootstrap closure. The remainder of this chapter changes its subject explicitly: from *what does low-A plasma physics change?* to *what engineering debt does the low-A solution create, and can HTS help pay it?*

113. The same geometry takes radial space away from the centre stack

The inboard radial ledger is the unavoidable counterargument. Let c_{cs} denote the radial build occupied by the centre-stack TF conductor, its structural case, insulation, cooling, and any allocated central-solenoid function. The innermost radius available to the toroidal field is

$$R_{\mathrm{cs}} = R_0 - a - b - c_{\mathrm{cs}} = R_0\left(1 - \frac{1}{A}\right) - b - c_{\mathrm{cs}}. \tag{189}$$

The same $BR =$ constant relation now gives

$$B_{\mathrm{cs}} = \frac{B_0 R_0}{R_{\mathrm{cs}}}, \qquad B_0 = B_{\mathrm{cs}}\left[1 - \frac{1}{A} - \frac{b + c_{\mathrm{cs}}}{R_0}\right]. \tag{190}$$

Every term in the square bracket has a physical meaning. Lowering A subtracts geometric room; b is demanded by neutrons; c_{cs} is demanded by current, structure, insulation, cooling, and possibly start-up flux. The bracket must be positive even before a usable central field can exist.

As $A \to 1$, the purely geometric factor $1 - 1/A$ tends to zero. No improved plasma stability can reverse that subtraction. A low-A reactor must instead choose among a larger R_0, thinner-but-credible inboard nuclear and magnet systems, a higher permissible B_{cs}, different start-up/current-drive architecture, or some combination. This is the centre-stack problem in its most elementary form.

PLACE IMAGE HERE — Cumulative construction state: low aspect ratio reveals the centre stack. Start with the complete conventional radial-build state, then reduce A while preserving the plasma, first wall/nuclear island b, and TF function. On the inboard side show the shrinking interval $R_{\mathrm{cs}} = R_0 - a - b - c_{\mathrm{cs}}$. Split c_{cs} into conductor, structure, insulation/cooling, and optional central- solenoid allocation. Label B_0 at the plasma and B_{cs} at the central column. The image must show that low A improves some plasma geometry while making this physical interval scarce; do not depict a solved reactor.

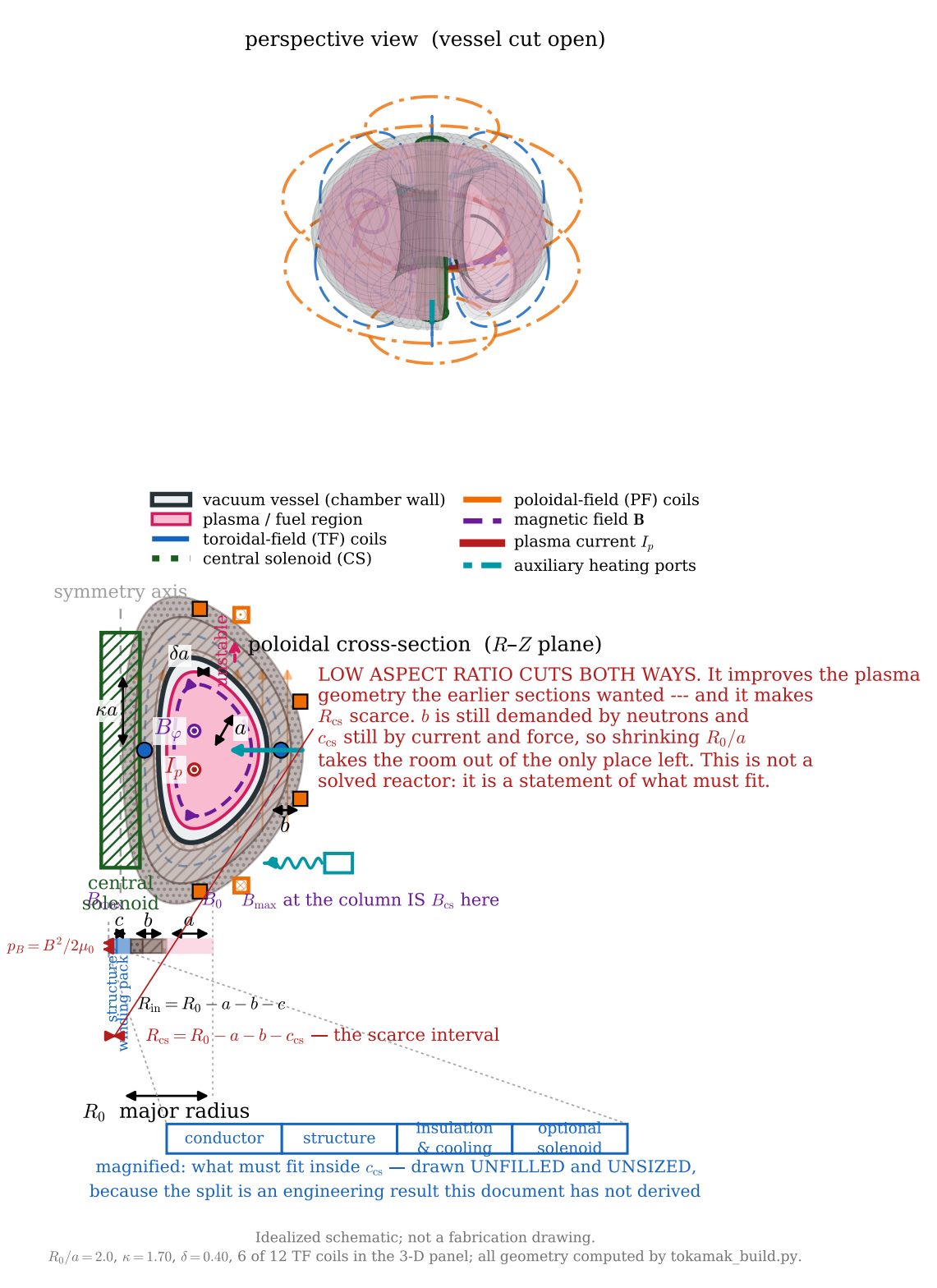


Figure 42: The same inventory as the complete radial build, at $A = 2$: the plasma, the nuclear island b and the TF function are all preserved — only the aspect ratio changes. What changes with it is the room left for R_{cs}. The magnified strip is drawn unfilled and unsized on purpose: that split is an engineering result this document has not derived.

114. HTS is a centre-stack lever, not a centre-stack eraser

Equation (190) explains why HTS becomes especially valuable in a low-aspect-ratio branch. For fixed desired B_0, reducing A raises the required central-column field B_{cs}. A conductor with credible operating current margin at higher field can make a smaller or more capable centre stack possible. It may therefore preserve some of the low-A gain that an ordinary conductor field ceiling would otherwise erase.

But the same equation returns us to page 47:

$$p_B = \frac{B_{cs}^2}{2\mu_0}.$$

The central column must carry that magnetic loading while remaining manufacturable, cooled, electrically insulated, protected, and maintainable in a neutron environment. HTS does not turn c_{cs} into zero; its conductor and support still occupy space, and higher field can increase the structural allocation. The central column also competes with shielding and tritium-breeding volume, which cannot be dismissed for a D–T plant.

> **Design consequence.** HTS can raise the credible field ceiling in the precise radial interval that low aspect ratio makes scarce. It is a powerful enabling lever for an ST, but it cannot by itself close the centre-stack equation.

115. What remains after the ST answers Freidberg's two questions

The narrowly posed Freidberg result is now established on page 111: the kink obstruction is closed within the declared finite-A model, and the bootstrap obstruction has been reduced to an explicit current-drive condition that still awaits an independent affordability calculation. That result does not settle every physical or engineering demand placed on a reactor. The following questions are separate and still require an integrated calculation:

1. a finite-aspect-ratio equilibrium and stability calculation, rather than the large-aspect-ratio kink and Troyon closures;
2. a finite-aspect-ratio neoclassical bootstrap and current-drive calculation, with explicit profiles and collisionality;
3. a transport and confinement basis applicable to the intended ST regime;
4. a radial centre-stack design that makes the bracket in Eq. (190) positive with credible stress, current density, shielding, cooling, protection, and maintenance margins;
5. a divertor, first-wall, blanket, and tritium-breeding calculation. This includes the exhaust problem Freidberg's four ratios intentionally did not close.

The centre-stack calculation begins next. Its task is not to reopen the two Freidberg ratios; one has been passed and the other has been reduced to a stated condition. Its task is to measure the exact geometric and mechanical debt created by the low-A solution. Only then can HTS, reduced shielding, larger radius, alternative current initiation, or modular maintenance be discussed as quantified responses rather than wishful labels.

116. The central column cannot evade Ampere's law

The toroidal field outside an axis-enclosing current path obeys

$$B_\varphi(R) = \frac{\mu_0 N I}{2\pi R}.$$

Evaluate it at the plasma centre:

$$NI = \frac{2\pi R_0 B_0}{\mu_0}. \tag{191}$$

This is a striking centre-stack fact. The required TF ampere-turns are set by the desired plasma product $R_0 B_0$, not by the small radius of the central column. Moving the current path inward raises the local field so that RB_φ remains constant; it does not make the required enclosed current disappear.

Let $J_{\mathrm{wp,cs}}$ be the allowed winding-pack current density after conductor, stabilizer, insulation, cooling, joints where relevant, and allocated structural fraction are counted honestly. The current-carrying cross-sectional area has the necessary lower bound

$$A_{\mathrm{wp,cs}} \geq \frac{NI}{J_{\mathrm{wp,cs}}} = \frac{2\pi R_0 B_0}{\mu_0 J_{\mathrm{wp,cs}}}. \tag{192}$$

It is a lower bound because a real column also needs support and non-conductor functions. The equation makes the danger of vague compactness claims visible: low aspect ratio shrinks the available column, while the TF current-carrying area is controlled by the plasma field target.

117. A first necessary radius inequality for the centre stack

Even in the unrealistically generous case that the entire circular column were available to the TF winding pack,

$$\pi R_{\mathrm{cs}}^2 \geq A_{\mathrm{wp,cs}}.$$

Substitute Eq. (192):

$$R_{\mathrm{cs}} \geq \left(\frac{2R_0 B_0}{\mu_0 J_{\mathrm{wp,cs}}} \right)^{1/2}. \tag{193}$$

This is not a design formula. It is a necessary condition with every real allocation set to zero except current-carrying area. A credible design needs a stronger inequality,

$$\pi R_{\mathrm{cs}}^2 \geq A_{\mathrm{wp,cs}} + A_{\mathrm{str}} + A_{\mathrm{ins/cool}} + A_{\mathrm{sol}} + A_{\mathrm{margin}}, \tag{194}$$

where A_{str} carries magnetic load, $A_{\mathrm{ins/cool}}$ provides electrical and thermal function, A_{sol} is any central-solenoid allocation, and the final term represents manufacturing, protection, and operational margin.

Combine Eq. (189) with Eq. (194). Lowering A reduces the left-side geometric radius, while the right side is set by field, materials, and operating functions. The centre-stack problem is therefore a geometric inequality test. It can fail before we ask whether the plasma is stable.

Design consequence. A proposed ST must publish a centre-stack area ledger, not merely an aspect ratio and a peak field. Without the conductor, structure, shielding, and start-up allocations, there is no check that the proposed central column exists.

118. Higher central field helps the plasma only by increasing mechanical demand

At the centre-stack radius, the local magnetic-pressure scale is

$$p_{B,\mathrm{cs}} = \frac{B_{\mathrm{cs}}^2}{2\mu_0}.$$

If a representative structural load path of area A_{load} is carried by material limited to σ_{allow}, force balance requires

$$A_{\mathrm{str}} \gtrsim \frac{p_{B,\mathrm{cs}}}{\sigma_{\mathrm{allow}}} A_{\mathrm{load}}. \tag{195}$$

The detailed geometry determines A_{load} and the actual stress distribution; the quadratic field dependence is already inescapable. As low A reduces R_{cs}, Eq. (190) pushes B_{cs} upward for a fixed B_0. The winding pack and its supporting structure then compete for the same shrinking area.

HTS can improve the conductor part of the ledger by providing useful current margin at higher field. It does not make the structural fraction vanish. In fact, if a design uses the higher field that HTS enables, Eq. (195) can require more support. That is why a high-field ST must report both conductor performance and a credible mechanical case.

Model and limits. Equation (195) is a lower-bound scaling, not a finite-element stress certification. Real centre columns need three-dimensional load paths, strain limits for the superconductor, cyclic-fatigue analysis, joints and terminations where used, electrical insulation, quench or fault protection, and neutron-degraded material properties.

119. The transformer also loses flux swing as the column shrinks

The transformer argument from page 20 gives a second, independent centre-stack constraint. A solenoid of effective radius R_{sol} changes linked magnetic flux by the scale

$$\Delta\Psi_{\mathrm{sol}} \sim \pi R_{\mathrm{sol}}^2 \Delta B_{\mathrm{sol}}. \tag{196}$$

Faraday's law converts this flux swing into the volt-seconds available to initiate and ramp plasma current. Since R_{sol} must fit inside the same central column, $R_{\mathrm{sol}} < R_{\mathrm{cs}}$, the available inductive flux falls at least quadratically as the column becomes narrow.

This fact connects directly to the Freidberg current ledger. A pulsed ST cannot claim steady operation simply because it can make an initial plasma current; it needs enough flux swing for its chosen pulse or a non-inductive start-up and sustainment route. Conversely, a steady-state ST with high bootstrap fraction has a second incentive to develop non-inductive current initiation: it can release central-column area otherwise reserved for a large solenoid.

The choice is an engineering-plasma co-design decision. Removing solenoid allocation from Eq. (194) may help the radial ledger, but transfers an obligation to radio-frequency, neutral-beam, helicity-injection, or other start-up/current-drive systems. It cannot be deleted from the thesis by a diagram.

120. The central-stack test is now ready to be made quantitative

The equations above turn the phrase centre-stack problem into five checkable questions:

1. What B_0, R_0, and A are demanded by the plasma mission?
2. Does Eq. (190) leave positive inboard radius after the required nuclear build b?
3. Does the available πR_{cs}^2 exceed the full area ledger in Eq. (194)?
4. Can the structural allocation carry the field-dependent load without exceeding stress or superconducting-strain limits?
5. If solenoid area is reduced, what explicit non-inductive start-up and current maintenance system replaces its flux swing?

The next step is not to assign convenient answers. We will define a transparent reference ST radial ledger and vary its field, radius, shielding, conductor current density, and solenoid allocation. That will identify whether an HTS-enabled central stack can close geometrically, where it fails, and which missing technologies would have to carry the remaining burden.

121. A published ST study shows why the ledger must be literal

It is useful to test the preceding geometry against a published allocation, provided we do not mistake one study's choices for universal constants. Menard *et al.* examined an illustrative FNSF configuration with $A = 2$, $R_0 = 1.87\,\mathrm{m}$, and a 30 cm centre-column radius. Their stated inboard nuclear protection included approximately 60 cm of water-cooled tungsten-carbide shielding; the corresponding radial picture also includes a 2 cm first wall. Their magnet assessment used a $70\,\mathrm{MA\,m^{-2}}$ winding-pack current-density assumption and imposed both support-stress and superconducting-strain constraints. [9]

The first calculation needs no plasma model. At $A = 2$,

$$a = \frac{R_0}{A} = 0.935\,\mathrm{m}, \qquad R_0 - a = 0.935\,\mathrm{m}.$$

Temporarily subtract only the stated shield and first-wall thickness:

$$0.935 - 0.600 - 0.020 = 0.315\,\mathrm{m}. \tag{197}$$

This rough geometry check is close to the study's 0.30 m centre-column radius. It is not an argument that 15 mm is spare room: it says that even before every cable, support, coolant channel, tolerance, and current-initiation function is drawn, the low-A inboard budget is already measured in centimetres.

> **Design consequence.** The centre stack is not an abstract penalty applied after plasma optimization. A realistic neutron shield can consume most of the inboard radius in an $A \simeq 2$ machine. A claimed compact ST must display its own radial ledger with the same honesty.

[9] Menard *et al.*, cited on page 105, Secs. 2.1 and 5.1, especially their Figs. 46–47. The numbers in this paragraph are inputs and results of that particular FNSF study, not universal requirements.

122. Why a D–T centre stack begins with shielding, not conductor

The inboard build b in Eq. (189) is not a decorative engineering margin. D–T fusion emits 14.1 MeV neutrons, which are uncharged. Magnetic confinement does not bend them. They stream into the first wall, blanket, shield, structural material, and magnets; a reactor must absorb their heat, breed tritium from lithium, and keep radiation damage to components within their allowed lifetime.

That is why the 60 cm shield in the illustrative calculation on page 121 cannot simply be traded for a better magnet. It was a neutron-protection choice for the study's mission and material assumptions. Changing it demands a new neutronics, heating, lifetime, and tritium-breeding calculation. The same study found that at low aspect ratio the inboard side can also be important to a tritium-breeding solution.[10]

This gives the radial order of operations:

plasma mission	⟶	neutron, heat, and tritium functions	⟶	space left for TF conductor, support, and st

Reversing that order—choosing a desired tiny centre column first and assigning the nuclear functions whatever space survives—would be the same category error as assuming a finished tokamak at page 1. The nuclear build is a design demand that helps reveal the permitted machine shape.

PLACE IMAGE HERE — Cumulative ST radial ledger: a neutron first, magnet second construction. Draw one clean outboard-midplane to inboard-midplane cross-section for the illustrative $A = 2$, $R_0 = 1.87$ m state. On the inboard side add components in causal order: plasma edge, first wall, approximately 0.60 m shielding as a study-specific allocation, then the approximately 0.30 m central-column envelope. Inside that envelope use unfilled subdivisions for TF winding, support, insulation/cooling, and optional solenoid–do not assert their final sizes. Include the arithmetic of Eq. (197) and a clear illustrative published-allocation, not-universal-reactor-template note.

[10] Menard *et al.*, cited on page 105, discuss the inboard shielding and breeding allocation in their Sec. 5.3. Their conclusion is configuration-specific; the general point here is only that breeding and shielding claim radial space before the magnet is sized.

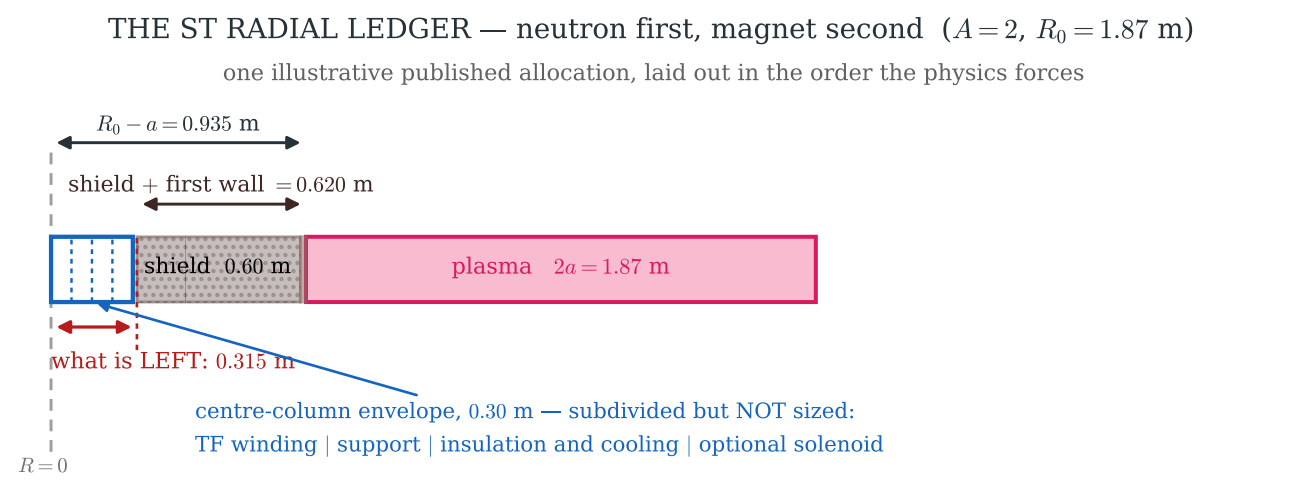


Figure 43: The published illustrative allocation, laid out in the order the physics forces: shield first, magnet into what is left. The arithmetic is Eq. (197) evaluated. The 15 mm difference from the study's stated column radius is not spare room — it is the residue of a check that has not yet paid for one cable, support, coolant channel or tolerance.

123. What HTS changes in this ledger

HTS changes a specific term: the credible current density and operating field margin of the TF winding pack. In the necessary bound,

$$A_{\mathrm{wp,cs}} \geq \frac{2\pi R_0 B_0}{\mu_0 J_{\mathrm{wp,cs}}},$$

raising a credibly usable $J_{\mathrm{wp,cs}}$ reduces the *minimum* current-carrying area for a prescribed $R_0 B_0$. It can therefore create room that a conventional conductor cannot. In the field relation

$$B_{\mathrm{cs}} = \frac{B_0 R_0}{R_{\mathrm{cs}}},$$

HTS can also make a larger central field technically plausible without immediately crossing the conductor's critical surface.

Those are real levers, but neither equation says that the whole right side of Eq. (194) scales down as $1/J_{\mathrm{wp,cs}}$. The support, insulation, cooling, protection, radiation, and manufacturing terms remain. Moreover, raising B_{cs} raises $p_{B,\mathrm{cs}} \propto B_{\mathrm{cs}}^2$, which can enlarge the structural term. A good HTS design may close the equation; HTS alone is not the missing calculation.

The published low-A work gives a concrete architectural direction rather than a shortcut. A later $A = 2$, $R_0 = 3\,\mathrm{m}$ HTS pilot-plant concept places all superconducting poloidal-field coils outside the TF system, uses continuous HTS TF coils, retains only a small central solenoid, and keeps inboard blanket/shield functions in the radial build.[11] Each choice answers one item in the ledger: outside PF coils avoid consuming central volume; a small solenoid makes non-inductive start-up more important; HTS improves the TF field option; blanket and shield remain non-negotiable.

Design consequence. HTS is the enabling material in a co-design, not an exemption from the co-design. Its success must be reported as a closed area, stress, radiation, and start-up ledger at the chosen R_0, A, and B_0.

[11] Menard *et al.*, cited on page 105, Sec. 5.4 and Fig. 53. It is a conditional systems study, not a demonstration that every one of its performance targets has been experimentally achieved.

124. What remains after the Freidberg answer

The causal argument has reached a useful, deliberately narrow conclusion. The finite-A calculation on pages 106–111 answered Freidberg's two current questions. The earlier first-principles geometry explains why that answer was possible:

1. Eq. (183) changes the current–safety-factor geometry in the favorable direction for the kink/current coupling.
2. Eq. (180) strengthens the trapping physics that underlies a route to bootstrap current.

HTS then supplies a plausible material lever for the adverse inboard field and winding-area problem created by the same geometry. Published integrated ST studies show that these ingredients can be put into a serious systems calculation.

That establishes the result relevant here: a specified finite-A ST model closes the kink constraint that failed in Freidberg's conventional baseline and reduces the bootstrap constraint to an explicit, quantified current-drive condition ($I_{\mathrm{CD,allow}} \geq 2.6$–$2.9\,\mathrm{MA}$, page 111), whereas merely raising H in that baseline did neither. It does not establish a theorem that every ST or every other reactor constraint is closed; exhaust, blanket, tritium, radiation, maintenance, and cost remain their own questions.

The next calculation must therefore have two ledgers, evaluated together:

finite-A plasma ledger	centre-stack reactor ledger
q, β_N, equilibrium/stability	b, R_{cs}, A_{wp}, A_{str}
bootstrap and driven current	shield, breeding, stress, start-up
transport and exhaust	HTS margin, protection, maintenance

The next section makes the centre-stack ledger into an explicit, auditable parameter scan. It does not relabel that separate engineering question as a failed Freidberg plasma constraint.

125. A dimensionless necessary test for a central TF winding

For a first scan, reserve the inboard nuclear build b before allocating the central column. The outer radius of the column envelope is then

$$R_{\mathrm{env}} = R_0 \left(1 - \frac{1}{A}\right) - b. \tag{198}$$

This is deliberately more generous than a finished design: it gives every remaining square millimetre to the central column before deciding how much is conductor, structure, cooling, or solenoid.

Let f_{wp} be the fraction of the envelope's circular cross-sectional area that the designer has explicitly allocated to the TF winding pack. It is an accounting input, not a material constant. Equation (192) then requires

$$\frac{2\pi R_0 B_0}{\mu_0 J_{\mathrm{wp,cs}}} \leq f_{\mathrm{wp}} \pi R_{\mathrm{env}}^2.$$

After cancelling π, define the dimensionless winding occupancy

$$\eta_{\mathrm{wp}} \equiv \frac{2R_0 B_0}{\mu_0 J_{\mathrm{wp,cs}} \left[R_0(1 - 1/A) - b\right]^2}. \tag{199}$$

The necessary test is simply

$$\eta_{\mathrm{wp}} \leq f_{\mathrm{wp}} < 1. \tag{200}$$

If $\eta_{\mathrm{wp}} > 1$, even a column made entirely of winding pack cannot carry the required TF current. If $f_{\mathrm{wp}} < \eta_{\mathrm{wp}} < 1$, a central column exists only on paper because the current pack has consumed space explicitly promised to the other functions.

Model and limits. This is a necessary electromagnetic-area test. It treats the winding pack as uniformly current carrying and does not certify field quality, stress, strain, quench protection, neutron lifetime, joints, penetrations, or manufacturability. Those omissions make passing the test easier, never harder.

126. Solving the test exposes a lower-aspect-ratio boundary

The test can be solved for the smallest aspect ratio permitted by the chosen area allocation. Take the positive square root of Eq. (200):

$$R_0\left(1-\frac{1}{A}\right)-b\geq\left(\frac{2R_0B_0}{\mu_0J_{\mathrm{wp,cs}}f_{\mathrm{wp}}}\right)^{1/2}.$$

Divide by R_0 and rearrange:

$$A\geq A_{\mathrm{min,wp}}\equiv\left[1-\frac{b}{R_0}-\left(\frac{2B_0}{\mu_0J_{\mathrm{wp,cs}}f_{\mathrm{wp}}R_0}\right)^{1/2}\right]^{-1}. \tag{201}$$

The bracket must be positive. If it is zero or negative, the stated R_0, B_0, nuclear build, winding-pack current density, and allocation cannot coexist at *any* aspect ratio in this simplified central-column arrangement.

Every trend in this equation has a physical reading:

- Increasing B_0 increases the required ampere-turns and raises the minimum allowed A.
- Increasing b/R_0 claims the inboard radius for nuclear function and raises the minimum allowed A.
- Increasing credible $J_{\mathrm{wp,cs}}$ or giving the winding pack a larger honest fraction f_{wp} lowers the electromagnetic area demand.
- Increasing R_0 helps twice: the nuclear build occupies a smaller fraction of the radius and the required winding area is less crowded into the column.

The last two points identify exactly what HTS and a less compact plant can improve. They also show why neither word automatically guarantees a viable ST.

127. A transparent teaching scan: field makes the low-A limit move

Use the deliberately declared teaching inputs

$$R_0 = 3.0\,\mathrm{m}, \quad b = 0.65\,\mathrm{m}, \quad J_{\mathrm{wp,cs}} = 70\,\mathrm{MA\,m^{-2}}, \quad f_{\mathrm{wp}} = 0.40.$$

The current-density scale is comparable to an assumption used in the illustrative Menard study on page 121; the radius, nuclear build, and 40% allocation here are *not* a proposed reactor. They merely make the algebra visible. Substitution in Eq. (201) gives:

B_0 (T)	$A_{\mathrm{min,wp}}$	η_{wp} at $A = 2$
4	1.97	0.378
6	2.24	0.567
8	2.54	0.756

At $A = 2$, the available envelope radius is $R_{\mathrm{env}} = 3(1 - 1/2) - 0.65 = 0.85\,\mathrm{m}$. The first row barely clears the assigned 40% winding allocation. The latter two do not: they require more winding area than has been allocated even before support, insulation, cooling, or a solenoid are tested. The table is reproducible from `centre_stack_ledger.py`; every input is displayed at the top of that file.

This is the exact sense in which HTS is a boost mechanism. If it raises the *credible system-level* $J_{\mathrm{wp,cs}}$ at the relevant field, temperature, strain, radiation, and protection margin, it can lower $A_{\mathrm{min,wp}}$. If the design instead spends the HTS benefit to raise B_0, the square-root field penalty pushes $A_{\mathrm{min,wp}}$ upward. The net result is a co-design decision, not a one-directional law.

Audit note. Passing the 4 T row does not make it a reactor. The table omits the structure term that grows with central field, the non-winding allocations, the neutron and tritium calculation behind b, and all plasma/stability/exhaust tests. It is intentionally a failure detector, not a feasibility certificate.